GE VERNOVA

ADMS

16.25.1

DNAF User Guide

Copyright

Change History

Date	Change Description
May 2, 2023	ADMS 3.16: • Removed section "Look-Ahead DPF Analysis". • Editorial and formatting updates.
July 21, 2023	ADMS 3.16.0.1: • Updated section Viewing Results.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 08, 2024	ADMS 16.7.0: • Removed section "DER Manual Dispatch Display". • Updated document format and styling.

Contents

1. Introduction

The GE Advanced Distribution Management System (ADMS) allows distribution operators and engineers to monitor, control, analyze, and plan the operation of the power distribution system. The Distribution Network Analysis Functions (DNAF) suite forms an integrated package of applications that operate in real-time mode and study mode. Real-time mode uses the current power system conditions (SCADA and Advanced Metering Infrastructure measurements) in evaluating the system. These measurements consist of analog and status measurements, including switch statuses. The status measurements can also be modeled as a multiphase point measurement which contains individual phase statuses in one point. Study mode solves operational studies using data retrieved either from a savecase or a real-time snapshot of the system.

The core network analysis application is the Distribution Power Flow (DPF). The power flow engine generates a large amount of output data that cannot be easily observed by the operator. For this reason, several distribution analysis functions are used to analyze the results of the power flow and to present summarized information to the operator.

Voltages and angles from the Energy Management System (EMS) State Estimator solution can also be used in the real-time DPF to initialize the distribution source voltage locations. These locations can be on the source side of the station transformer or at locations in the sub-transmission network.

The GE DNAF suite includes the following applications:

- **Distribution Power Flow (DPF):** This function finds the complex voltages at all nodes and the power flows through all feeder segments in the distribution system.

  The following functions are executed in conjunction with DPF:

  - **Bus Load Allocation (BLA):** This function provides estimates for the loads (kW and kVAR) at the distribution feeder nodes.
  - **Limit Monitor (LIM):** This function checks for branch flows and voltages that are outside of acceptable operating limits.
  - **Power Quality Index Evaluation (PQI):** This function determines the extent of feeder voltage violations (including voltage imbalance).
  - **Loss Analysis (LA):** This function calculates a device's technical losses and aggregates the losses by feeder and by station.
- **Protection Validation (PRV):** This function calculates phase-to-ground, double-phase-to-ground, and three-phase fault currents for each application protection zone, where a PRV zone is bounded by feeder circuit breakers and field reclosers (fuses may also be considered in study mode).
- **Fault Location (FL):** This function uses fault amp measurements and faulted phase status measurements to determine potential fault locations along distribution feeders.
- **Short Circuit Calculation (SCC):** This function calculates single phase-to-ground, double phase-to-ground, phase-to-phase, and three-phase currents for a selected location (power

transformer, line, and switch devices).

- **Load and Volt/VAR Management (LVM):** This function optimizes and coordinates capacitors, voltage regulating transformers (regulators and LTCs) and distributed generation VAR output to maintain system operation within voltage limits. LVM can be used as a least-intrusive demand management tool.
- **Fault Isolation and Service Restoration (FISR):** This function provides switching plans to isolate faulted sections and restore service to the non-faulted sections.
- **Automatic Feeder Reconfiguration (AFR):** This function provides plans to unload overloaded segments, and it also generates Return to Normal (RTN) switching plans.
- **Planned Outage Study (PLOS):** This function provides plans to isolate a component, which minimizes the number of interrupted customers. This application is only available in study mode.
- **Fault Isolation and Service Restoration Contingency Analysis (FISRCA):** This function executes a FISR analysis for every switchable section in the station. This application is only available in study mode.
- **Faulted Section Determination (FSD):** This function provides switching plans to identify a fault and isolate faulted sections. The FSD function applies a switching-based approach with step-by-step opening and closing switching cycles. At stage 1, a faulted feeder downstream of a faulted power transformer is identified and isolated. At stage 2, the faulted sections are identified and isolated.
- **Surgical Load Shed (SLS):** This function provides DMS data (critical loads, load types, transfer/throwover schemes, reserved capacity commitments, etc.) to the Scada Load Shed function to accurately prioritize and shed circuits, to obtain a desired target reduction in system load.

1.1. Users

The users of ADMS are distribution power system operators and engineers (analysts). This user guide provides a high-level explanation of each distribution network analysis function, along with details on how to manually execute the network applications.

The operators work with a real-time version of the analysis functions, which enables them to continuously monitor and analyze the present condition of the distribution system and perform any actions necessary to maintain satisfactory system operation.

Operators and engineers use a study-mode version of the network analysis functions, to evaluate study cases with user-defined changes or forecasted conditions, including loading changes, device parameter changes, capacitor switching, reconfiguration studies, and planned outage requests.

For a detailed explanation of the application solution engine(s) and configuration information, consult the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide* and the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

1.2. Terminology

For a list of terms and acronyms, and their definitions, refer to the *ADMS Glossary of Terms*.

2. Running the DNAF

This chapter describes general usage of DNAF functions.

The standard DNAF applications (DPF, PRV, FL, and SCC) are typically used to analyze the system to estimate load, locate faults, estimate bus voltages and losses, etc. The advanced DNAF applications (LVM, FISR, PLOS, AFR, and FISRCA) find solutions for improving network characteristics, restoring de-energized customers, isolating a component, etc.

The advanced DNAF applications can operate in the following modes:

- **Advisory:** In advisory mode, no output commands are sent after a solution. Switching plans are performed manually by the operator. This option is available for all real-time applications. In study mode, applications operate only in advisory mode.
- **Semi-Automatic:** In semi-automatic mode, output commands are sent after operator approval. This option is available for real-time FISR and AFR only.
- **Closed Loop:** In closed loop mode, output commands of a top-ranked plan are sent directly (without human interaction) to field devices to execute switching orders through SCADA. This option is only available for real-time LVM, FISR, and AFR.

  > **Note**
  >
  > In study mode, the operator can automatically implement or restore (undo) a switching plan by clicking the relevant button on the plan tabular display.

- **Closed Loop - Calculated Voltage Limit Violation Not Worse:** Works similarly to the Closed Loop mode, but also considers plans with voltage violations. A top-ranked plan can be executed automatically if the calculated voltage violations are not made worse by the plan. A voltage margin specified in the DNAF Configuration Editor is used to determine if a plan resulting in voltage violations is acceptable. This option is only available for real-time FISR and AFR.

The default plan execution modes for LVM, FISR, and AFR are defined on the application configuration pages in the DNAF Configuration Editor. At run time, the operator can change the plan execution mode in the Real-Time DNAF System Configuration display.

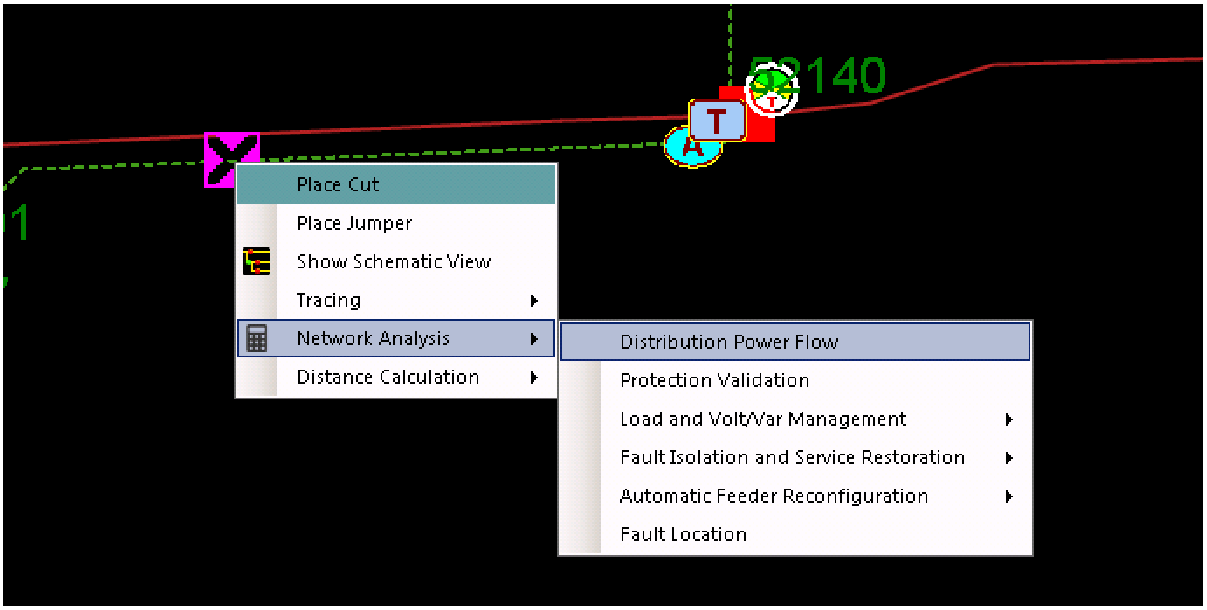
2.1. Executing DNAF Applications Manually

You can manually invoke a network application from the network view or from the DNAF Solution display. Depending on the configuration specified in the DNAF Configuration Editor, different applications are available in real-time mode and in study mode. (For example, Planned Outage Study is a study-mode only application.) The manual request is the highest priority request, followed by event requests and then periodic requests.

To execute a network application from the network view (available for real-time applications only):

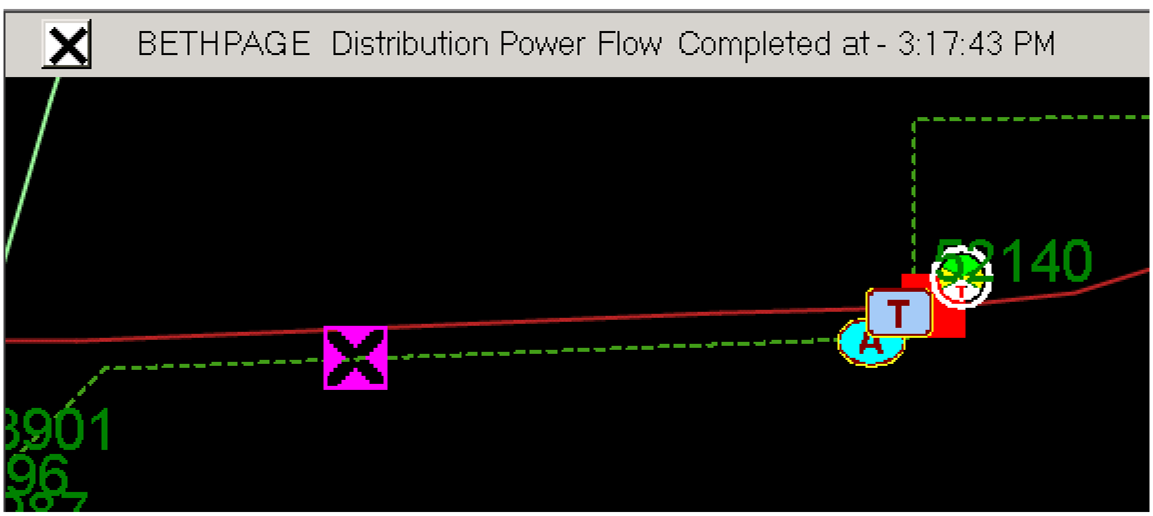
1. On a geographic or schematic network view, right-click a line.

   The Line context menu appears.



2. Point to Network Analysis and then select the application.

   A DNAF status bar appears, showing the current solution status.



To open the DNAF Solution display:

1. On a geographic or schematic view, select a substation, a device, or a line.
2. On the device-specific toolbar, click the Run Network Analysis button.

By default, the station that is initially selected in the DNAF Solution display is the energizing station of the selected device when the DNAF Solution display was opened. This is the station that will be solved unless you clear the check box. You can select additional stations as needed by checking the station boxes.

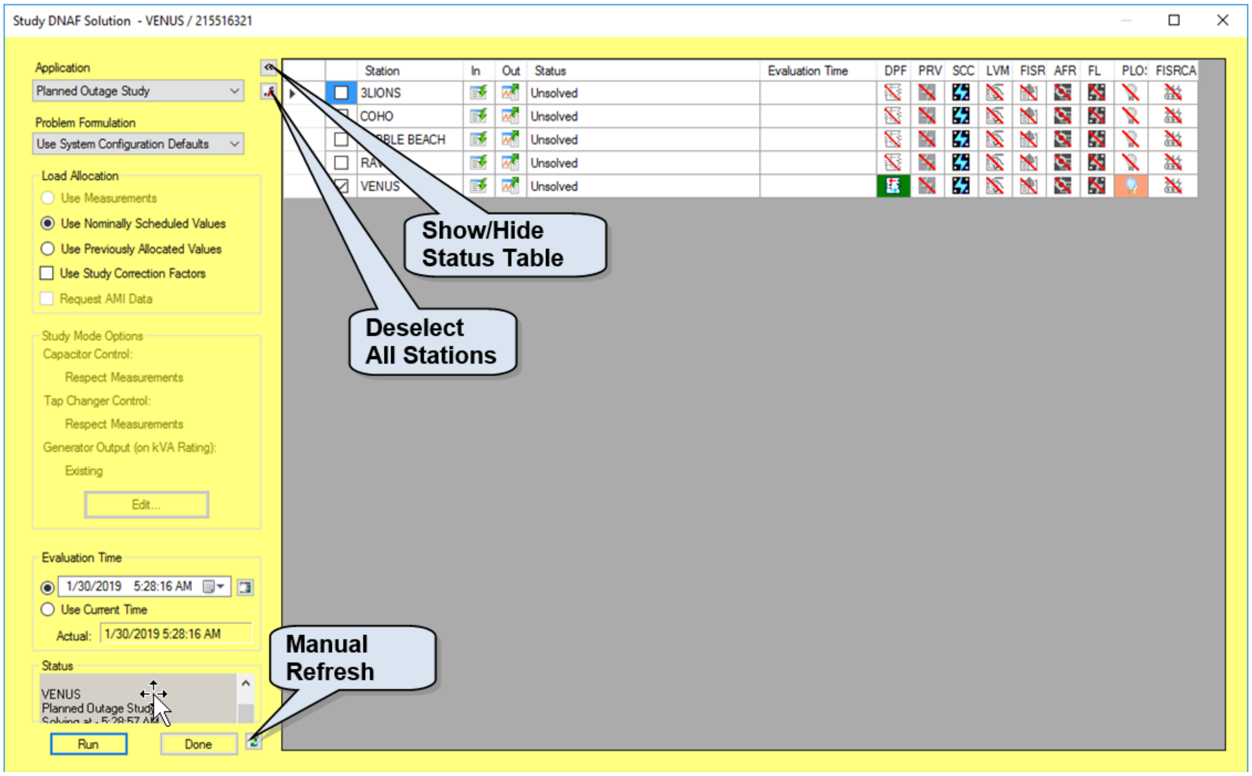


Figure 1. DNAF Solution Display (Study Mode Version)

The background color for the DNAF Solution display is similar to the current mode background color, which can be configured through the Net Dynamics Configuration Editor. Generally, different colors are used for real-time, study, and simulator modes to clearly distinguish between the operating environments.

In study mode, the display provides additional options for load allocation, capacitors, tap changers, generator/DER Loading, and evaluation time.

The left side of the DNAF Solution display contains application options and controls and shows the solution status. The right side of the display contains a status table that shows solution statuses for stations. The In/Out buttons and the application icons are clickable and navigate to the relevant data input and solution result displays.

2.1.1. DNAF Solution Display - Left Pane

Application: Contains available DNAF applications. The Distribution Power Flow application is selected by default. The DNAF Solution display retains the last application that was selected.

Problem Formulation: Contains problem formulations that are available for the selected application. The Use System Configuration Defaults option provides the default problem formulation for the selected application. The Problem Formulation field is available for the advanced DNAF functions (FISR, AFR, PLOS, FISRCA, and LVM). These functions are described in later chapters.

Load Allocation: Contains options that are associated with the Bus Load Allocation algorithm.

- The following three options specify the load allocation approach. The options are configurable only in study mode. (In real-time mode, load allocation uses the actual measurements.) Only one of the options can be selected for a solution.
    - **Use Measurements:** If selected, the load allocation uses all valid analog measurements. If no valid measurements are available in study mode, this option is not available.
    - **Use Nominally Scheduled Values:** If selected, the load allocation uses the nominal customer load multiplied by the scaling values from the appropriate load profile (based on season, day type, and evaluation time) for setting the demand values (kW and kVAR).
    - **Use Previously Allocated Values:** If selected, the load allocation uses the load values that were determined from a previous power flow solution. This option is commonly used when performing switching studies (after the loads have been allocated, based on an earlier measurement set using the Use Measurements option).
- **Use Study Correction Factors:** If selected, the study load correction factors that are calculated and persisted in the real-time power flow and stored in the station model file are applied during the load allocation. This option does nothing if the Use Previously Allocated Values option is selected.
- **Request AMI Data:** If selected, the customer meter measurement data is used. This option is only available for the DPF and LVM applications.

Study Mode Options: Contains options that allow the operator to study "what if" scenarios. The options are available only for DPF, FISR, and AFR in study mode.

- For DPF, click the Edit button to show the Study Mode Options display, where you can edit study-mode options.
- For FISR and AFR, you can select the "Make All Switching Devices Eligible" option to consider all switching devices for switching plans.

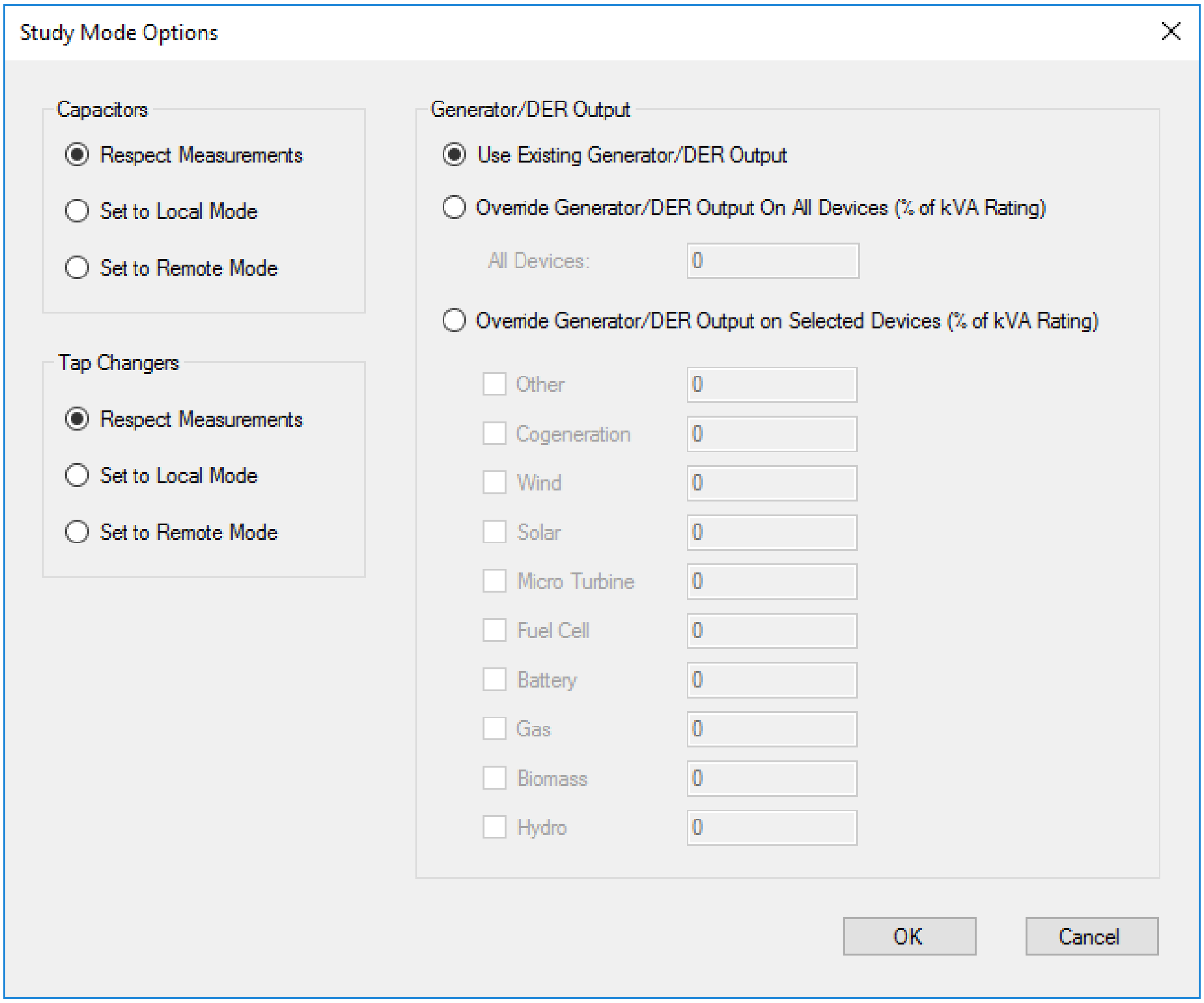


Figure 2. Study Mode Options Display

This display contains three groups of options: Capacitors, Tap Changers, and Generator/DER loading.

A study mode Distribution Power Flow (DPF) execution triggered from the line's right-click menu in the geographic display will always use the default Generator/DER Output setting, Use Existing Generator/DER Output.

Capacitors and Tap Changers: These groups contain options that allow the operator to study "what if" scenarios in different control modes. The options are available only in study mode and apply only to devices that have both remote and local control capability. Only one option can be selected.

- **Respect Measurements:** If selected, DPF emulates the local controller behavior for devices that do not have a valid SCADA remote/local measurement, or the remote/local measurement value is local. Step/position of other respective devices will not change.
- **Set to Local Mode:** If selected, DPF emulates the local controller behavior for all the respective devices.

- **Set to Remote Mode:** If selected, DPF is prohibited from changing the step/position of any of the respective devices.

Generator/DER Output: This group contains options that allow the operator to study "what if" scenarios by setting the generation output target to from 0% to 100% of the nameplate rating of generators and energy resources by different types. The types supported are: Wind, Solar, Micro Turbine, Fuel Cell, Battery Storage, Gas, Biomass, Cogeneration, Hydro, and Other.

If enabled, the real and reactive power target of individual devices can be determined from the nameplate rating, rated power factor, and the manually entered loading. This will override the scheduled or telemetered injection values. However, this will not override the manually entered real and reactive output values set through the Manual Input Parameter Entry display for individual devices.

The options are available only in study mode. Only one option can be selected.

- **Use Existing Generator/DER Output:** If selected, the scheduled or telemetered injection values are used.
- **Override Generator/DER Output on All Devices (% of Rating):** If selected, the manually entered output target will be used for all devices of all types.
- **Override Generator/DER Output on Selected Devices (% of Rating):** If selected, the manually entered output target will be used for devices of the selected types. Scheduled or telemetered injection values will be used for devices of the types not selected.

Evaluation Time: Contains options that allow setting the study time for which the solution is run. The solution uses the time and date to determine the allocated loads, based on dynamic profiles/schedules or statically modeled load profiles. The Evaluation Time options can be set only in study mode. (In real-time mode, the application is solved for the current time.)

- **Manual Time:** Use this option to set the solution date and time manually. The button to the right of the Date/Time field toggles between the date picker and the up/down control.
- **Current Time:** If selected, the solution is run for the actual time, based on current conditions.

Status: This field shows the status of the application execution. It refreshes every time the Run button is clicked.

Buttons:

- **Run:** Executes the DNAF application for the selected stations.
- **Done:** Closes the display.
- **Refresh the Status Table:** Clears the Status text field and refreshes the solution status table at the right pane of the display.

2.1.2. DNAF Solution Display - Right Pane

The table on the right of the display provides navigation to input and output displays, and it also provides a summary of the applications that have been solved for each substation and navigation to the results.

Station: Lists all stations in the system. For study mode, lists the stations that are loaded in the study case.

In: Opens the appropriate data input display for the application selected in the Application list.

Out: Opens the summary results display for the application selected in the Application list.

Status: Shows the latest evaluation status.

Evaluation Time: Shows the time when the study was completed.

Application Columns: In the application columns, each button is colored to graphically reflect the time when an application was last solved, or the solution status. The colors are:

- green for recent solutions
- light green for old solutions
- blue for very old solutions
- orange for suspect and rejected solutions

Clicking a button under a specific application column brings up the latest summary results display for the selected station. The selected buttons are shown in dark blue.

Buttons:

- **Select:** Selects or deselects a station from the list.
- **Deselect All Stations:** Deselects all currently selected stations.
- **Show/Hide the Status Table:** Toggles the display of the status table.

2.2. Executing DNAF Applications Automatically

In real-time mode, network analysis functions can be configured to execute periodically (DPF, PRV, and LVM) or based upon specific events, such as a fault or network topology change. This way, the operator knows that the results are always current. In study mode, all network applications can only be solved manually, using the DNAF Solution display.

The periodic and event triggers are configured in the DNAF server configuration file. For more information, refer to the `NetServerDnafConfig.txt` section in the *ADMS Software Installation and Maintenance Guide*.

For example, event triggers include the following:

- **Topology Change:** A topology change includes a change in switching device status or the insertion/removal of a temporary modification including a cut, temporary ground, or jumper. A capacitor status change is not considered an event trigger for LVM, but it is an optional trigger for DPF.

  For a topology event, main and lateral triggers are separated (options include no topology change triggers, main only, lateral only, or both main and lateral). Substation internal equipment is considered part of the circuit main for processing topology event triggers.

- **Sudden Measurement Change Trigger:** A feeder head sudden measurement change trigger may also trigger an application. This trigger includes a flow trigger (Amps) and a voltage trigger.

  If the measurement change exceeds a user-configurable threshold (in Amps for the flow trigger and percent Volts for the voltage trigger), and if a specified time interval has elapsed since the last sudden measurement change solution, an application is triggered and solved. The elapsed time interval is used to prevent too many solutions from being triggered if there is an error in the measurement that is causing its value to change dramatically between measurement scans.

  The two sudden measurement change threshold values (Amps and Volts) are set using the DNAF Configuration Editor; refer to DNAF Configuration Editor.

  When DNAF_DPF_CHECK_MEAS_CHANGE is also set to **1**, an additional validation check is performed before the DPF measurement change event trigger is issued:

  Measurement age check: The measurement timestamp must be recent, i.e., within the maximum age defined by the DNAF_MAXTIME_CHECK_MEAS_CHANGE parameter (default 300 seconds). If the measurement is outdated, the DPF event trigger is skipped.

  These check prevent redundant DPF evaluations caused by unchanged or stale SCADA voltage measurements.

> **Note**
>
> The SCADA sudden measurement change trigger currently applies only to those voltage and current measurements that are attached to the station internal objects (such as reference bus interfaces, station transformers, breakers, and feeder heads). The trigger does not apply to the measurements of any device outside of the station internals (such as reclosers and sensors).

- FISR can be triggered automatically by a fault or by a loss of voltage condition.
- AFR can be triggered automatically when DPF detects an overload in a power transformer or a feeder main (backbone) line.

The following table describes how measurement and topology change events trigger DNAF applications. The event trigger is enabled if the parameter is set to True (=1).

Parameter	Description	Trigger DPF?	Trigger PRV?	Trigger LVM?
DNAF_EVENT_TRIGGER_TOPOLOGY_BACKBONE	The backbone switch, cut, temporary ground, or jumper status change event trigger.	Yes	Yes	Yes

Parameter	Description	Trigger DPF?	Trigger PRV?	Trigger LVM?
DNAF_EVENT_TRIGGER_TOPOLOGY_LATERAL	The lateral switch, cut, temporary ground, or jumper status change event trigger. A "lateral switching device" is a device that does not connect to the feeder main at either terminal.	Yes	Yes	No
DNAF_EVENT_TRIGGER_MEAS_CHANGE	The SCADA sudden measurement change event trigger.	Yes	No	No

The DNAF_EVENT_DELAY parameter in the DNAF server configuration file specifies the time delay in seconds for triggering DNAF applications by event. This time delay allows the system to reach a steady-state condition before running a DNAF application.

In addition, the DNAF applications use the Scada system's alarming and event logging functionality to alert operators about specific conditions, such as flow violation, backfeed conditions, and plan implementation.

In real-time mode, DNAF applications and alarm events can be enabled or disabled, such as to isolate problematic portions of the model. For more information, refer to DNAF System Configuration Display.

2.3. DNAF System Configuration Display

On the DNAF System Configuration display, the operator can enable or disable DNAF applications and alarm events, change problem formulations and plan execution modes, and select SLS rule sets for system elements at run-time. This allows the operator to disable problematic portions of the model that are experiencing solution issues. The operator can also temporarily disable events (alarms) on a given substation when topology changes may be occurring.

The settings on the DNAF System Configuration display override the default settings in the DNAF Configuration Editor.

To open the DNAF System Configuration display: from the RT or ST menu, click DNAF System Configuration.

The cell color on the DNAF System Configuration display shows the following:

- **White:** Editable value.
- **Blue:** Selected by the user.
- **Light Gray:** Primary editable value.
- **Medium Gray:** Not a primary value; it just reflects the values of cells below it in the hierarchy. For example, it shows station execution mode for FISR.
- **Dark Gray:** Read-only value; it cannot be edited.

Note

For FISR and AFR, execution mode can be set only for feeders. Transformer, substation, and system level cells are colored medium gray. They reflect the status of the feeders and show Mixed if feeders contain different settings.

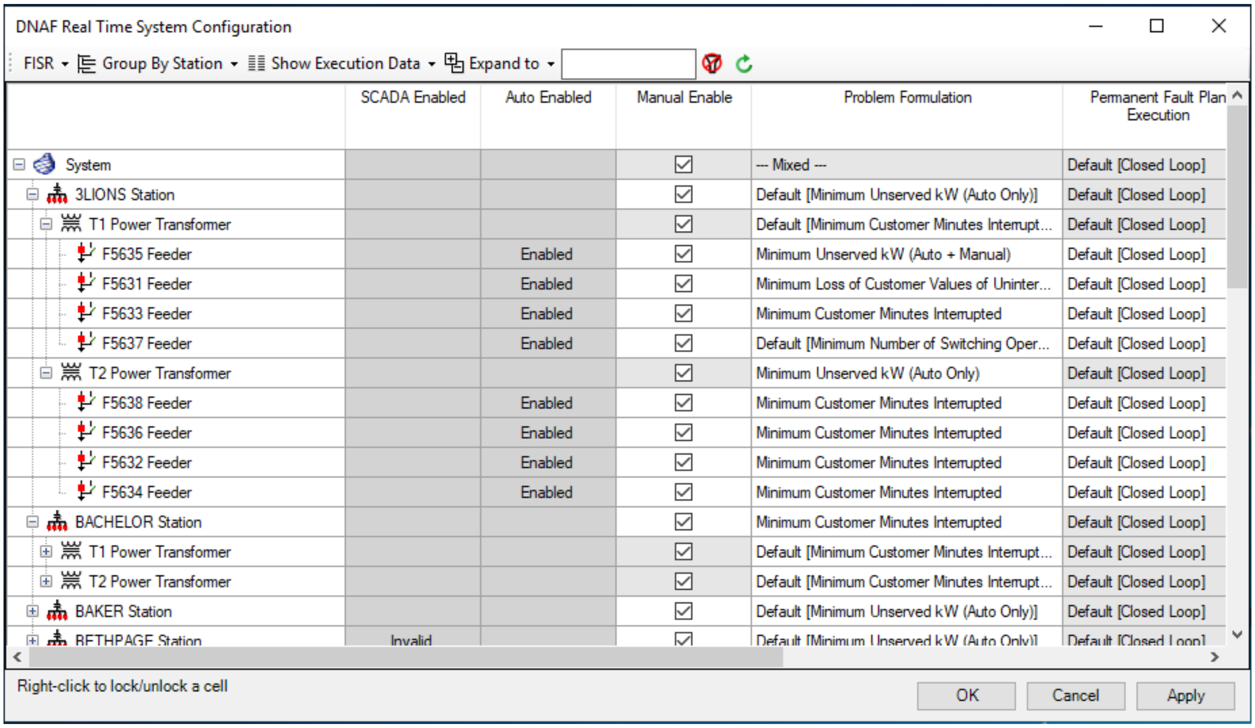


Figure 3. DNAF Real Time System Configuration Display

The DNAF System Configuration toolbar includes the following fields and controls:

- **DNAF Application:** Allows selecting an application (DPF, PRV, FL, SCC, FISR, AFR, LVM, or SLS). In study mode, you can also select PLOS, and FISRCA.
- **Display:** Specifies how stations are displayed in the window. This allows a large number of stations, power transformers, or feeders to be managed with minimal user action. It also allows the operator to easily locate stations or groups of stations in the window.
  - **Group by Stations:** All stations are listed in alphabetical order.
  - **Group by Division:** All stations grouped by divisions. The divisions are listed in alphabetical order.
  - **Group by Optimization Group:** All stations grouped by optimizations groups. The optimizations groups are listed in alphabetical order.
- **Show Data:** Allows selecting the data to be displayed in the grid.
  - **Show Execution Data:** Shows options for enabling and disabling the manual and periodic execution of the application. For advanced applications (AFR, FISR, and LVM), also shows the SCADA/Auto Enabled settings and options for selecting problem formulations and plan execution modes.

    - **Show Alarm Data:** Shows options for enabling and disabling alarm events.
    - **Show All Data:** Shows both the execution and the alarm data.
- **Expand To:** Expands the elements tree to show optimization groups, divisions, stations, power transformers, or feeders.

> **Note**
>
> Power transformer and feeder options are available only for AFR, FISR, and LVM.

- **Filter:** Filters the elements tree (system, optimizations groups, divisions, stations, transformers, or feeders) by the text entry in the filter box. Only those items whose name matches the search criteria are displayed.
- **Clear Filter Text:** Clears the filter box and returns to the full system display.
- **Refresh:** Immediately refreshes the display to reflect any recent system changes. By default, the display is automatically refreshed every 60 seconds.

The DNAF System Configuration window data grid contains the following fields:

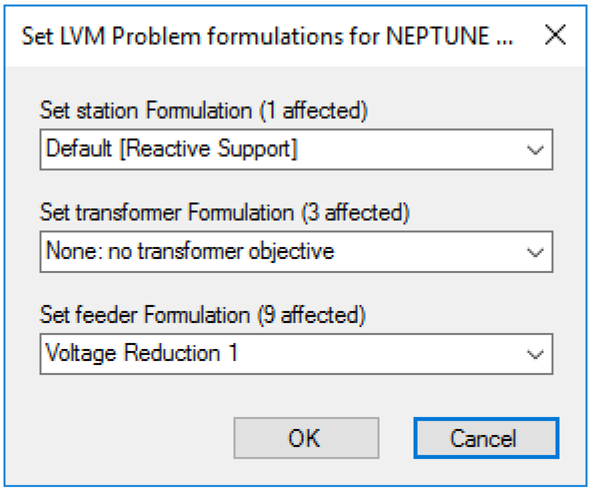
- **System Elements:** Optimization groups, divisions, stations, power transformers, and feeders are presented in the leftmost column of the grid as a hierarchy under "System". Some settings are only applied to a specific level. For example, DPF application enabling/disabling settings are only applied to stations (options are not shown at the power transformer and feeder level).
- **Manual Enable:** Allows enabling or disabling applications for system elements. Disabled applications cannot be executed manually using the DNAF Solutions display or automatically due to periodic and event triggers. System elements (for example, feeders) where an application is disabled are not considered in application solutions.

  The SCADA Enabled and Auto Enabled values can override the Manual Enable values. For more information, refer to the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

- **SCADA Enabled:** For LVM, FISR, and AFR, this shows the SCADA Enabled status (Enabled, Disabled, or Invalid - for a suspect SCADA status measurement). The cell is blank if there is no SCADA status measurement for this system item.
- **Auto Enabled:** For LVM, FISR, and AFR, this shows the Auto Enabled status: Enabled or Disabled based on the automatic disabling checks. (See Automatically Disabling Applications for more details).
- **Problem Formulation:** Allows selecting the problem formulation that is used when the DNAF application is automatically executed from a periodic or event trigger. In addition, the Use System Configuration Defaults option in the Problem Formulation menu on the DNAF Solution display represents the problem formulation specified in this field.

> **Note**
>
> This setting overrides the default application’s problem formulation defined using the DNAF Configuration Editor (see DNAF Configuration Editor).

For FISR, AFR, and LVM, it is possible to have different problem formulations for the station, transformer, and feeder levels. Clicking the Problem Formulation field for the system, optimization group, division, station, or transformer opens the Set Problem Formulation dialog box. The operator can set problem formulations for the element and lower elements in the hierarchy.



Note

For sub-elements of stations and power transformers with fixed regulation problem formulations, you can select only fixed regulation problem formulations.

- **Plan Execution:** Allows selecting the execution mode for periodic and event triggers.

  For AFR and FISR specifically, you can select execution modes for feeder levels only. By default, the most restrictive execution mode among the affected feeders is used (for example, Advisory).

  For LVM specifically, it is possible to have different execution modes for the station and transformer. When the execution mode for a station is set to Advisory, execution modes for the station's transformers are also set to Advisory and cannot be modified. If the station is set to Closed Loop, the most restrictive mode from all device moves in the plan is selected based on the devices' associated transformers or station.

- **FISR FL Event Trigger:** If selected, the Fault Isolation and Service Restoration (FISR) application event triggering requires both Set fault indicator measurements and FL application results. This option is only available for the FISR application.
- **LVM DA Support Mode:** If selected, the LVM integration with AFR/FISR is enabled. If AFR/FISR cannot generate a plan without flow or voltage violations, LVM may be able to generate plans to correct the violations in the new system configuration after the FISR/AFR switching.
- **Alarms/Events:** Allows enabling and disabling particular alarm events.

The following example shows how to disable executing the Protection Validation (PRV) application manually and due to periodic and event triggers for some stations:

1. Open the DNAF System Configuration display.

2. Select PRV from the application drop-down list.
3. Select the Group By Station option on the toolbar.
4. In the Manual Enable column, clear the check boxes for stations where the PRV application is to be disabled.

   This disables running PRV for the selected stations. Similarly, to disable PRV for all stations, select the check box at the system level.
5. Click OK or Apply to save the changes.

The following example shows how to enable alarms for the Fault Location (FL) application:

1. Open the DNAF System Configuration display.
2. Select FL from the application drop-down list.
3. Select the Show Alarm Data option on the toolbar.
4. Select check boxes in the alarm columns (FL Started Event, FL Success Event, and FL Failed Event) for the system element.
5. Click OK or Apply to save the changes.

2.3.1. Automatically Disabling Applications

Auto Disabling refers to automatically disabling feeders, substations or transformers for analysis based on certain conditions that are defined in the DNAF Configuration Editor (see the following images) for specific applications. If a system element is automatically disabled, the value of the Auto Enabled cell is set to Disabled and a tooltip describes the condition that is causing this status (see DNAF Real Time System Configuration Display). The cell is blank if auto disabling is not relevant for this system item.

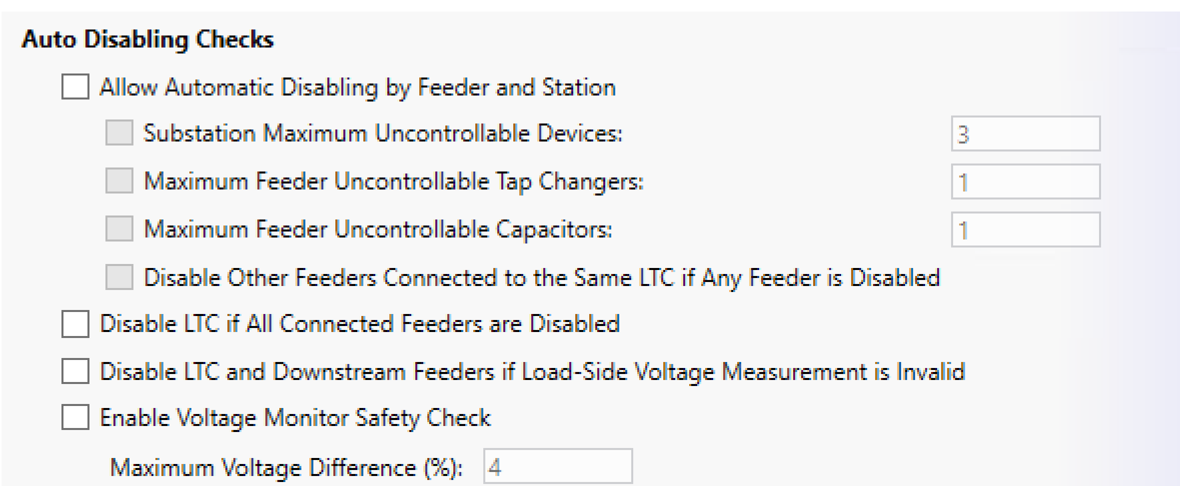


Figure 4. Auto-Disabling Options for LVM

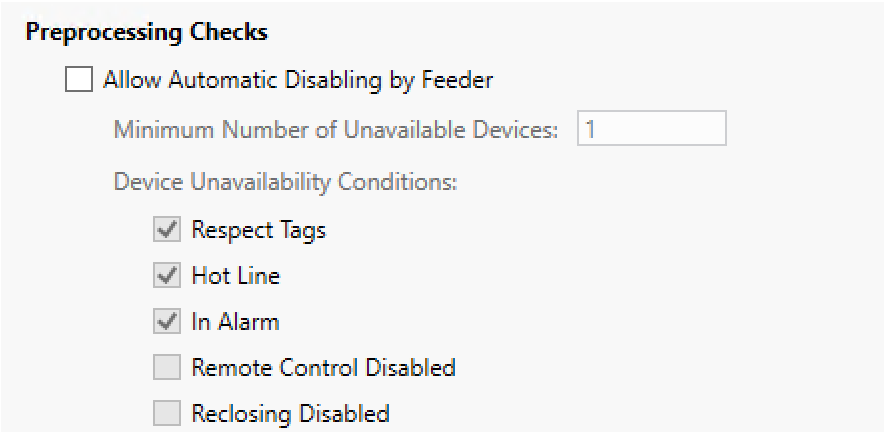


Figure 5. Auto-Disabling Options for FISR and AFR

2.3.2. Locking User Settings

The setting locking feature on the DNAF System Configuration display enables you to "freeze" the setting and prevent it from being updated when changes are made higher in the hierarchy. For example, if some feeder problem formulations are locked, changing the problem formulation at the system level will change the problem formulation for all stations, transformers, and feeders except for the locked cells.

> **Note**
>
> LVM Failsafe Operation may override a locked problem formulation on transformers and feeders if the problem formulation is defined to inherit from the substation problem formulation.
>
> For example, consider the configuration shown in the following image. If a power transformer has a "Default [None: No Transformer Objective]" Problem Formulation locked by the user, and the substation has a Problem Formulation with no lock, the LVM failsafe functionality (when enabled) on the substation will override the power transformer Problem Formulation with the pre-configured failsafe Problem Formulation. This is because the power transformer "Default [None: No Transformer Objective]" Problem Formulation inherits the substation formulation for LVM when failsafe is executed.

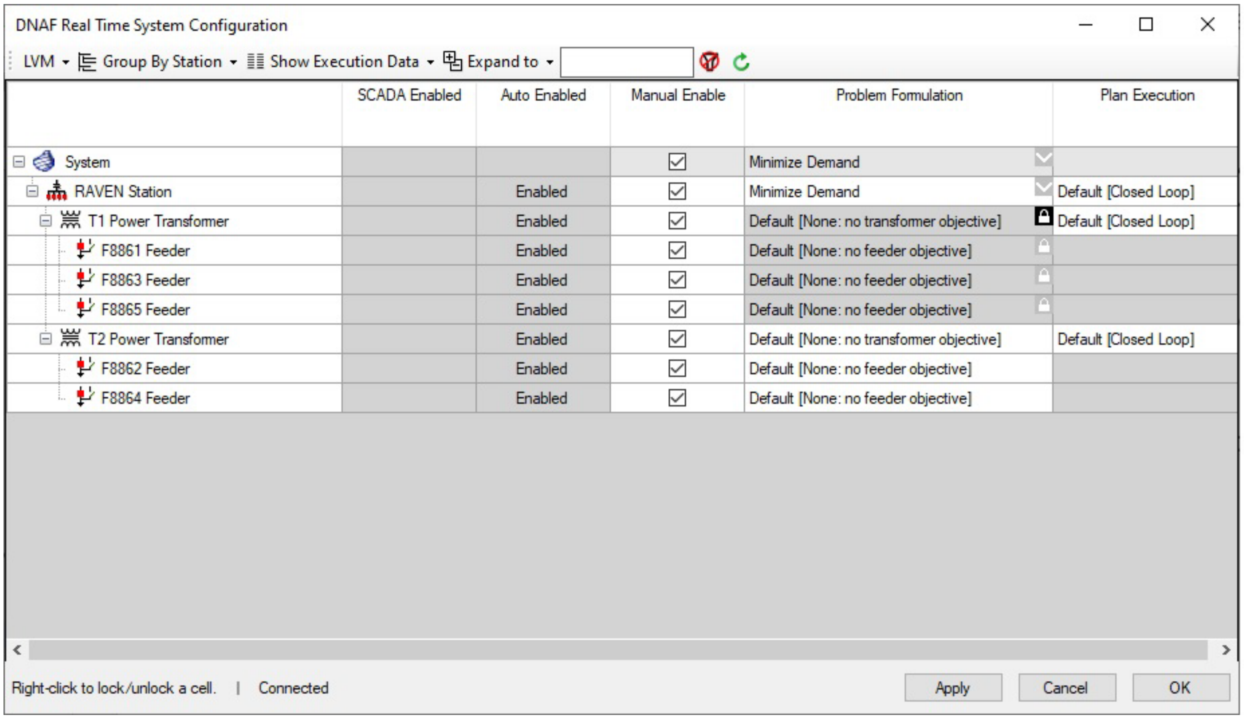


Figure 6. Locking default problem formulation for LVM in the DNAF System Configuration Display

Right-click a cell to lock/unlock the associated setting. When locking a setting, you must also enter a comment with the reason for locking it.

Locked cells are shown with a black lock icon .

To help identify the locked cells, cells higher in the hierarchy are shown with a down chevron . For example, a locked feeder cell is shown with a black lock icon and the upstream transformer, station, and system cells have chevron icons.

Cells downstream of a locked cell are also locked. These dependently locked cells are shown with a gray lock icon . For example, locking a cell at the station level will lock all bank and feeder cells below it. Unlocking a cell will automatically unlock all cells beneath it unless those cells contain a separate locked cell (that is, with a black icon).

When hovering over a cell with either a black or gray lock icon, a tooltip will show the lock reason for all locks affecting that cell.

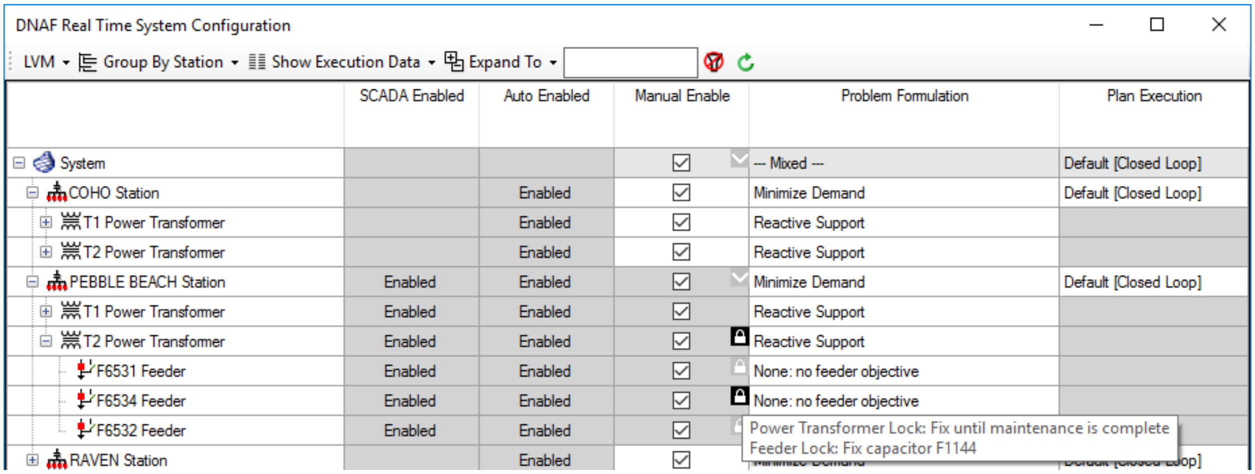


Figure 7. Locking Settings on the DNAF System Configuration Display

2.4. Study Mode Operations

In study mode, all network solutions are requested manually by the user as described in Executing DNAF Applications Manually. There are no periodic or event-triggered solutions.

The DNAF System Configuration window can also be used in study mode to enable applications and configure problem formulations. For more information, refer to DNAF System Configuration Display.

The application solution engines are the same in the real-time and study mode environments, but there may be differences in the network application configuration that create differences between the real-time and study mode results.

The study mode environment provides the ability to conduct "what if" studies, including topology changes, capacitor switching changes, loading level changes, and voltage regulating device changes (blocking, voltage reduction, line drop compensation setting changes, and others). Other options are also available, including the historical load data and temperature adjustment schedules, which are used to allocate the loads.

There are several methods available for initializing the study mode environment. The user can manually set up a case by copying station mode files, dynamics files, and network application files (`alg.dat`, `msgdef.dat`, `optalg.dat`, `phid.dat`, `tftype.dat`, and `DNAF_PARAMS.xml`) to the appropriate study server and study client folders. It is expected that this manual method will not be employed very often.

Three other methods also exist for initializing the study environment:

- **Initialize to Real-Time:** This function copies the latest model files (located in the real-time client folder on the local computer), real-time server dynamics files, real-time server manual entry files, real-time SCADA measurement data, and real-time server application data files to the appropriate folder on the local computer. This option allows the current network configuration to be analyzed in an offline environment.

- **Initialize to Nominal:** This option is similar to the Initialize to Real Time function, except that the dynamics, manual entries, and measurement data are not copied from the real-time server to the local computer. With this option, the nominal topology is used as a starting point for the study, and no other user-made changes are included.
- **Savecase Retrieval:** In study mode, a package can be created that combines all the station model files, dynamics data, measurement data, application data, and application results files into a single file called a "savecase". The data that is stored in the savecase file is user-configurable.

Also, additional stations can be loaded or unloaded from the study case using the Study Case Setup - Advanced Initialization popup, which can be accessed from the Study Case Setup window.

2.5. Simulator Mode Operations

Only the DPF and FL applications are supported in simulator mode. These applications in simulator mode work similarly as in the real-time mode. The applications can be triggered manually, periodically, and by event.

There are some differences when manually changing system and device parameters:

- The load values that are entered via the simulator Manual Input Parameter Entry Display appear as changes in the feeder current on the real-time substation one-line (assuming that the station has telemetry on the feeders).
- Device parameters entered in the simulation system via data-entry tabular and pop-up displays are not applied to devices in the real-time system. These changes are used for DNAF calculations in the simulated power system.

2.6. Application Input Displays

Operators can modify system-wide settings and device-specific parameters using the Manual Input Parameter Entry display and manual entry tabular and pop-up displays. These changes will be considered in subsequent DNAF solutions.

Tabular displays that allow setting device input data (also called "manual entries") can be accessed from the Application Input Summary display, which can be opened from the RT, ST, or SIM menu. You can also access these displays from the Network Analysis Inputs option on the station toolbar.

- **Measurement Data:** This display provides a summary of the measurements modeled in ADMS, including the current measurement value, measurement quality, and whether a measurement point is a multi-phase point measurement. This display allows a DMS measurement quality to be changed. In study mode, this display also allows a SCADA analog measurement to be changed.
- **Load Category Input:** This display allows station-wide load-to-voltage sensitivity coefficients to be entered for all load types (categories).

- **Station Feeder Input:** This display shows station-wide feeder and transformer parameters entered via the Manual Input Parameter Entry display. This display also allows station-wide options to be implemented, such as blocking all LTCs/regulators and enabling voltage reduction for all LTCs/regulators associated with the substation.

  The Feeder Input display can be accessed from the Station Feeder Input. It allows the feeder head flow and power factor to be set for an individual feeder.

- **Reference Bus Input:** This display allows the voltage magnitude and angle for a reference bus interface object to be set.
- **LVM Device Eligibility:** This display shows key data for all LVM optimization devices (capacitors, tap changing transformers, generators, and distribution energy resources) in one place, including the current step and control availability data.
- **Capacitor Input:** This display allows capacitor options including the on and off levels, maximum number of daily operations, and hours between consecutive switching, steps, and size to be set (study mode only).
- **Tap Changer Input:** This display allows tap changer options including the primary, secondary, and tertiary tap positions, voltage target, bandwidth, and line drop compensation settings to be set.
- **Generator/ Distribution Energy Resource Input:** This display allows local control settings including the voltage target and real and reactive power targets to be set.
- **Fault Location Input:** This display allows fault location fault current Amp values and faulted phases to be entered.

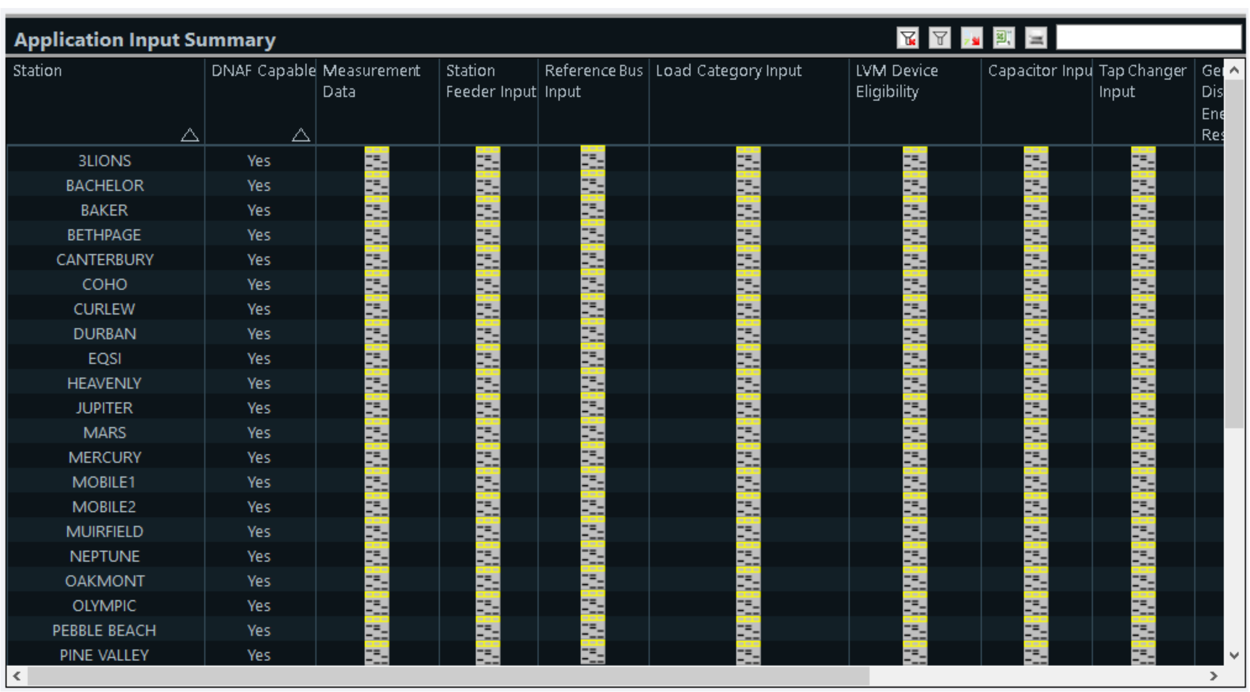


Figure 8. Application Input Summary Display

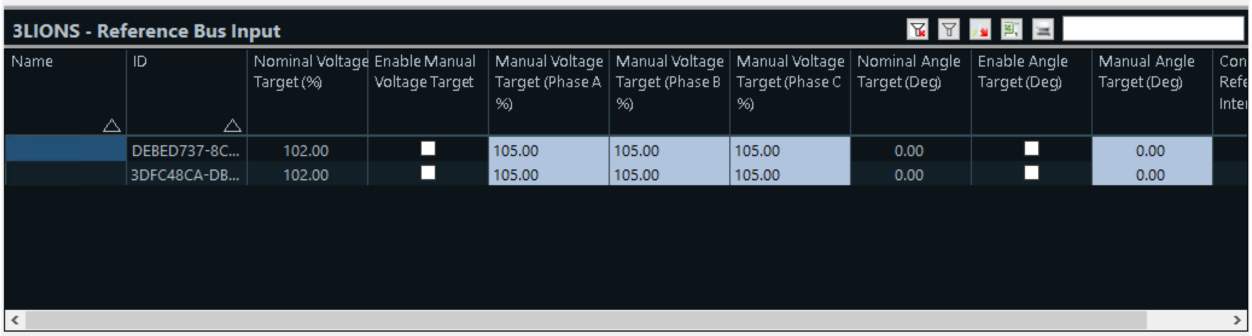


Figure 9. Reference Bus Input Display

In addition to the manual entry tabular displays, device-specific data can be entered via data-entry pop-up displays, which can be accessed from device-specific toolbars. Data-entry displays are available for the following devices:

- **Switching Devices:** The display allows setting the switch flow limit and load break rating.
- **Shunt Capacitors:** The display allows setting the local control settings.
- **Lines:** The display allows setting per-phase line flow limits.
- **Buses:** The display allows setting high voltage and low voltage operational and optimization limits.
- **Transformers:** The display allows setting local control settings including the voltage target, bandwidth, and line drop compensation settings, as well as the per-phase transformer static flow limit for regulators and load tap changers.
- **Generators:** The display allows setting real and reactive power targets.
- **Distribution Energy Resources:** The display allows setting real and reactive power targets as well as the Contribute To Fault Current flag.
- **Loads:** The display allows setting the voltage limits and per-phase nominal load values.
- **Series Reactors:** The display allows setting series reactor flow limits.
- **Reference Bus Interface Objects:** This display allows setting voltage and angle targets and positive, negative, and zero sequence resistance and reactance.

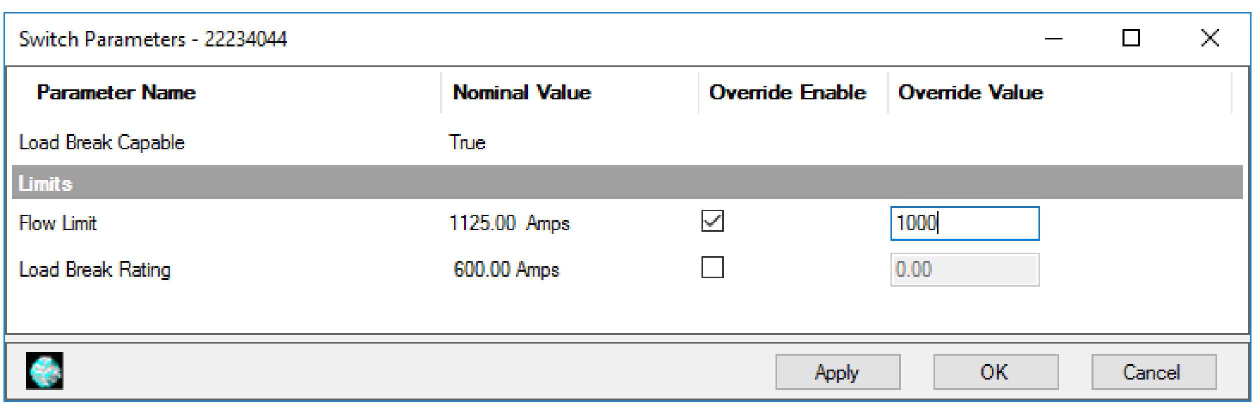


Figure 10. Switch Parameters Display

2.6.1. Manual Input Parameter Entry

The Manual Input Parameter Entry display allows an operator to manually set parameters for the whole system and system sub-elements. This display can be accessed from the RT, ST, or SIM menu.

2.6.1.1. Overview

The Manual Input Parameter Entry display is used to manually override parameters that affect more than a single device, such as feeders, transformers, stations, and overall system parameters. The display allows parameters to be defined explicitly for a given system element (station, feeder, and so on) or to be inherited from a higher system element (for example, parameters for a station could be inherited from parameters set at the system level).

This concept can be best understood by an example. Consider a case where the display might be used to set a Load Scaling value (a feeder level parameter).

- **Change system scaling:** By changing the system-level value for Load Scaling, all feeders for all stations that do not have explicitly set manual overrides will inherit this setting.
- **Reducing scaling for a specific station:** By changing the Load Scaling value at the station level and by setting all sub-feeders to inherit this value, all feeders in the station will take on the new load scaling value.
- **Clearing all system scaling:** By changing the system-level setting for Load Scaling to Auto and setting all sub-elements to inherit this value, it is possible to clear a mixture of manual settings on different feeders or stations and return all load scaling values to the default value.

The display allows the following types of parameters to be set:

- **General:** Parameters for station temperature and feeder load scaling are included in this group.
- **Violation Limits:** Ampacity limits for transformers and voltage limits for buses and loads are exposed in each group. However, the values only inherit from a station (for station internal buses and loads) or feeder (for station external buses and loads).
- **Optimization Limits:** Voltage limits for buses and loads are exposed in each UI group. However, the values only inherit from a station (for station internal buses and loads) or feeder (for station external buses and loads).
- **LVM Targets:** Minimum and maximum power factor targets for stations, power transformers, and feeders are included in this group.
- **Real/Reactive Power Throttling:** The rates at which real/reactive power increases and decreases are throttled during an LVM plan execution.
- **Fault Impedance:** Fault impedance resistance and reactance values for FL, SCC, and PRV.

2.6.1.2. Main Display Options

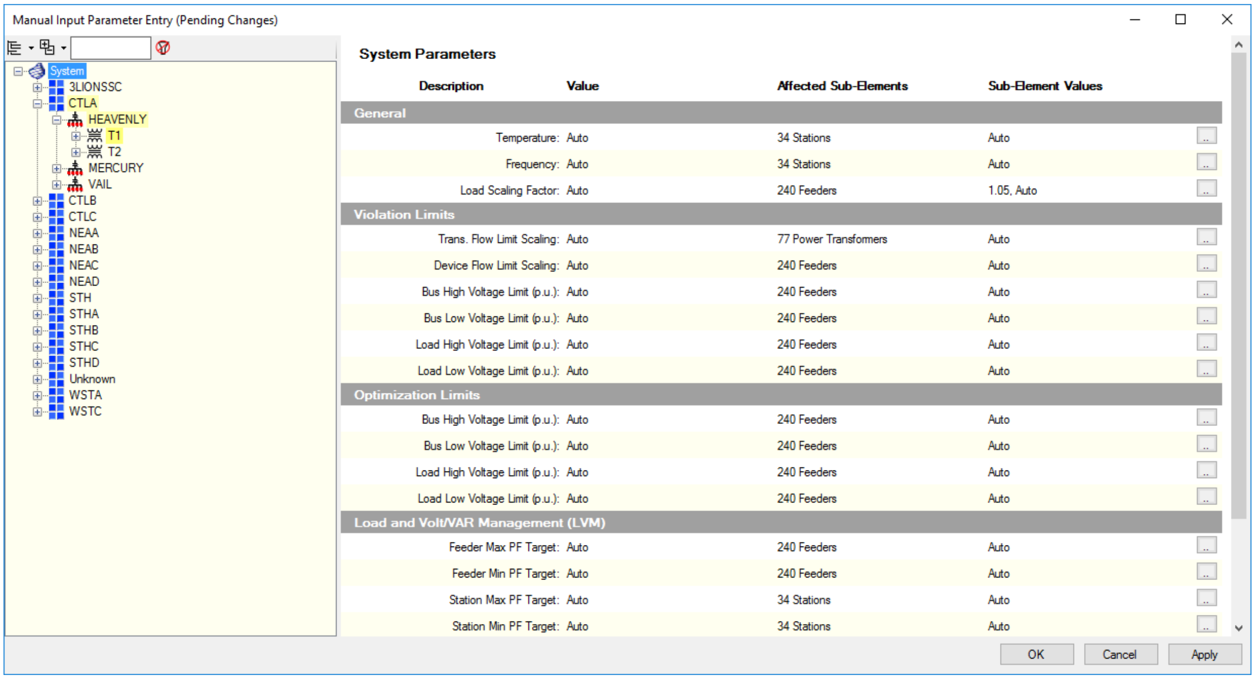


Figure 11. Manual Input Parameter Entry Display

The main display provides navigation to the parameters that are to be modified. The left pane includes a tree view showing the structure of the system. The tree can be organized either by System > Division > Station, by System > Optimization Group > Station, or just by Station. Elements with a bright yellow background show items where parameters have been modified, and elements with a pale yellow background show that elements below that item have modifications.

The right pane shows the different parameters for the element currently selected in the tree. The Value column shows current values of the parameters, while the following columns show exactly which sub-elements will be affected by changes to this parameter and the values of those sub-elements. Buttons in the right-most column open the Edit Parameter display, allowing each parameter to be modified.

2.6.1.3. Edit Parameter Display

Clicking an Ellipsis button on the Main Manual Input Parameter Entry display opens the Edit Parameter display.

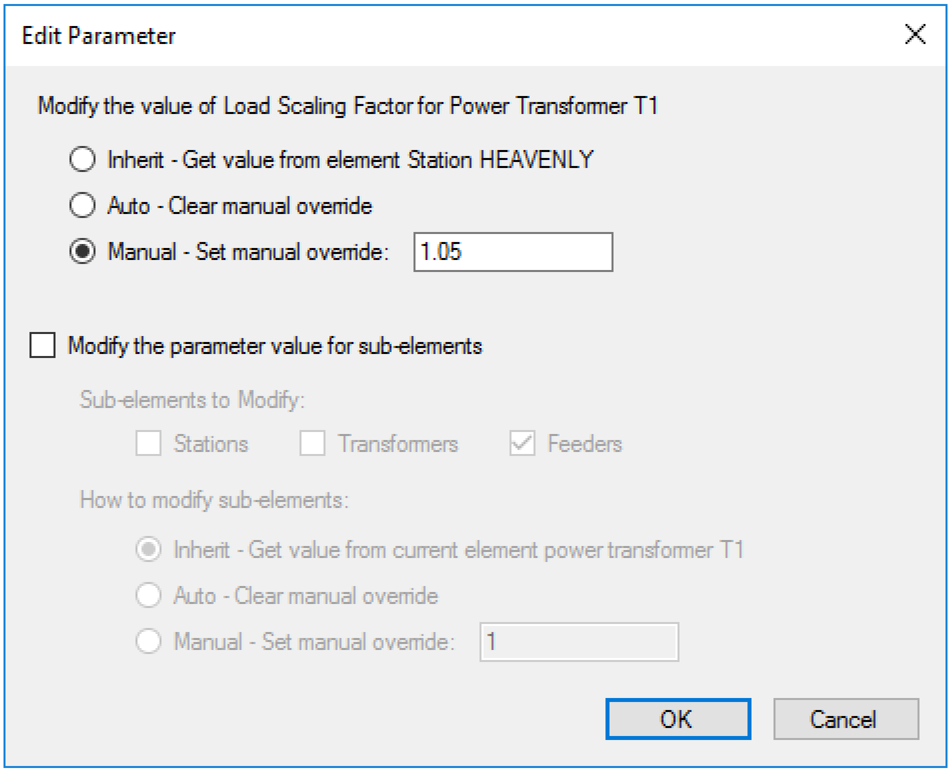


Figure 12. Edit Parameter Display

The top half of the Edit Parameter display provides options for setting the current parameter value. The options include the following:

- **Inherit:** This option means that the value of the parameter for this element will be determined by the value of a parent element. For example, if the current element is a feeder, the parameter value would be retrieved from the station (or if that value was also set to inherit, it would be retrieved from the system level).
- **Auto:** If this option is selected, the parameter value has no manual override and the default or model-set value for the parameter will be used.
- **Manual:** Use this option to set an explicit manual override value for the parameter.

The bottom half of the display provides the option for setting/overriding sub-elements. For example, if a transformer parameter is being edited, the option to set the value of the parameter for all feeders belonging to that transformer is provided. The "Sub-elements to Modify" group determines which sub-elements can be changed. The options for modification follow the options described above.

A common usage scenario for this section of the display is to use the Inherit option to clear any manual override parameters set on individual sub-elements.

2.7. Viewing Results

Network application results can be viewed on tabular and attribute displays. Some results, such as flow and voltage violations, are also shown on the geographic display.

Tabular displays provide both summarized and detailed information, such as real and reactive power, and current and voltage data for all feeders calculated by the power flow. These displays can be accessed from the RT, ST, or SIM menu, device-specific toolbars, other tabular displays, or the DNAF Solution display.

To view results for an individual device, such as the flow through a cable, select a device and bring up an attribute display from the device-specific toolbar.

The following tabular displays can be accessed from the RT menu:

- **Advanced Application Plan Status System View:** This display shows the summary for advanced application plan execution.
  - **Plan Control Moves:** This display shows switching steps for the selected plan.
- **Station Exception:** This display provides station summary topology information (de-energized customer count and abnormal topology conditions) along with summary power flow results, including high voltage violation, high load or bus voltage, low voltage violation, low load or bus voltage, load voltage imbalance for three-phase loads, flow imbalance for three-phase switching devices, branch loading, highest ground return current, reserved capacity information, and the latest DPF solution status.
  - **Feeder Exception Summary Display:** This display provides summary topology information along with summary power flow results for all feeders in a given station. The Feeder Exception Summary display has similar content to the Station Exception Summary display.
- **Station Output Summary:** This display provides navigation to all individual application output summary displays. From this display, navigation to the following output summary displays is available:
  - **DPF Results:** This display provides summary DPF results for the substation and feeders. From the Substation pane, summary device displays are available for many devices including power transformers and regulators, capacitors, generators, loads, and distribution transformers.
  - **SCC Results:** The Short Circuit Calculation display provides the short circuit summary information for the last analyzed device.
  - **PRV Results:** The Protection Validation display provides summary protection validation results for breakers and reclosers, and fuses depending on the application configuration.
  - **FL Results:** The Fault Location display provides a summary of potential feeder fault locations.
  - **LVM Solution:** The LVM Plan Control Moves display provides the latest LVM plan.
  - **FISR Solutions:** The FISR Results Summary display provides the latest set of FISR solutions with reconfiguration plans for the previous FISR execution.
  - **FISRCA Solutions:** The FISR Contingency Analysis Solution Overview display provides the latest set of reconfiguration plans for the previous FISRCA execution. This display is available in Study mode only.

- **AFR Solutions:** The AFR Results Summary display provides the latest set of reconfiguration plans for the previous AFR execution.
- **PLOS Solutions:** The Planned Outage Service display provides the latest set of reconfiguration plans for the previous PLOS execution. This display is available in Study mode only.
- **Application Diagnostic Summary:** The Application Diagnostic Information display provides the diagnostic details for all DNAF applications executed for a substation.

- **Base Application Status:** This display provides the latest solution time and status for the standard DNAF applications (DPF, PRV, FL, and SCC).
- **Advanced Application Status:** This display provides the latest solution time and status for the advanced DNAF applications (LVM, FISR, PLOS, AFR, and FISRCA).
- **BLA Station Summary:** This display provides detailed solution information about the latest DPF load allocation results. Navigation to the BLA Results Summary display is also available on this display.
- **LVM Station Summary:** This display summarizes LVM application results by station.
- **LVM Device Exclusions List:** This display provides a summary of devices that are excluded for LVM based on manual exclusions and device exclusion settings in the DNAF Configuration Editor. In addition, the LVM Control Failure flag can be removed from the devices and the Control Processed flag can be toggled for devices.
- **Load Shed Overview:** This display provides the load shed status for all stations in the system.
- **DPF Analyst Results:** This display provides voltage change summary information for each station, including data for conductors, transformers, and feeder main switches. It also provides the maximum phase angle difference for currently open feeder main switching devices.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

2.7.1. DNAF Display Plug-in

The DNAF Display Plug-in provides a centralized source for viewing DNAF results. It contains navigation links to the LVM System Monitor, Implementation Details, System Monitor Stations, DNAF Aggregation Summary, Station Exception, and other DNAF summary displays. The DNAF Display Plug-in can be accessed from the RT menu.

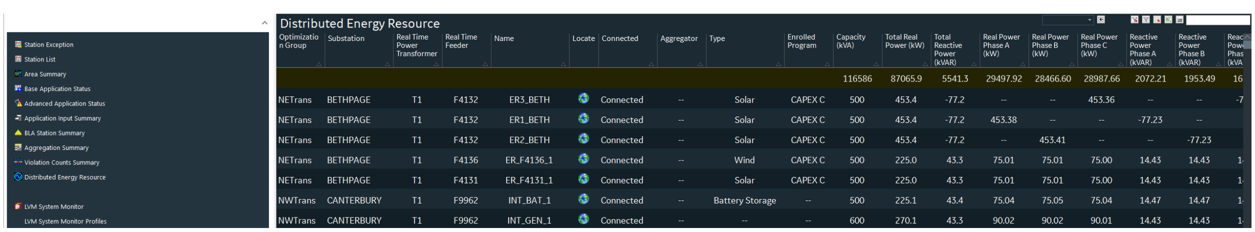


Figure 13. DNAF Display Plug-in

The left side of the DNAF Display Plug-in contains control panels for accessing DNAF displays. The right side of the display shows the appropriate DNAF data in grid tabular form.

2.7.2. DNAF Aggregation Summary Display

The DNAF Aggregation Summary display shows the following load and voltage aggregation values for the entire distribution system and for individual station optimization groups, divisions, substations, transformers, and feeders:

- Real and reactive power demand.
- Power factor.
- Technical losses.
- Feeder voltage at station buses and regulator buses.
- Capacitor online and offline reactive power capacity.

The power flow and voltage values are obtained from SCADA measurements and DPF calculations. The display also shows the difference between the measured and calculated load values and the weighted average for the measured and calculated voltage values.

To call this display, select the DNAF Display Plug-in option from the RT DMS drop-down menu, and then select DNAF Aggregation Summary.

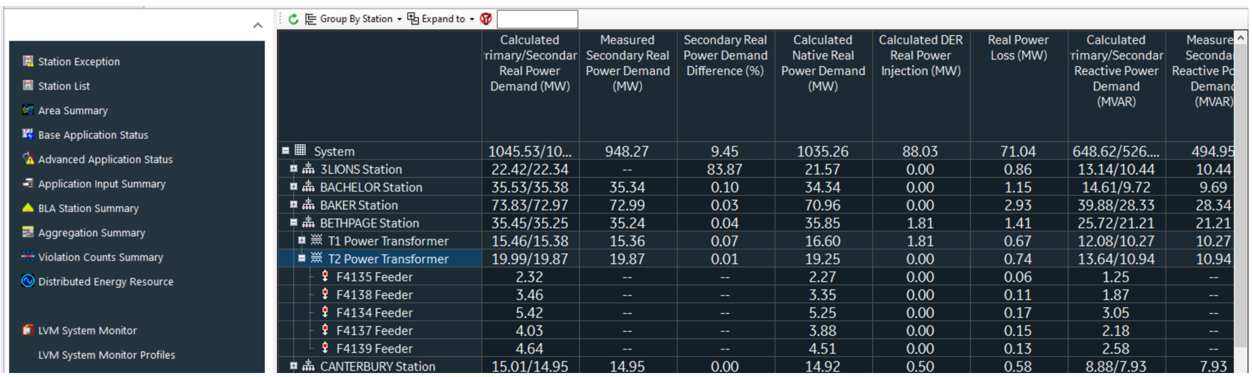


Figure 14. DNAF Aggregation Summary Display

The DNAF Aggregation Summary toolbar includes the following controls:

- **Display:** Specifies how stations are displayed in the window. This allows a large number of stations, power transformers, or feeders to be managed with minimal user action. It also allows the operator to easily locate stations or groups of stations in the window.
    - **Group by Stations:** All stations are listed in alphabetical order.
    - **Group by Division:** All stations grouped by divisions. The divisions are listed in alphabetical order.
    - **Group by Optimization Group:** All stations grouped by optimizations groups. The optimizations groups are listed in alphabetical order.
- **Expand To:** Expands the elements tree to show stations, power transformers, or feeders.

- **Filter:** Filters the elements tree (system, divisions, optimizations groups, stations, transformers, or feeders) by the text entered in the filter box. Only those items that match the search criteria are shown.
- **Clear Filter Text:** Clears the filter box and returns to the non-filtered view.
- **Refresh:** Immediately refreshes the display to reflect any recent system changes. By default, the display automatically refreshes every 1 minute.

The DNAF Aggregation Summary display shows the following information:

- **Calculated Primary/Secondary Real Power Demand:** Calculated real power at the source side (primary) and at the load side (secondary), if applicable, in MW. For power transformer level, the value is on the secondary side.
- **Measured Secondary Real Power Demand:** Measured secondary real power in MW.

  For transformer and feeder levels, the value is shown only if the measurement exists. For system and station levels, the value is aggregated from downstream transformers because there are no associated physical measurements. If any of the transformers do not have a measured value, the calculated value is used in the summary.
- **Secondary Real Power Demand Difference:** Difference between the calculated and measured secondary real power, in percentage.
- **Calculated Native Real Power Demand:** Calculated native real power demand in MW.
- **Calculated DER Real Power Injection:** Calculated DER/DG real power injection in MW.
- **Real Power Loss:** Calculated real power loss in MW. Losses include losses in lines, transformers, regulators, series reactors, and other equipment.
- **Calculated Primary/Secondary Reactive Power Demand:** Calculated reactive power at the source side (primary) and at the load side (secondary), if applicable, in MVAR.
- **Measured Secondary Reactive Power Demand:** Measured secondary reactive power in MVAR.

  For transformer and feeder levels, the value is shown only if the measurement exists. For system and station levels, the value is aggregated from downstream transformers because there are no associated physical measurements. If any of the transformers do not have a measured value, the calculated value is used in the summary.
- **Secondary Reactive Power Demand Difference:** Difference between the calculated and measured secondary reactive power, in percentage.
- **Calculated Native Reactive Power Demand:** Calculated native reactive power demand in MVAR.
- **Calculated DER Reactive Power Injection:** Calculated DER/DG reactive power injection in MVAR.
- **Reactive Power Loss:** Calculated reactive power loss in MVAR. Losses include losses in lines, transformers, regulators, series reactors, and other equipment.
- **Calculated Primary/Secondary Power Factor:** Calculated power factor at the source side (primary) and at the load side (secondary), if applicable.

- **Measured Secondary Power Factor:** Measured power factor at the load side, if the measurement exists.
- **Calculated Average Voltage, Station Internal Regulated Bus Only:** Calculated average voltage in per unit. Only station internal regulated MV buses are included.
- **Measured Average Voltage, Station Internal Regulated Bus Only:** Measured average voltage in per unit. Only station internal regulated MV buses are included.
- **Calculated Weighted Average Voltage, Station Internal Regulated Bus Only:** Weighted calculated average voltage in per unit. Only station internal regulated MV buses are included. The weighting factor is the calculated current value on the same bus.
- **Measured Weighted Average Voltage, Station Internal Regulated Bus Only:** Weighted measured average voltage in per unit. Only station internal regulated MV buses are included. The weighting factor is the measured current value on the same bus.
- **Calculated Average Voltage, All Regulated Bus:** Calculated average voltage in per unit, including both station MV buses and regulator buses on feeders.
- **Measured Average Voltage, All Regulated Bus:** Measured average voltage in per unit, including both station MV buses and regulator buses on feeders.
- **Calculated Weighted Average Voltage, All Regulated Bus:** Weighted calculated average voltage in per unit, including both station MV buses and regulator buses on feeders. The weighting factor is the calculated current value on the same bus.
- **Measured Weighted Average Voltage, All Regulated Bus:** Weighted measured average voltage in per unit, including both station MV buses and regulator buses on feeders. The weighting factor is the measured current value on the same bus.
- **Calculated Online Capacitor Reactive Power:** Calculated total reactive power injection of online capacitors in MVAR.
- **Calculated Offline Capacitor Reactive Power:** Calculated total reactive power injection of offline capacitors in MVAR.
- **Calculated Total Capacitor Reactive Power:** Calculated total reactive power injection of online and offline capacitors in MVAR.
- **Calculated Real Power Generation Capacity:** Calculated real power generation capacity in MW.
- **Calculated Reactive Power Generation Capacity:** Calculated reactive power generation capacity in MVAR.

2.7.3. DNAF Violation Counts Summary Display

The DNAF Violation Counts Summary display shows the aggregated count of the following violations based on different components (switches, lines, distribution transformers and loads) for the entire distribution system and for individual station optimization groups, divisions, substations, transformers, and feeders:

-

Flow imbalance.

- Voltage imbalance.
- Flow limit.
- Voltage limit.

The violations are identified by the LIM function, which detects all limit violations following each DPF execution. For more information about value calculation for this display, refer to section “DNAF Violation Counts” in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

To call this display, select the DNAF Display Plug-in option from the RT DMS drop-down menu, and then select Violation Counts Summary.

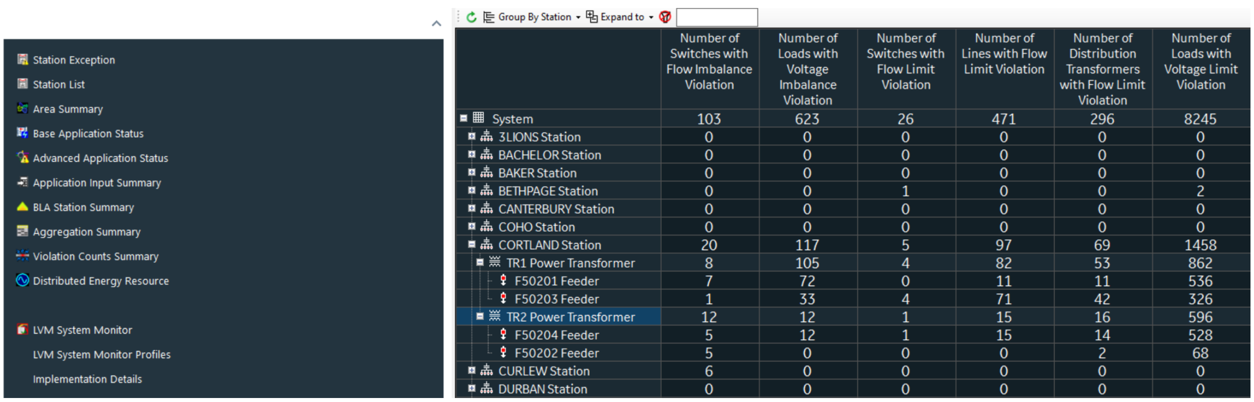


Figure 15. DNAF Violation Summary Display

The DNAF Violation Counts Summary toolbar includes the following controls:

- **Display:** Specifies how stations are displayed in the window. This allows a large number of stations, power transformers, or feeders to be managed with minimal user action. It also allows the operator to easily locate stations or groups of stations in the window.
  - **Group by Stations:** All stations are listed in alphabetical order.
  - **Group by Division:** All stations grouped by divisions. The divisions are listed in alphabetical order.
  - **Group by Optimization Group:** All stations grouped by optimizations groups. The optimizations groups are listed in alphabetical order.
- **Expand To:** Expands the elements tree to show stations, power transformers, or feeders.
- **Filter:** Filters the elements tree (system, divisions, optimizations groups, stations, transformers, or feeders) by the text entered in the filter box. Only those items that match the search criteria are shown.
- **Clear Filter Text:** Clears the filter box and returns to the non-filtered view.
- **Refresh:** Immediately refreshes the display to reflect any recent system changes. By default, the display automatically refreshes every 1 minute.

The following information is shown on the DNAF Violation Counts Summary display:

- **Number of Switches with Flow Imbalance Violation:** Number of three phase switching devices with a flow imbalance limit violation.
- **Number of Loads with Voltage Imbalance violation:** Number of three phase loads with a voltage imbalance limit violation.
- **Number of Switches with Flow Limit Violation:** Number of switching devices with a flow limit violation.
- **Number of Lines with Flow Limit Violation:** Number of lines with a flow limit violation.
- **Number of Distribution Transformers with Flow Limit Violation:** Number of distribution transformers with a flow limit violation.
- **Number of Loads with Voltage Limit Violation:** Number of loads with a high or low voltage limit violation.

2.8. Managing Distributed Energy Resources (DERs)

2.8.1. Distributed Energy Resource Display

The Distributed Energy Resource display shows the details about distributed energy resources in the system.

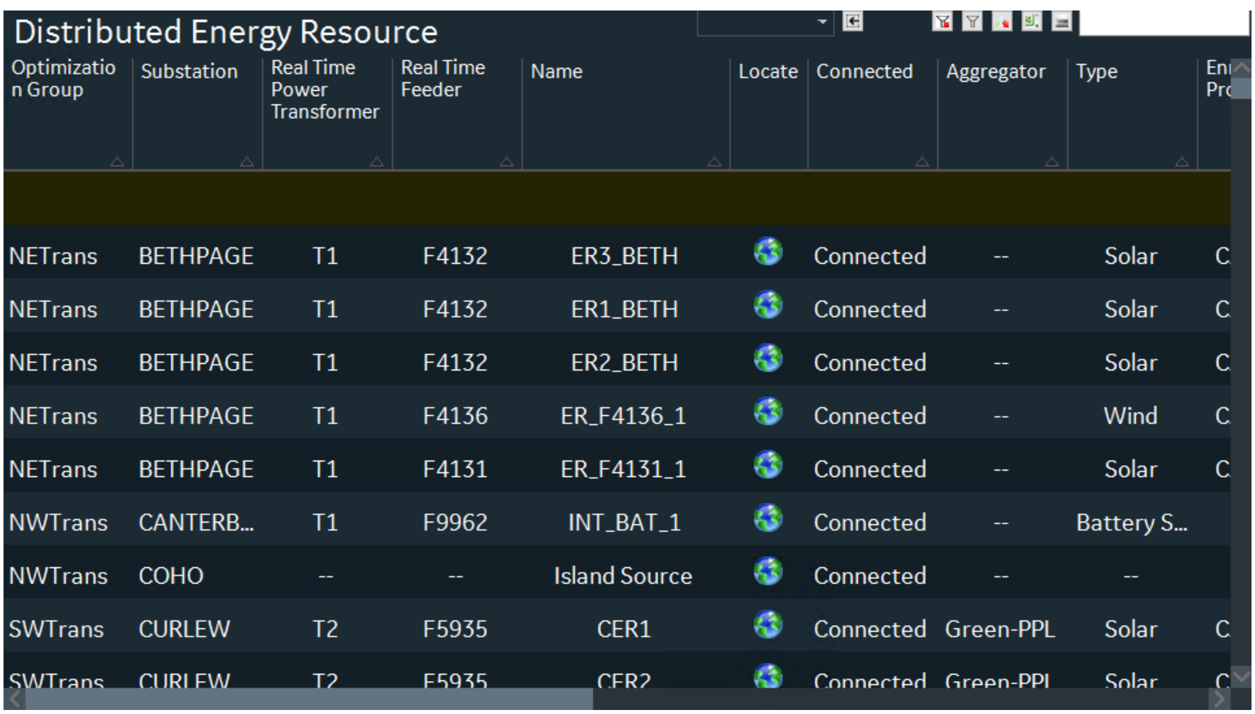


Figure 16. Distribution Energy Resource Display

To open this display, select the DNAF Display Plugin option from the RT DMS drop-down menu, and then select Distributed Energy Resource.

The following information is shown on this display:

- **Optimization Group:** Optimization group name.
- **Substation:** Substation name.
- **Real Time Power Transformer:** Energizing power transformer name.
- **Real Time Feeder:** Energizing feeder name.
- **Name:** Name of the energy resource.
- **Connected:** Connection status (Connected or Disconnected).
- **Aggregator:** Aggregator name.
- **Type:** DER device type, such as Wind or Solar.
- **Enrolled Program:** Enrolled programs of the DER.
- **Capacity (kVA):** Capacity in kVA.
- **Real Power Phase A/B/C:** The real power value at phases A, B, and C in kW.
- **Total Real Power (kW):** The total real power in kW.
- **Total Reactive Power (kVAR):** The total reactive power in kVAR.
- **Real Power Target (kW):** Real power target in kW.
- **Reactive Power Target (kVAR):** Reactive power target in kVAR.

Navigation:

- "Locate" navigates to the energy resource on the network view.
- "Analyst Attributes" displays the Analyst Attributes pop-up.
- "Analyst Outputs" displays the Analyst Output pop-up.
- "Operator Attributes" displays the Operator Attributes pop-up.
- "Operator Outputs" displays the Operator Output pop-up.

2.8.2. SCADA/DER Devices Tree View

The SCADA/DER Devices Tree View display lists SCADA devices and distributed energy resource objects that exist in the system in the hierarchical element tree view. The devices are grouped by substations and feeders. To open this display, select the SCADA/DER Devices Tree View Display option from the RT DMS drop-down menu.

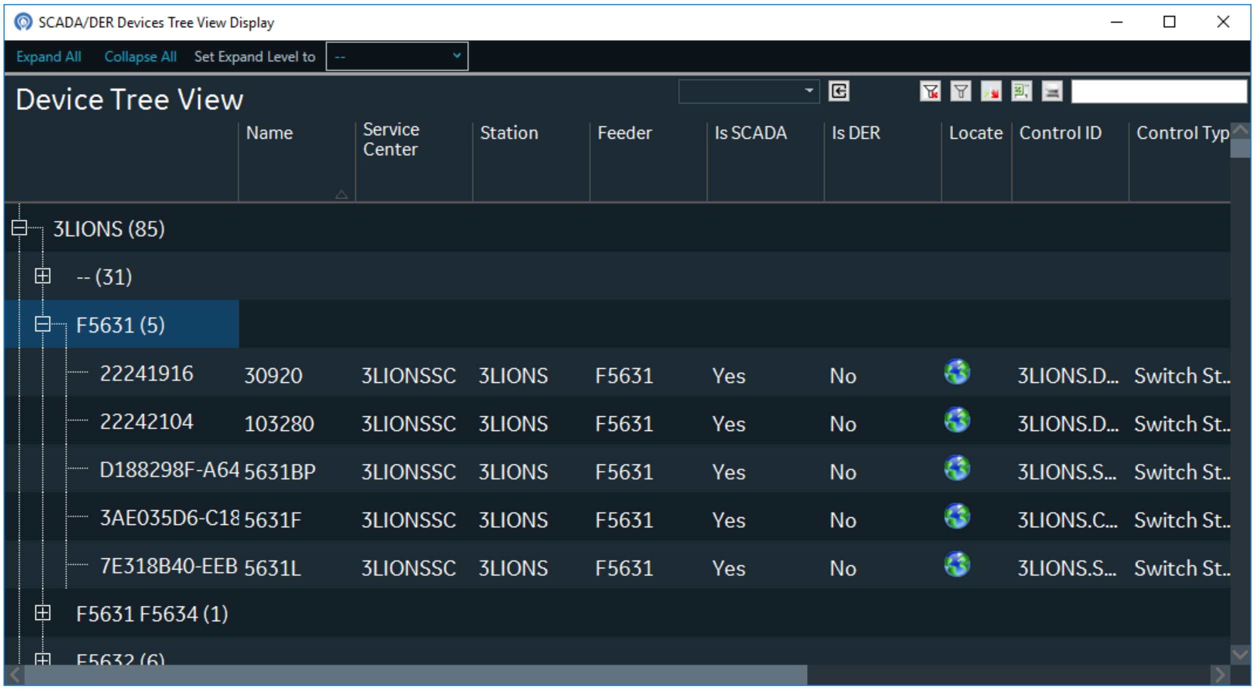


Figure 17. SCADA/DER Device Tree View Display

The following information is shown on this display:

- **Element tree:** Hierarchical element tree, which includes substations, feeders, and devices. The elements are listed in alphabetical order.
- **ID:** Device ID.
- **Name:** Device name.
- **Service Center:** Service center that the device belongs to.
- **Station:** Substation that is associated with the device.
- **Feeder:** Feeder that is associated with the device.
- **Is SCADA:** Specifies whether this is a SCADA device.
- **Is DER:** Specifies whether this is an energy resource.

2.8.3. Downstream DER Display

The Downstream DER display shows the details about distributed energy resources downstream of a distribution transformer. To open this display, select the Downstream DER option from the distribution transformer context menu on the network view.

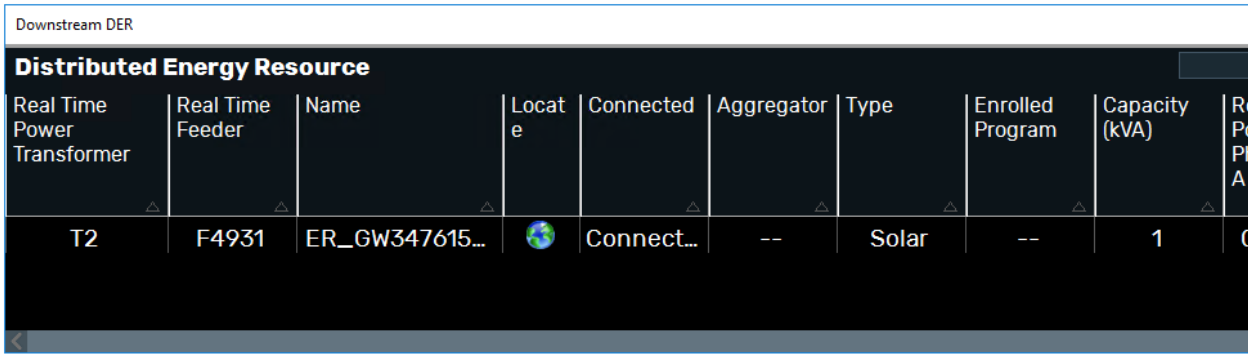


Figure 18. Downstream DER Display

For the description of the display fields, refer to Distributed Energy Resource Display.

2.8.4. Traced Load and Generation Display

The Traced Load and Generation display shows total load and generation for the traced part of the network or downstream of the selected switching device or transformer.

To open the Traced Load and Generation display for the traced part of the network:

1. Trace the required part of the network.
2. Right-click an empty space on the geographic network view and select the Traced Load and Generation menu option.

The display header shows the trace color for the last trace. The display also shows the last traced component type, name, feeder, and substation.

> **Note**
>
> When multiple traces appear on the network view, the Traced Load and Generation display shows data for all traces.

To open the Traced Load and Generation display for a switching device or transformer:

1. Select a switching device or transformer.
2. Right-click the device and select the Downstream Load and Generation menu option.

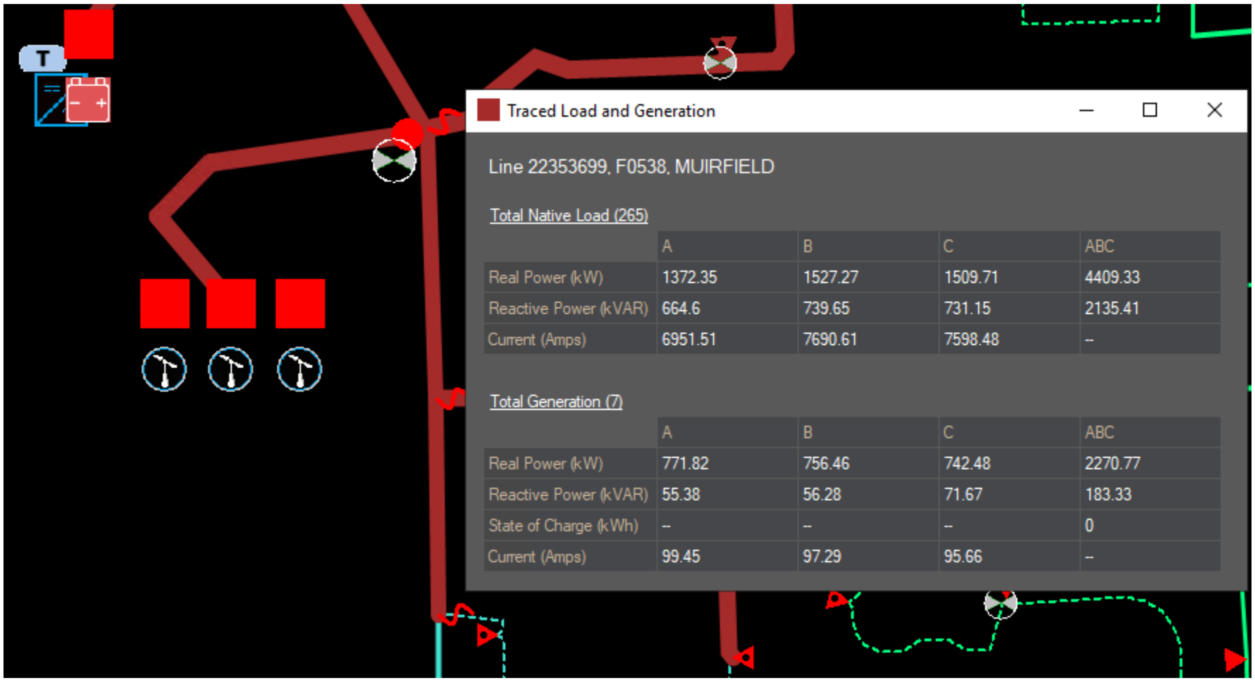


Figure 19. Traced Load and Generation Display

The following information is shown on this display:

- **Total Native Load:** Total native load for traced loads. The counter next to the Total Native Load field shows the number of traced loads. Click Total Native Load to open the Load display, which shows load data for traced loads.
  - **Total Native Load Real Power:** Total native load real power for phases A, B, C, and ABC in kW.
  - **Total Native Load Reactive Power:** Total native load reactive power for phases A, B, C, and ABC in kVAR.
  - **Current Phase A/B/C:** Phase A, B, and C current in Amps.
- **Total Generation:** Total generation for traced generators, energy resources, and loads with embedded generation. The counter next to the Total Generation field shows the number of traced devices. Click Total Generation to open the Generation display, which shows generation for traced devices.
  - **Total Real Power:** Total generation real power for phases A, B, C, and ABC in kW.
  - **Total Reactive Power:** Total generation reactive power for phases A, B, C, and ABC in kVAR.
  - **Total State of Charge:** Total energy resource state of charge for phases A, B, C, and ABC in kWh.
  - **Current Phase A/B/C:** Phase A, B, and C current in Amps.

2.9. DNAF Configuration Editor

Each network application has a set of configurable parameters that can be set by the user (analyst). These parameters include logging parameters (for writing out server output files), convergence tolerances, definitions of problem formulations for the switching reconfiguration applications, definitions of problem formulations for the LVM application, and other key parameters. These parameters are stored in the `DNAF_PARAMS.xml` file.

To make it easier to edit the `DNAF_PARAMS.xml` file, the DNAF Configuration Editor tool was created. This tool provides a graphical user interface for editing the file and automatically placing the file online in ADMS.

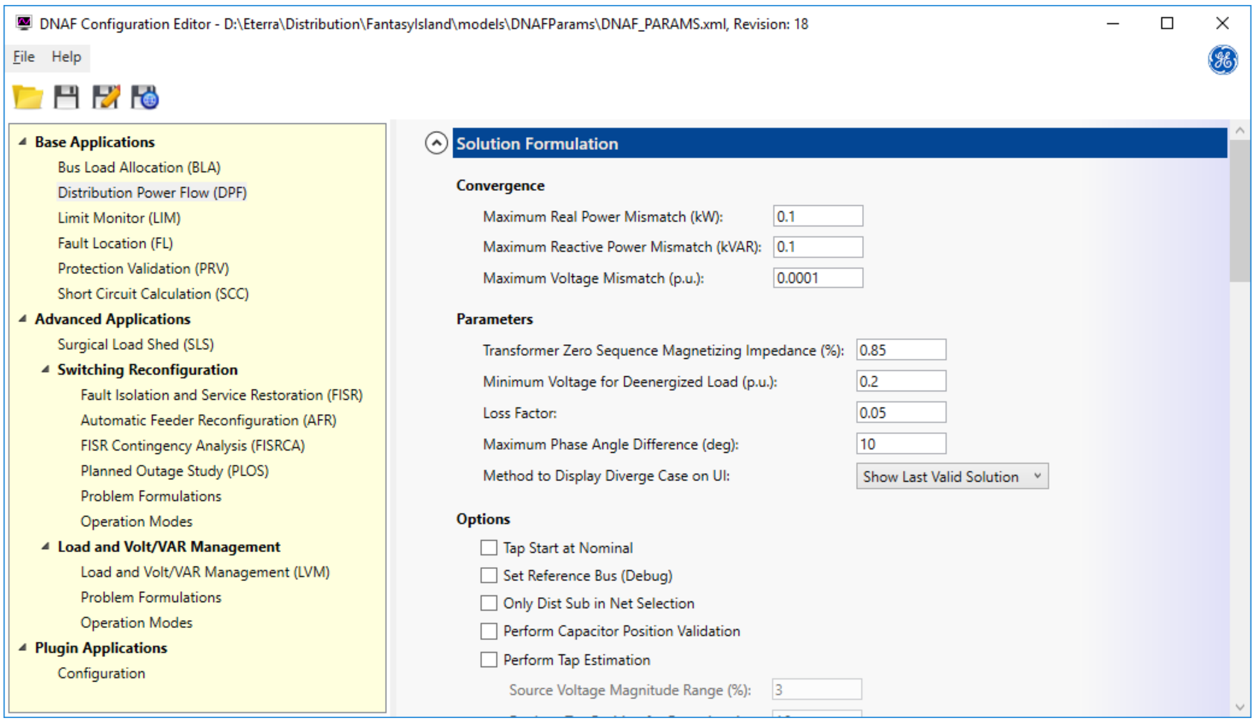


Figure 20. DNAF Configuration Editor

For details on the DNAF Configuration Editor and the application configuration parameters, consult the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide* and the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

3. Distribution Power Flow

The Distribution Power Flow (DPF) application is an integral part of the DMS. It provides a phase domain solution to unbalanced power system network models. The DPF calculates individual bus phase voltages, network power flows, and network losses.

Solution input and output are in terms of phase quantities. There are no assumptions of a radial network, and radial networks are solved just as an interconnected network. The solution handles multiple PV (power and voltage) buses and any physically realizable combination of network connections.

DPF provides a solution based on convergence tolerances for real power, reactive power, and voltage. The power flow solution is obtained using a nodal admittance matrix analysis technique, where the unknowns are the phase bus voltages. Individual phase voltages are treated as independent variables. Symmetrical component analysis is not used during the solution, but some sequence quantities can be computed and output in the final states of an analysis.

Bus Load Allocation

Within the DPF (and other DNAF, including PRV, LVM, and all reconfiguration applications), a load estimation procedure named "Bus Load Allocation" (BLA) is executed to determine the nodal load values. The BLA procedure uses the real-time measurements from SCADA and pre-defined customer type information to scale the distribution system loads to the real-time measurements. The expected value at a given date/time is obtained by multiplying the nominal load by a scaling factor that reflects the energy usage at a specific time.

The BLA function allocates loads (kW and kVAR) at distribution nodes to support other analysis functions. Loads are allocated using the load model data, and they take into account actual or predicted feeder loads and the relevant day type and time of day. The BLA ensures that the load model and current network topology are consistent with metered flows, current breaker statuses, and metered loads along the distribution feeders.

The input data for the BLA is taken from the Network Topology Processor (NTP) function and the distribution system model. The NTP determines the substation and feeder connectivity, and the distribution system model provides all load data, including the typical load shapes.

DPF Post-Processing Functions

After the DPF has been solved, several post-processing functions are run, including Limit Monitor (LIM), Loss Analysis (LA), and the Power Quality Index (PQI) evaluation.

The Limit Monitor function identifies all limit violations after an execution of the DPF. All load and bus voltages and branch flows calculated by the DPF are checked against their limits. If values are outside an acceptable range, a limit violation has occurred. Limit violations can be displayed to the operator and in real-time mode, alarms can optionally be sent to the centralized alarm facility.

The Loss Analysis function calculates individual device losses and aggregates losses for selected groups of devices, including feeders and substations. The losses are available per phase and for every device involved in a DPF solution.

The Power Quality Index application function is responsible for estimating the quality of service. The function calculates voltage quality based on accumulated power quality indices, which are based on load voltage excursions from quality limits.

3.1. Running DPF in Real-Time Mode

In real-time mode, DPF can be configured to solve for event triggers, such as a switch status change or the insertion/removal of a temporary modification, including cuts and jumpers. If event triggers are enabled, the DPF solves the impacted station and the connected surrounding portions of the network after a user-configured time delay. This time delay allows the system to reach a steady-state condition.

In addition to event triggers, a periodic trigger can also be configured. At each minimum periodic time interval, if the station load has changed by more than the maximum load change percentage from the previous solution, then another solution is executed for the station. If a solution has not been performed within the maximum periodic time interval, then a forced solution is executed for the station. The maximum load change percentage is set using the DNAF Configuration Editor, and the maximum DPF periodic time interval and the minimum DPF periodic time interval are configured in the DNAF server configuration file.

Also, a deadband for periodic triggers is implemented to avoid excessive triggering. If DPF was last solved for a station within a configurable time frame, the DPF periodic trigger for that station is ignored. For more information, refer to the "NetServerDnafConfig.txt" section in the *ADMS Software Installation and Maintenance Guide*.

A DPF solution can be manually requested at any time using the DNAF Solution display, as described in Executing DNAF Applications Manually. This manual request is the highest priority request, followed by event requests, and then periodic requests.

3.2. Running DPF in Study Mode

After study mode has been initialized and the desired changes have been made, the DPF solution can only be triggered manually, as described in Executing DNAF Applications Manually.

3.3. Running DPF with a Specific Evaluation Time

You can run Distribution Power Flow (DPF) with a specific evaluation time. This feature enables you to calculate DPF for the current network conditions for different dates with different load schedule values. This feature is designed for the simulator mode, but it is also supported for the real-time mode.

To enable this feature, set the SPECIFIC_TIME_FOR_DPF_EVALUATION_ENABLED parameter to 1 in the NetServerSimConfig.txt configuration file (and in NetServerDnafConfig.txt, if needed).

To apply a different system time for running DPF in mode, use the Distribution Operator Training Simulator (DOTS) Evaluation Time window. You can open this window from the SIM menu.

> **Note**
>
> The DPF evaluation time for the real-time mode can be defined using the offset file specified by the DOTS_DPF_EVALUATION_TIME_DEFINITION_FILE parameter in the NetServerDnafConfig.txt file. If the offset file path is the same as that configured by the DOTS_DPF_EVALUATION_TIME_DEFINITION_FILE parameter in the NetServerSimConfig.txt file, you can use the DOTS Evaluation Time window to specify the evaluation time.

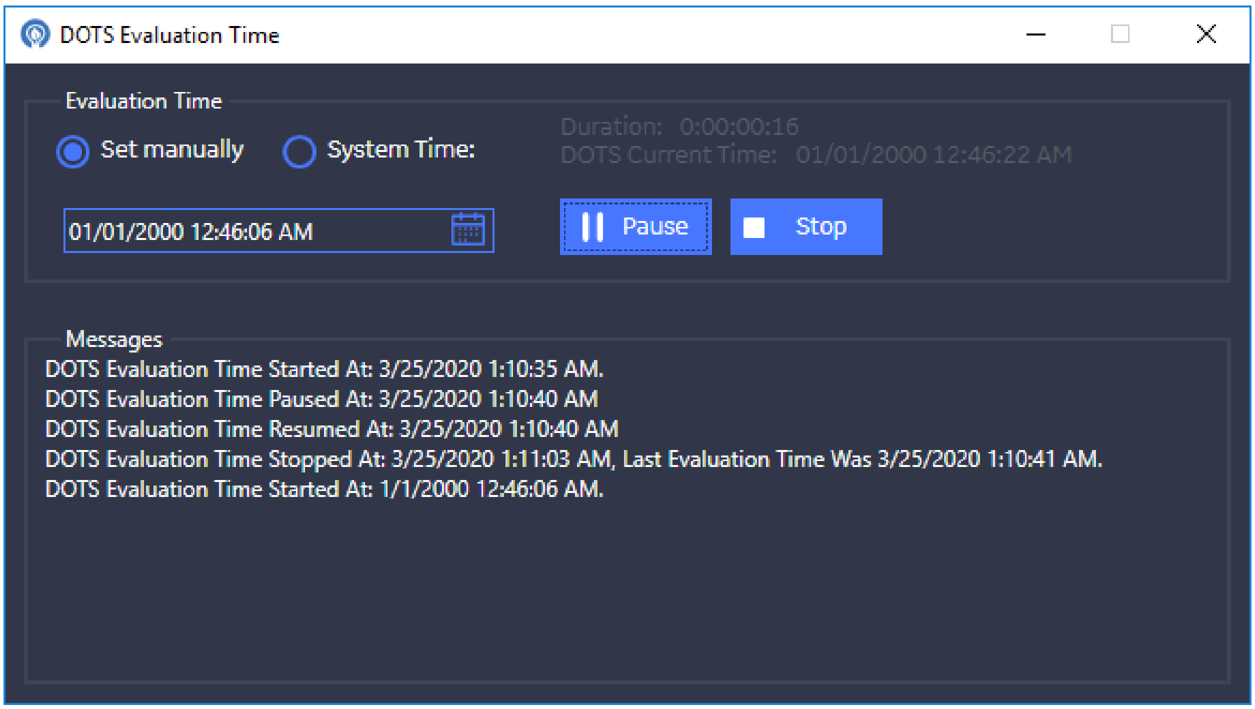
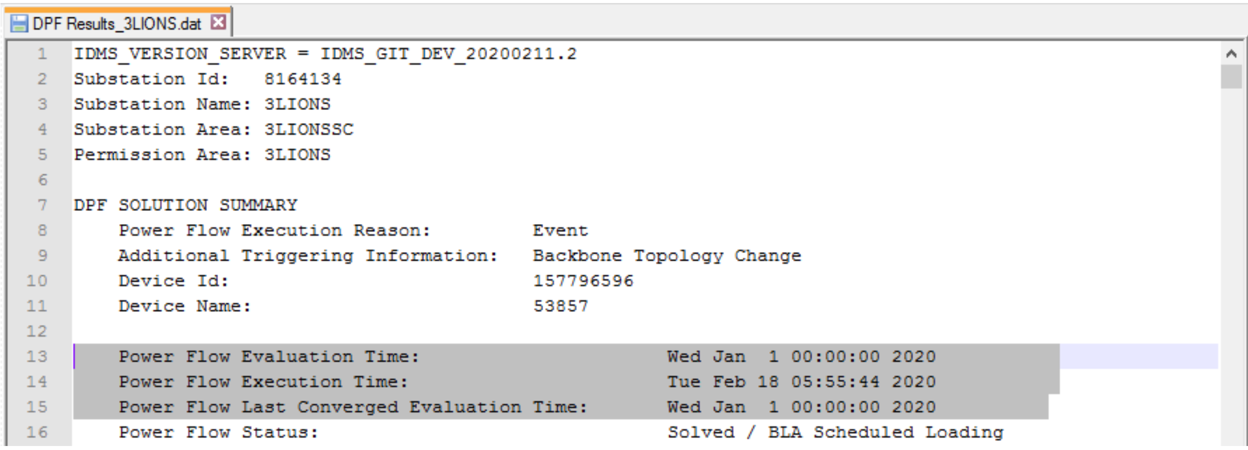


Figure 21. DOTS Evaluation Time Window

The DOTS Evaluation Time window includes the following fields and controls:

- **Evaluation Time:** Specifies the system time for DPF.
  - **Set Manually:** Select this option to use a specific system time for running DPF.
  - **System Time:** Select this option to use the actual system time for running DPF.
- **Start:** Applies the selected system time for DPF. All subsequent DPF solutions are run with the specified system time.
- **Pause:** When DOTS evaluation is paused, the system time for running DPF is fixed at a certain time. If a DPF solution occurs (due to a manual trigger or periodic trigger or event trigger), it is solved with the evaluation time equal to the paused system time.
- **Resume:** Resumes running DPF for the selected system time.
- **Stop:** Resets the time settings and re-applies the actual system time for DPF.

- **Duration:** Duration (in seconds) for applying a specific system time for running DPF.
- **DOTS Current Time:** The currently applied system time.
- **Messages:** Log messages that are associated with running DPF with a specific evaluation time.




```
IDMS_VERSION_SERVER = IDMS_GIT_DEV_20200211.2
Substation Id:   8164134
Substation Name: 3LIONS
Substation Area: 3LIONSSC
Permission Area: 3LIONS

DPF SOLUTION SUMMARY
    Power Flow Execution Reason:          Event
    Additional Triggering Information:    Backbone Topology Change
    Device Id:                            157796596
    Device Name:                          53857

    Power Flow Evaluation Time:                       Wed Jan  1 00:00:00 2020
    Power Flow Execution Time:                        Tue Feb 18 05:55:44 2020
    Power Flow Last Converged Evaluation Time:        Wed Jan  1 00:00:00 2020
    Power Flow Status:                                Solved / BLA Scheduled Loading
```

Figure 22. DPF Results File - Running DPF with a Specific Evaluation Time

3.4. Distribution Power Flow (DPF) Results

DPF results can be viewed on the following tabular displays:

- **DPF Station Results:** This display provides high-level DPF output for the substation and its devices.

  To access this display:

  - Click the DPF Results button on the Station Output Summary display.
  - Click the DPF Status button on the Station Exception display.
  - Click the button under the DPF application column on the DNAF Solution display.

- **DPF Feeder Results:** This display shows Distribution Power Flow output related to feeders associated with the substation. It shows the current (Amps) and real/reactive power with respect to each phase of the feeder. The feeder high and low load/bus voltages are also shown.

  To access this display:

  - Click the DPF Results button on the Station Output Summary display.
  - Click the DPF Status button on the Station Exception display.
  - Click the button under the DPF application column on the DNAF Solution display.

- **Substation DPF Convergence Data:** This display provides information about DPF convergence for the substation.

  To access this display, click the Convergence Data button on the DPF Station Results display.

- **Transformer and Regulator DPF Results:** This display provides a summary of the power flow results for each of the tap-changing transformers and regulators.

To access this display, click the Transformer and Regulator Summary button on the DPF Station Results display.

- **Capacitor DPF Results:** This display provides a summary of the power flow results for each capacitor in the selected station.

  To access this display, click the Capacitor Summary button on the DPF Station Results display.

- **Generator/Distributed Energy Resource DPF Results:** This display provides information for each generator and energy resource object in the selected station.

  To access this display, click the Generator/Distributed Energy Resource Summary button on the DPF Station Results display.

- **Load DPF Results:** This display shows Distribution Power Flow output related for each load in the selected station.

  To access this display, click the Load Summary button on the DPF Station Results display.

- **DPF Bus Voltage Summary:** This display shows the voltages for each bus in the selected station.

  To access this display, click the Bus Voltage Summary button on the DPF Station Results display.

- **Interface Summary:** This display provides the Amp, Volt, and phase angle for each phase of the interface. It also contains real and reactive power injections for each phase.

  To access this display, click the Interface Summary button on the DPF Station Results display.

- **Reserved Capacity DPF Results:** This display provides detailed information for the reserved capacity objects in a station.

  To access this display, click the Reserved Capacity Output button on the DPF Station Results display.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

A DPF Results output file is generated whenever DPF is triggered. This file provides a solution summary, application configurations, alarms, manual entry values, violations, measurements, bus load allocation results, and device power flow results.

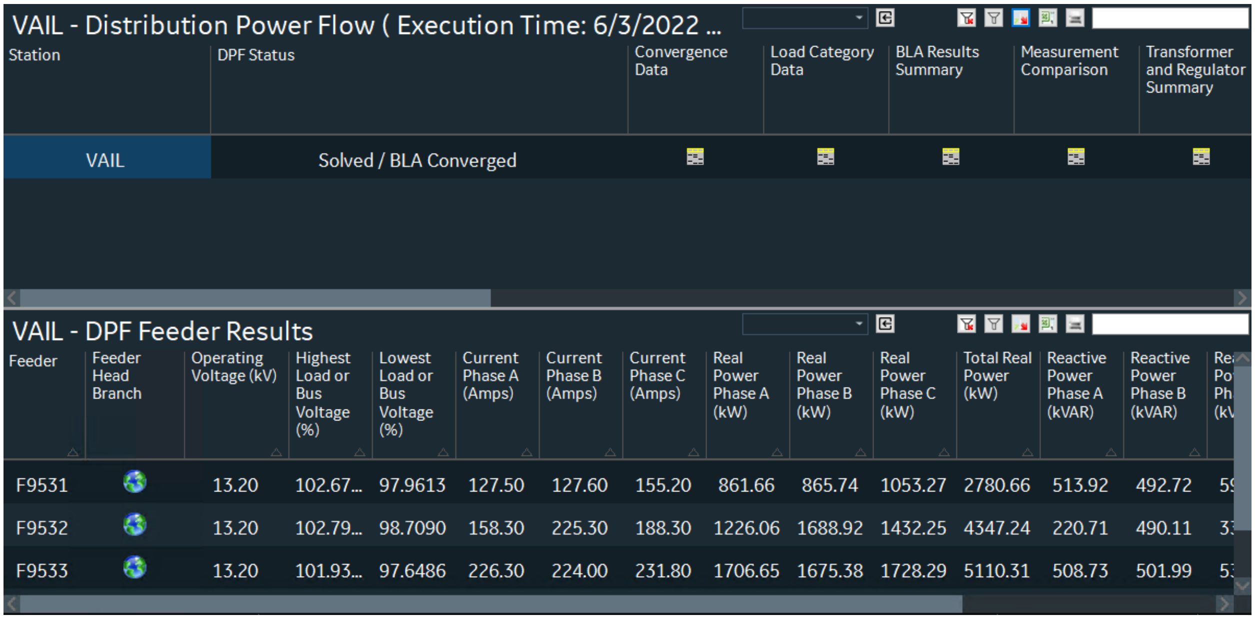


Figure 23. Station DPF Results Display

3.4.1. DPF Results for Devices

The following devices have DPF-associated output displays:

- Substations
- Lines/Cables
- Switching Devices (includes disconnects, fuses, breakers, reclosers, sectionalizers, elbows, permanent jumpers, and network protectors). The Analyst Output display for switches (breakers, reclosers, fuses (three-phase), and switches) shows the Distribution Power Flow (DPF) calculated total downstream real and reactive power DER injections per phase. These values are displayed only for three-phase feeder main (backbone) switching devices.
- Transformers (includes distribution transformers, auto-transformers, step up/down transformers, station power transformers, regulators, and zig-zag grounding transformers)
- Shunt Capacitors
- Series Reactors
- Loads
- Generators and distributed energy resources

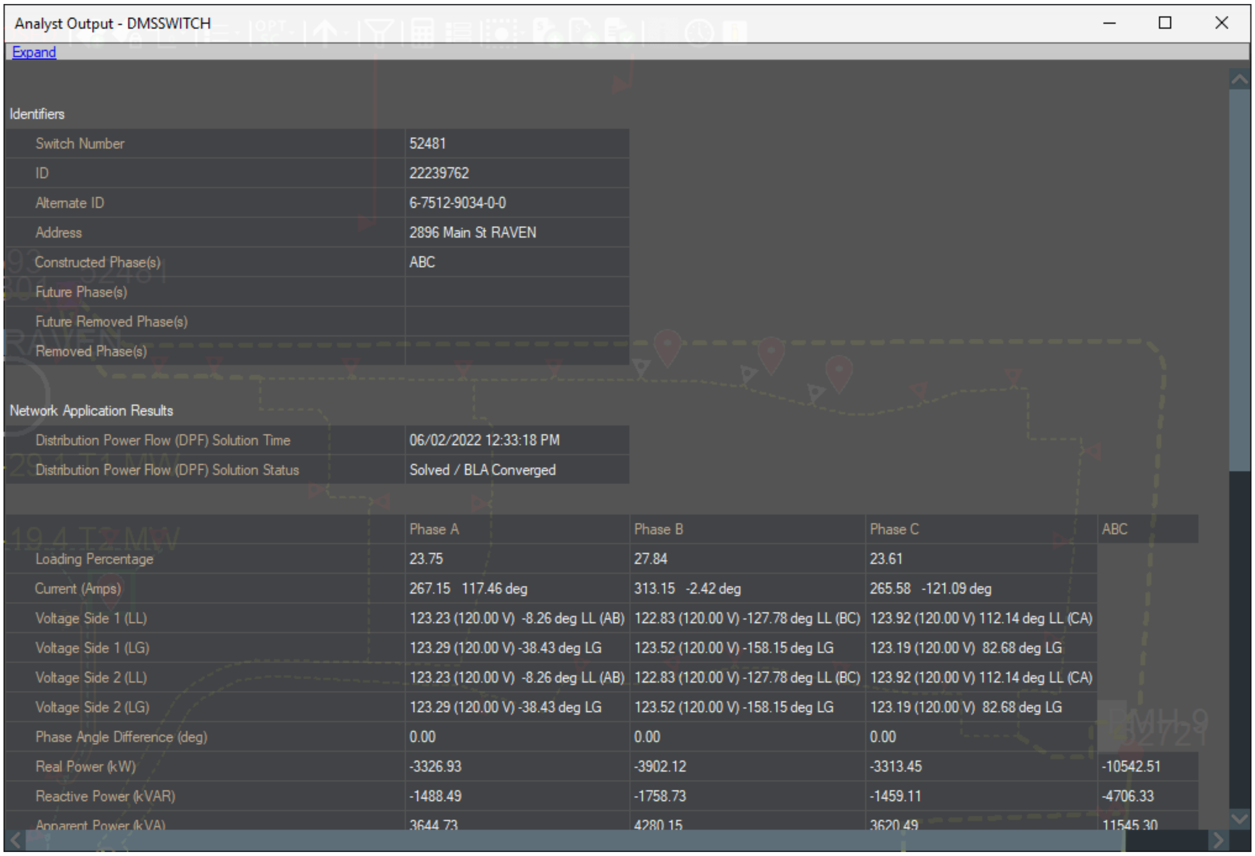


Figure 24. Switch Analyst Output Display

Because DPF provides a phase domain solution to unbalanced power system network models, the individual device results displays show per phase quantities (phases A, B, and C) for voltages and flows, including real power, reactive power, apparent power, and Amps. Summations of the individual phase flows are also shown on some displays.

For primary network voltage results, the values are shown either in per-unit, actual voltage values, or on a user-selected base (such as 120.0, which is commonly used in the United States). Voltages can be shown line to ground or line to line, depending on the network connection (wye or delta) and the number of phases that are present.

Results can also be shown on nominal switchable sections. A "switchable section" is the portion of the network that is bounded by feeder main switching devices (most commonly disconnect switches, reclosers, and breakers).

To view the results for a switchable section, select a line and then click the Attributes > Analyst Output on the toolbar. The line Analyst Output display shows the net flows or load (Amps, Watts, and VARs) within the section.

3.4.2. BLA Results

Bus Load Allocation results can be viewed on the following tabular displays:

- **BLA Results Summary:** This display provides Bus Load Allocation calculations for each measurement island.

  To access this display, click the BLA Results Summary button on the DPF Station Results display.

- **BLA Feeder Summary:** This display provides information about measured and calculated current for the feeders of a station.

  To access this display, click the Feeder Summary button on the DPF Station Results display.

- **Load Category Data:** This display provides the load to voltage dependency coefficients for each load category.

  To access this display, click the Load Category Data button on the Station DPF Results display.

- **BLA Station Summary:** This display provides Bus Load Allocation calculations by station.

  To access this display, click BLA Station Summary on the RT menu.

- **BLA Data:** This display shows parameters and values that have been used for the bus load allocation calculations.

  To access this display, use the following Netview URL:

  `Netview://RT/gridtabular/BLAdata_GridDisplay/<Substation Name>`.

- **Non-Converged BLA and Scheduled Loading Measurement Islands:** This display is available for FISR, AFR, and PLOS. It provides a summary of non-converged and scheduled loading measurement islands in the plan reconfiguration path.
- **BLA Consistency Check Summary:** This display provides information about the inconsistencies detected in a measurement island after evaluating the telemetered switching device's analog measurements (current, voltage, real power, reactive power, and apparent power).

  To access this display, click the BLA Consistency Check Summary button on the BLA Results Summary display. The BLA Results Summary display can be accessed from the BLA Station Summary, which can be accessed from the RT menu.

3.4.3. Loss/PQI Results

The Loss Analysis and Power Quality Index function results can be viewed on the following tabular displays:

- **Loss/PQI Station Losses Summary:** This display provides the total feeder and substation losses as well as the total losses of substation elements such as power transformers, station regulators, and station series reactors.

  To access this display, click the Loss/PQI Summary button on the DPF Station Results display.

- **Loss/PQI Station Power Transformer and Regulator Losses:** This display provides the transformer and regulator core and copper losses.

To access this display, click the Loss/PQI Summary button on the DPF Station Results display.

- **Loss/PQI Feeder Losses and Power Quality Indices:** This display provides the aggregated losses for elements along the feeder.

  To access this display, click the Loss/PQI Summary button on the DPF Station Results display.

3.4.4. LIM Results

The flow and voltage violation results calculated by the Limit Monitor (LIM) function within the DPF can be viewed on the following tabular displays:

- **High Voltage Violations:** This display shows the high voltage violations. To access this display, click the High Voltage Violation value on the Station Exception display.
- **High Voltage Violations per Feeder:** This display shows high voltage violations by feeder. This display can be accessed by clicking the High Voltage Violation value on the Feeder Exception Summary display.
- **Low Voltage Violations:** This display shows low voltage violations. To access this display, click the Low Voltage Violation value on the Station Exception display.
- **Low Voltage Violations per Feeder:** This display shows low voltage violations by feeder. To access this display, click the Low Voltage Violation value on the Feeder Exception Summary display.
- **Voltage Imbalance Violations:** This display shows voltage imbalance violations. To access this display, click the Largest Voltage Imbalance value on the Station Exception display.
- **Voltage Imbalance Violations per Feeder:** This display shows voltage imbalance violations for the selected feeder. To access this display, click the Largest Voltage Imbalance value on the Feeder Exception Summary display.
- **Flow Imbalance Violations:** This display shows flow imbalance violations. To access this display, click the Largest Flow Imbalance value on the Station Exception display.
- **Flow Imbalance Violations per Feeder:** This display shows the flow imbalance violations for the selected feeder. To access this display, click the Largest Flow Imbalance value on the Feeder Exception Summary display.
- **Station Branch Flows:** This display shows branch loading percentages including overload violations. To access this display, click the Maximum Loading value on the Station Exception display.
- **Station-Feeder Branch Flows:** This display shows branch loading percentages including overload violations for the selected feeder. To access this display, click the Maximum Loading value on the Feeder Exception Summary display.

Voltage violations are evaluated for all loads and buses, flow imbalance limit violations are evaluated for all three-phase switching devices, and voltage imbalance violations are evaluated for all three-phase loads. Overload violations are evaluated for all switching devices, transformers and lines, and capacity margin violations are evaluated for lines, transformers, and switching devices.

The violations are ordered on the relevant displays such that the most severe violations are presented first.

The voltage, overload, and capacity margin violations can also be shown graphically. To view the flow violation halos, click the Show Halos button on the common toolbar and select the relevant options. In Switch Analyst Output Display, the entire switchable section is highlighted to indicate that at least one device in the section has an overload violation. After zooming in further, the halo on the switchable section disappears, and only the individual device overloads are displayed, as shown in the following image. A distribution transformer is also shown in this image as overloaded. The flow violation halo colors (which represent one to three limit levels) are configurable through the Viewer Configuration Editor.

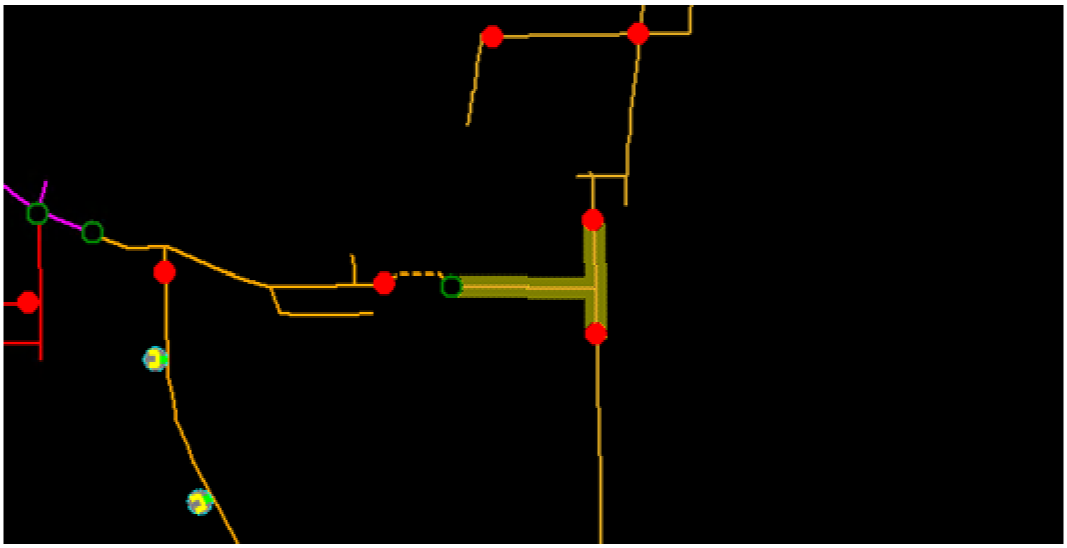


Figure 25. Switchable Section Overload Violation Halos

Figure 26. Individual Device Overload Violation Halos

Similar to the overload halos, the voltage halos are shown on a switchable section, and on the individual transformer or load that has the violation. In the above image, the entire switchable section is highlighted to indicate that at least one transformer/load in the section has an overload violation. After zooming in further, the halo on the switchable section disappears, and only the individual transformer/load violations are displayed, as shown in the following image. The low and high voltage violation halo colors are configurable through the Viewer Configuration Editor. In the following image, the high voltage violations are shown at the top and the low voltage violations are shown at the bottom.

Figure 27. Switchable Section Voltage Violation Halos

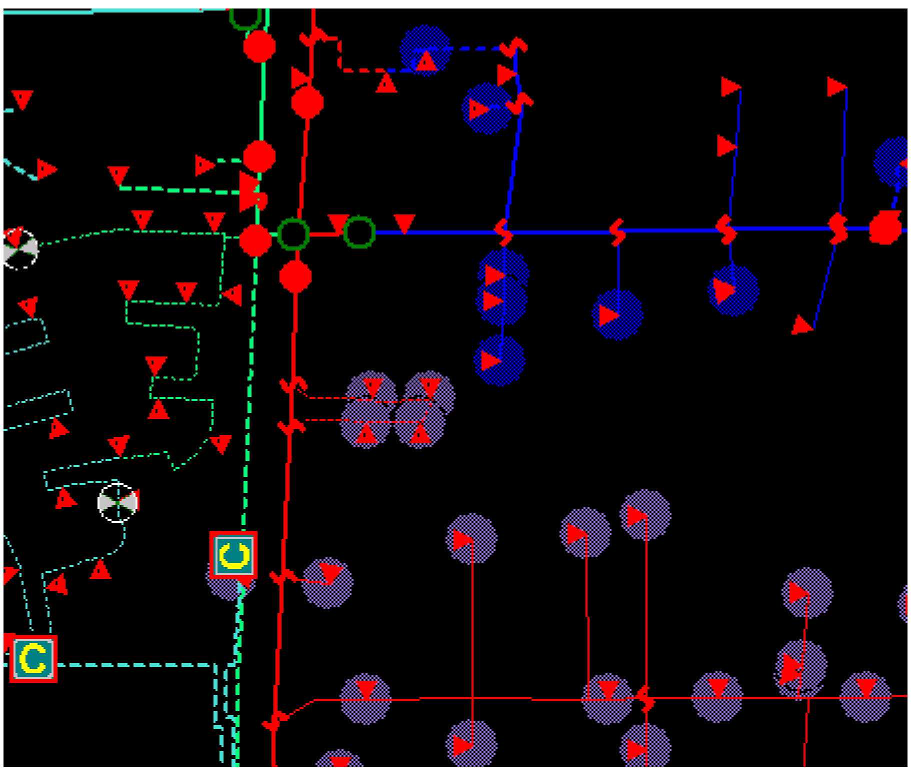


Figure 28. Transformer Voltage Violation Halos

3.4.5. DPF Check Before Operate

The Check Before Operate (CBO) functionality issues notifications when the operator attempts to issue a switching step or perform an operation that affects customers, crews, specific equipment, or network topology. For more information, refer to the *Distribution Management System User Guide.*

If the Enable Distribution Power Flow option is enabled in the Viewer Configuration Editor, the Check Before Operate (switching validation) functionality can show results from a Distribution Power Flow analysis of the post operation conditions. The following checks are considered before and after operation, on either side of a device, if applicable:

- Minimum and maximum voltages and violation state on the feeder.
- Maximum loading and violation state on the feeder.
- Real power, reactive power, and current flows through the device and violation state.

The DPF analysis of the post operation conditions is supported for switching operations, tap changing transformer tap position changes, and capacitor status changes. To run this check, click Evaluate on the device state change dialog.

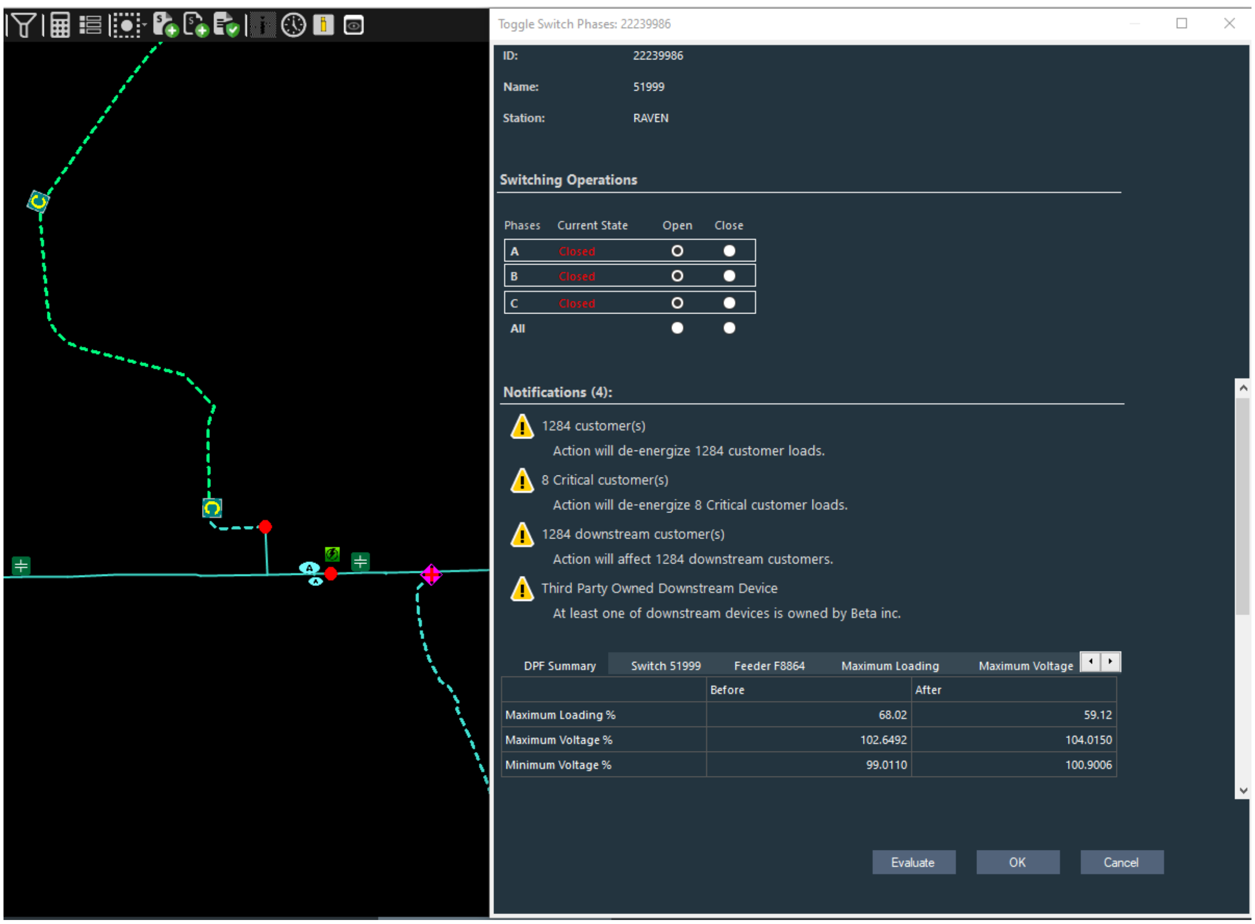
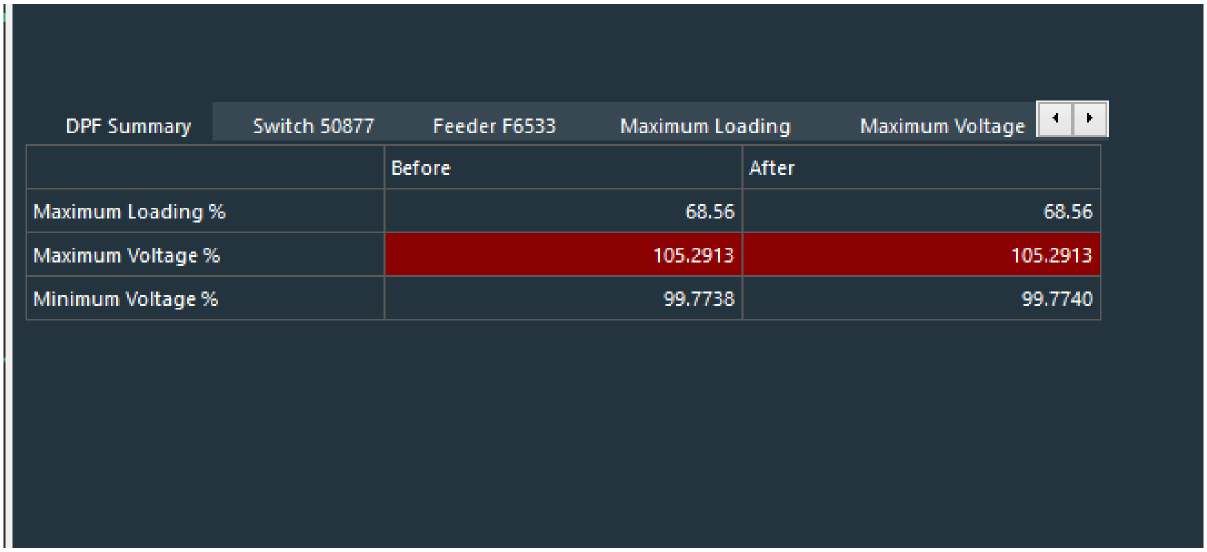


Figure 29. Check Before Operate Notifications

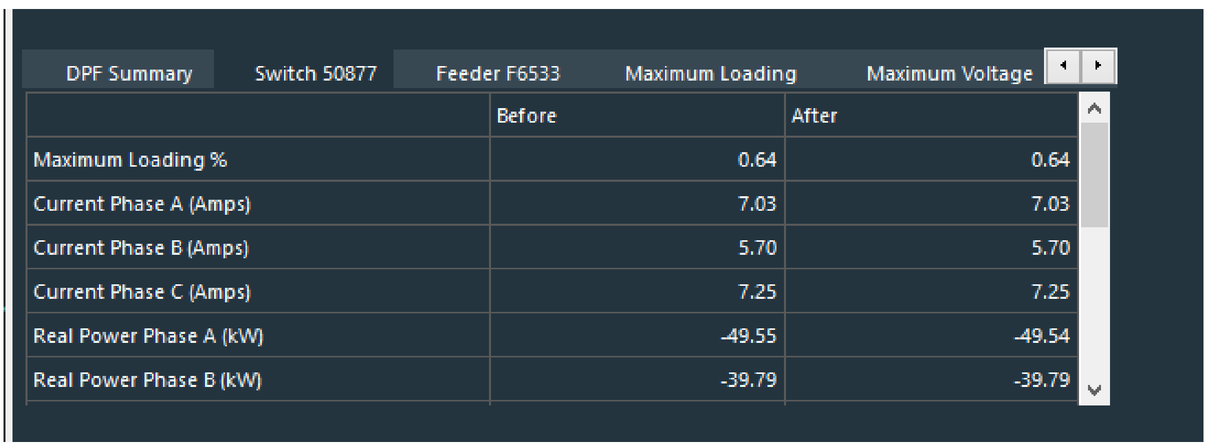
The DPF CBO Results are organized into multiple tabs displayed under Notifications:

- **DPF Summary:** Shows the maximum loading, maximum voltage, and minimum voltage on the feeder(s) before and after operation. Loading and voltage violations are shown in red. For the operation of tie switches between two feeders, the information for the two feeders will be summarized. For other device operations, only the information for the device's associated feeder will be summarized.



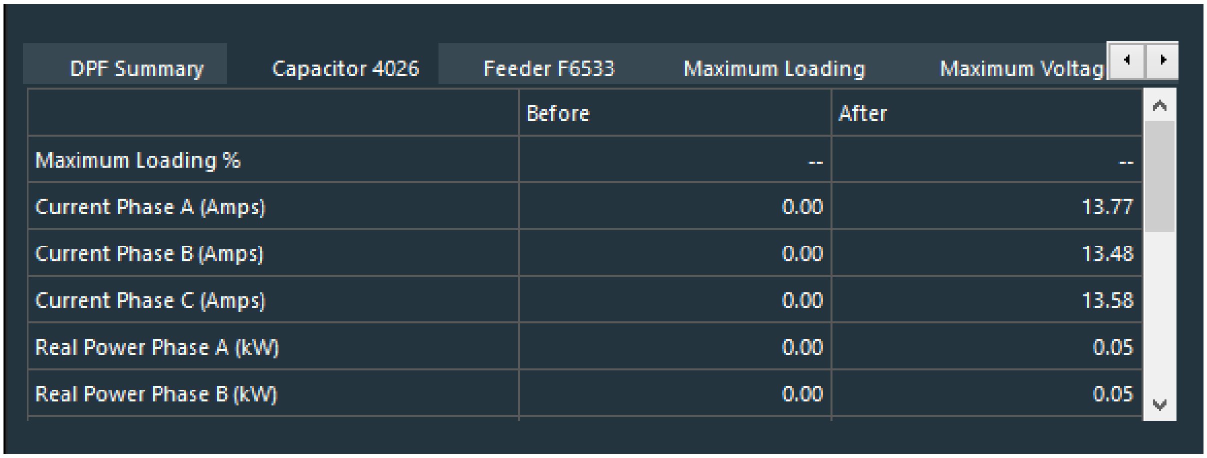
	Before	After
Maximum Loading %	68.56	68.56
Maximum Voltage %	105.2913	105.2913
Minimum Voltage %	99.7738	99.7740

- **Switch:** Shows the maximum loading, current, real power, and reactive power through the switch and voltages at the two sides of the switch before and after operation. This tab appears for switching device status changes.



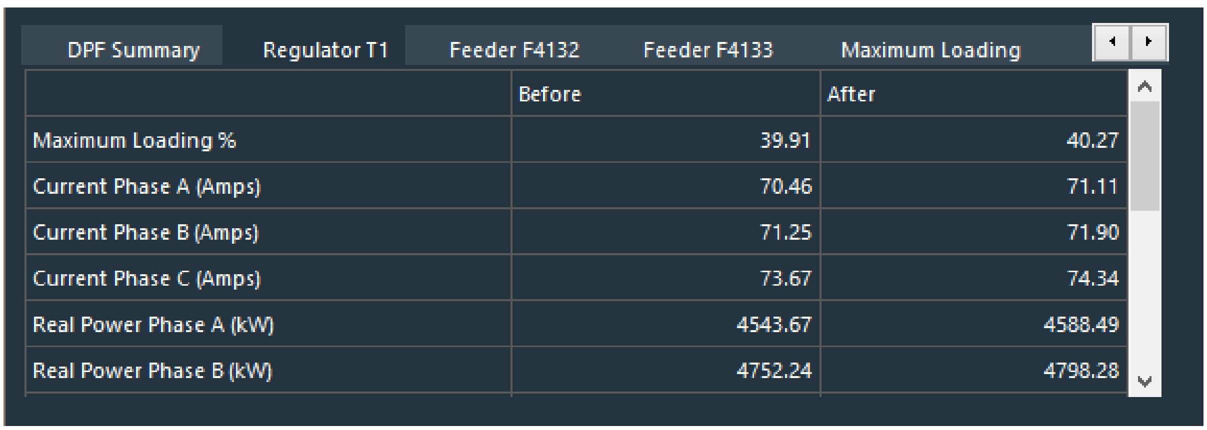
	Before	After
Maximum Loading %	0.64	0.64
Current Phase A (Amps)	7.03	7.03
Current Phase B (Amps)	5.70	5.70
Current Phase C (Amps)	7.25	7.25
Real Power Phase A (kW)	-49.55	-49.54
Real Power Phase B (kW)	-39.79	-39.79

- **Capacitor:** Shows the real power, reactive power, and current output from the capacitor and voltages at the connection node of the capacitor before and after operation. This tab appears for capacitor status changes.



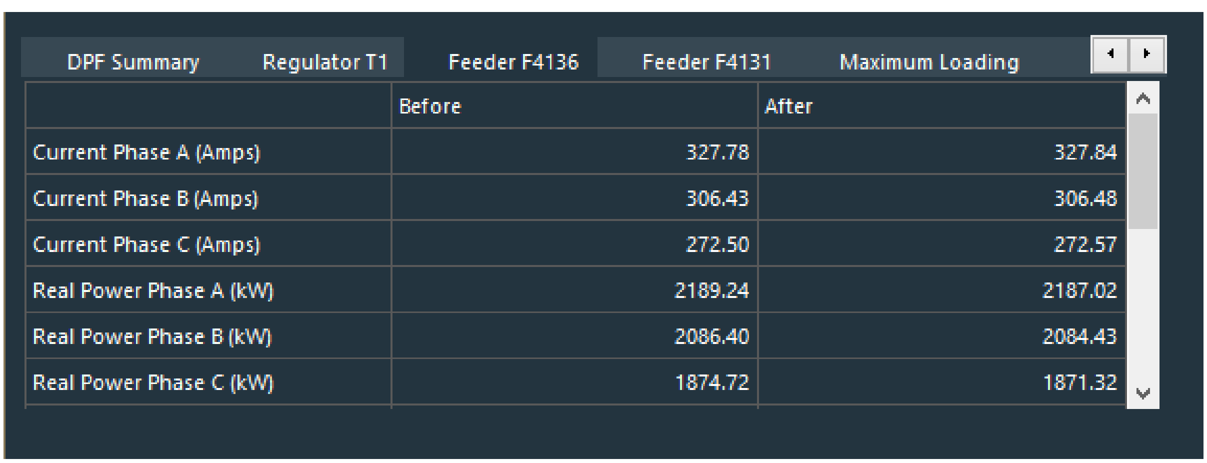
	Before	After
Maximum Loading %	--	--
Current Phase A (Amps)	0.00	13.77
Current Phase B (Amps)	0.00	13.48
Current Phase C (Amps)	0.00	13.58
Real Power Phase A (kW)	0.00	0.05
Real Power Phase B (kW)	0.00	0.05

- **Regulation Tap:** Shows the maximum loading, real power, reactive power, and current through the tap changing transformer and voltages at the two sides of the tap changing transformer for the regulation tap before and after operation. This tab appears for tap position operation of tap changing transformers



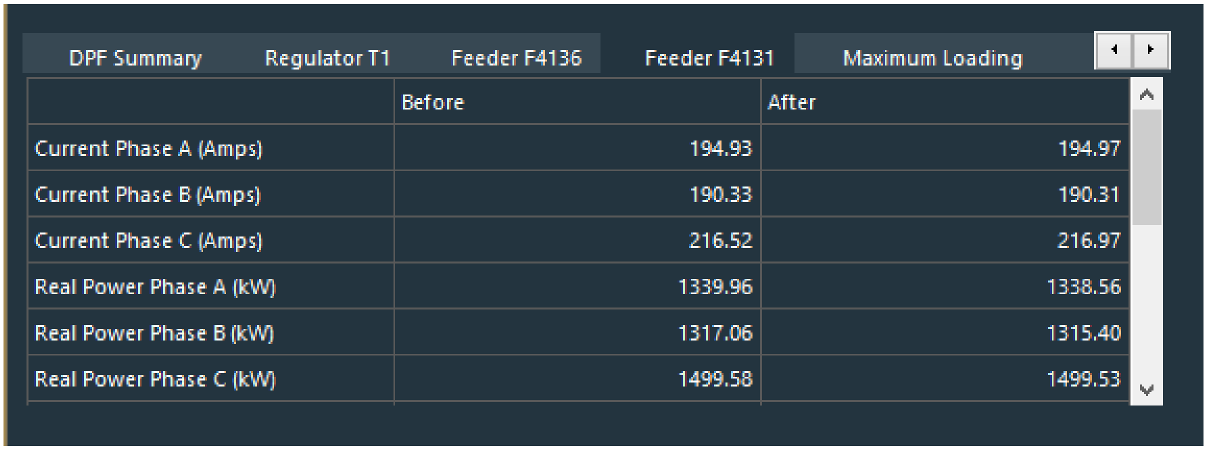
	Before	After
Maximum Loading %	39.91	40.27
Current Phase A (Amps)	70.46	71.11
Current Phase B (Amps)	71.25	71.90
Current Phase C (Amps)	73.67	74.34
Real Power Phase A (kW)	4543.67	4588.49
Real Power Phase B (kW)	4752.24	4798.28

- **Feeder 1:** Shows the current, real power, and reactive power through the feeder head branch and the maximum and minimum voltages on the first feeder before and after the device operation.



	Before	After
Current Phase A (Amps)	327.78	327.84
Current Phase B (Amps)	306.43	306.48
Current Phase C (Amps)	272.50	272.57
Real Power Phase A (kW)	2189.24	2187.02
Real Power Phase B (kW)	2086.40	2084.43
Real Power Phase C (kW)	1874.72	1871.32

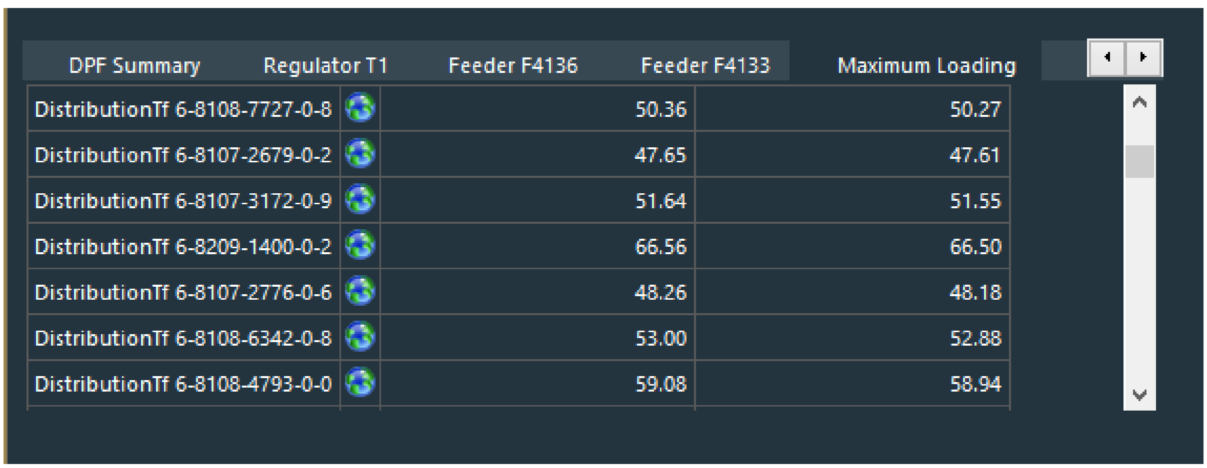
- **Feeder 2:** Shows the current, real power, and reactive power through the feeder head branch and the maximum and minimum voltages on the second feeder before and after the device operation. This applies to open tie switches between two feeders.



DPF Summary | Regulator T1 | Feeder F4136 | Feeder F4131 | Maximum Loading

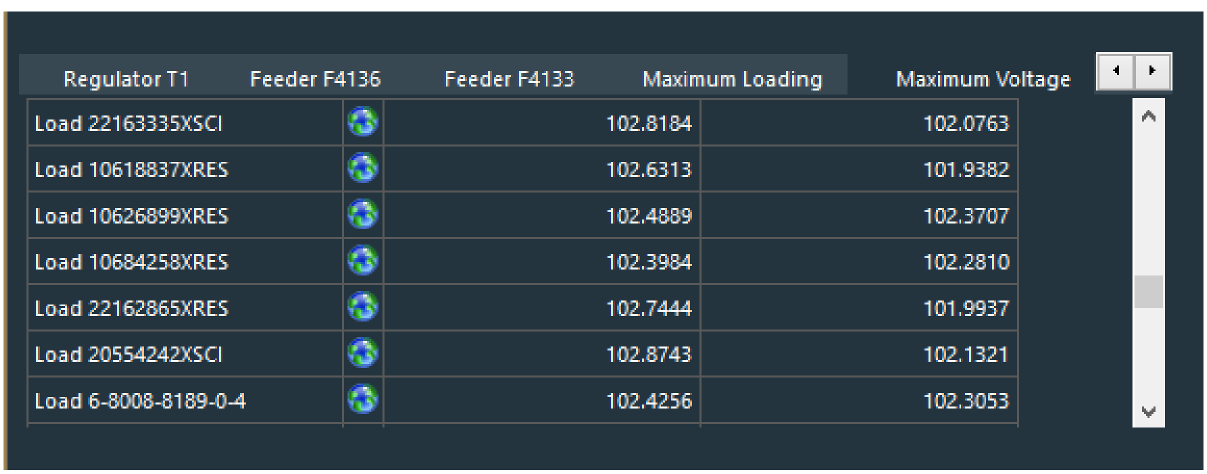
	Before	After
Current Phase A (Amps)	194.93	194.97
Current Phase B (Amps)	190.33	190.31
Current Phase C (Amps)	216.52	216.97
Real Power Phase A (kW)	1339.96	1338.56
Real Power Phase B (kW)	1317.06	1315.40
Real Power Phase C (kW)	1499.58	1499.53

- **Affected Devices:** Shows details for a user-configured number of devices that are affected, including the maximum branch loading levels, high voltages, and low voltages before and after the device operation. These results are displayed in three different tabs, "Maximum Loading", "Maximum Voltage", and "Minimum Voltage".



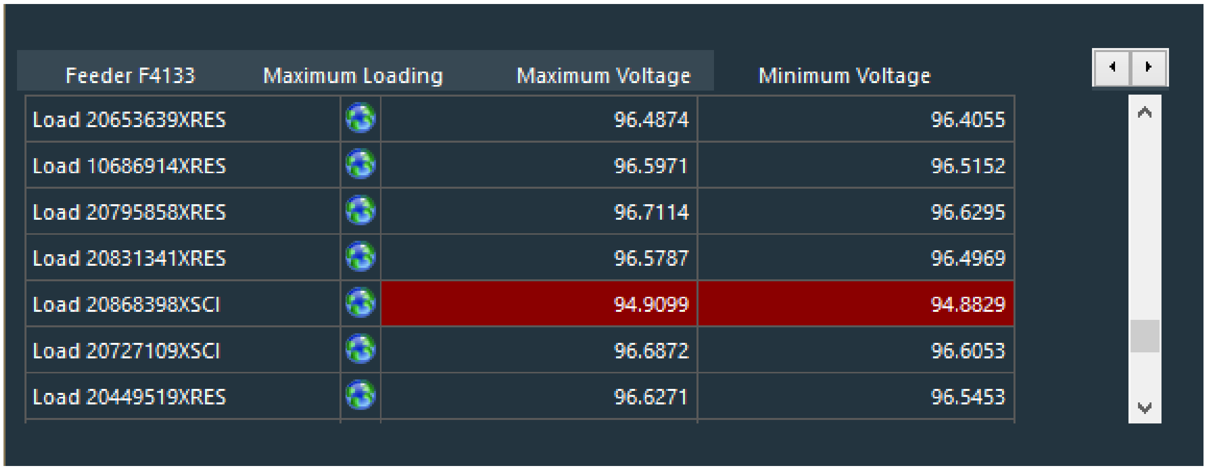
DPF Summary | Regulator T1 | Feeder F4136 | Feeder F4133 | Maximum Loading

DistributionTf 6-8108-7727-0-8		50.36	50.27
DistributionTf 6-8107-2679-0-2		47.65	47.61
DistributionTf 6-8107-3172-0-9		51.64	51.55
DistributionTf 6-8209-1400-0-2		66.56	66.50
DistributionTf 6-8107-2776-0-6		48.26	48.18
DistributionTf 6-8108-6342-0-8		53.00	52.88
DistributionTf 6-8108-4793-0-0		59.08	58.94



Regulator T1 | Feeder F4136 | Feeder F4133 | Maximum Loading | Maximum Voltage

Load 22163335XSCI		102.8184	102.0763
Load 10618837XRES		102.6313	101.9382
Load 10626899XRES		102.4889	102.3707
Load 10684258XRES		102.3984	102.2810
Load 22162865XRES		102.7444	101.9937
Load 20554242XSCI		102.8743	102.1321
Load 6-8008-8189-0-4		102.4256	102.3053



Feeder F4133	Maximum Loading	Maximum Voltage	Minimum Voltage
Load 20653639XRES		96.4874	96.4055
Load 10686914XRES		96.5971	96.5152
Load 20795858XRES		96.7114	96.6295
Load 20831341XRES		96.5787	96.4969
Load 20868398XSCI		94.9099	94.8829
Load 20727109XSCI		96.6872	96.6053
Load 20449519XRES		96.6271	96.5453

3.5. DPF Manual Inputs

Manual input parameters for DPF can be entered as described in Application Input Displays.

In addition, the BLA function takes into account static and dynamic schedules for feeder heads, generators, Distribution Energy Resources (DER), and loads.

3.6. DPF Configuration

The DPF configuration attributes are described in the "Distribution Power Flow" chapter of the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

Additional parameters specified in the application data files (`alg.dat`, `msgdef.dat`, `optalg.dat`, `phid.dat`, and `tftyp.dat`) for all of the DNAF are also described in the "Parameter Files" chapter of the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

4. Protection Validation

The Protection Validation (PRV) application simulates short-circuit scenarios for each PRV zone and calculates the fault currents. (A "PRV zone" is the portion of the distribution network that is bounded by breakers, reclosers, and ends of laterals and circuit mains.) In real-time mode, the Protection Validation application identifies unprotected or improperly protected PRV zones. The PRV application compares the fault currents with protection relay settings and the interruption capacity of breakers and reclosers.

The PRV application uses the same analysis engine as DPF, but it includes faults in the model to simulate the placement of faults throughout the network.

Three additional PRV options are available only in study mode. These options can be selected using the DNAF Configuration Editor.

- **Check Backup Protection for a PRV Zone (Protection Presence Validation)**

  For protection presence validation, the PRV zone is the portion of the distribution network bounded by breakers, reclosers, ends of laterals and circuit mains, and optionally fuses (fuses are never considered in real-time mode).

  Protection presence validation starts by checking if the PRV zone is properly protected by the protective device at the head of the zone. If the device does not protect the PRV zone, the function checks each protective device it encounters during an upstream trace (toward the feeder breaker) until it determines if there is a protective device with a minimum trip point or rating that operates for a fault at the point being considered. If no protective device is found to meet these conditions, up to and including the feeder breaker, the PRV zone being evaluated is considered to be unprotected and is flagged in the results.

- **Check Fuses**

  Use this option to include fuses in the protection validation study. Fuses are never considered during a real-time analysis.

- **Check Upstream Coordination Between Adjacent Devices (Protection Coordination)**

  The protection coordination option starts from the farthest downstream protection zone. Each protection zone supply side protection device minimum trip point or rating is compared to the next upstream (toward the feeder breaker) protection zone supply side protection device minimum trip point or rating. If the downstream supply side protection device minimum trip point or rating is not less than the upstream protection zone supply-side protection device minimum trip point or rating by a user-defined coordination factor, the two protective devices are considered to be uncoordinated.

  When a downstream protection zone supply side protection device is a fuse and the upstream protection zone supply side protection device is a device with an adjustable trip point (recloser, breaker, etc.), the fuse rating must be less than or equal to the rating contained within the user-supplied "Largest Fuse That Will Coordinate" value. If the fuse does not have a rating less than or equal to the rating contained in the user-supplied "Largest Fuse That Will Coordinate" value, the two protection devices are considered to be uncoordinated and are flagged in the results.

The following image represents an example of a feeder with a breaker, a three-phase recloser, a two-phase recloser, a single-phase recloser with no ground trip enabled, and three fuses. The data for the breakers and reclosers includes the phase trip current, the ground trip current, and the largest fuse that will coordinate. The PRV zones are shown as dotted red lines.

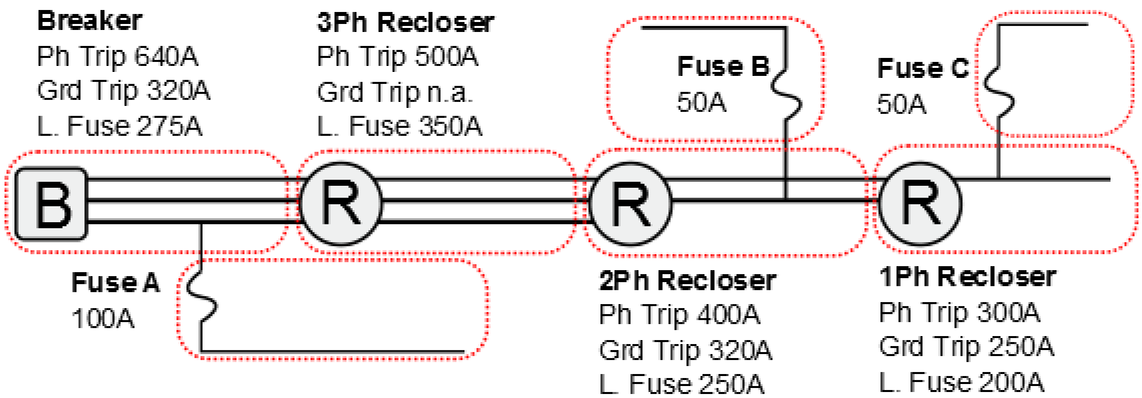


Figure 30. PRV Zones for Example of Upstream Protection Coordination

The calculations in the table below are based on the sample feeder and use a coordination factor of 0.8.

Device	Upstream Device	Comparison Type	Comparison	Result
Fuse C	1Ph Recloser	Rating / L Fuse	2 * 50 < 200	Ok
Fuse B	2Ph Recloser	Rating / L Fuse	2 * 50 < 250	Ok
Fuse A	Breaker	Rating / L Fuse	2 * 100 < 275	Ok
1Ph Rec	2Ph Recloser	Gr Trip / Gr Trip	250 < 320 * 0.8	Ok
2Ph Rec	3Ph Recloser	Gr Trip / Ph Trip	320 < 500 * 0.8	Ok
3Ph Rec	Breaker	Ph Trip / Gr Trip	500 < 320 * 0.8	Fail

4.1. Running Protection Validation in Real-Time Mode

In real-time mode, the PRV application can be configured to solve for event triggers, such as a switch status change or the insertion/removal of a temporary modification, including cuts and jumpers. If event triggers are enabled, the PRV application solves the impacted station and the connected surrounding portions of the network.

In addition to event triggers, a periodic trigger can be configured. If a solution has not been performed within the maximum periodic time interval, then a forced solution is executed for the station. The maximum PRV periodic time interval is configured in the DNAF server configuration file. For more information, refer to the "NetServerDnafConfig.txt" section in the *ADMS Software Installation and Maintenance Guide*.

A PRV solution can be manually requested at any time using the DNAF display, as described in

Executing DNAF Applications Manually. The PRV function is also executed when running reconfiguration applications (FISR, AFR, PLOS, and FISRCA) to analyze fault currents after switching plans.

4.2. Running Protection Validation in Study Mode

After study mode has been initialized and the desired changes have been made, the PRV solution can only be triggered manually, as described in Executing DNAF Applications Manually.

4.3. Protection Validation Results

The protection validation results can be viewed on the following tabular displays:

- **Protection Validation:** On this display, protection validation results are shown for all breakers and reclosers within a substation.

  This display can be accessed from the Station Output Summary display, from the substation toolbar, and from the DNAF Solution display.

- **Protection Validation Results:** This display provides protection validation results for switching plans recommended by the advanced switching reconfiguration applications.

  This display can be accessed from the FISR Plan, FISRCA Plan, PLOS Plan, and AFR Plan displays by clicking the Protection Validation Results button.

- **Protection Validation - Maximum Interruption Results:** This display provides information about the maximum interruption current, maximum fault current, maximum capacity ratio, and maximum interruption violation for the device. The results are obtained by simulating a three-phase fault on the load side of the breakers and reclosers involved in the reconfiguration plan.

  This display can be accessed from the Protection Validation Results display.

- **Protection Validation - Minimum Phase Pick-Up Results:** This display provides information about the minimum phase pick-up current, minimum phase fault current, and minimum phase pick-up ratio. The results are obtained by simulating a phase-to-ground fault at the highest impedance distance from the protection device.

  This display can be accessed from the Protection Validation Results display.

- **Protection Validation - Minimum Ground Pick-Up Results:** This display provides information about minimum ground pick-up current, minimum ground fault current, and minimum ground pick-up ratio. The results are obtained by simulating a phase-to-ground fault at the highest impedance distance from the protection device.

  This display can be accessed from the Protection Validation Results display.

- **Protection Validation and Coordination:** This display provides information about the minimum phase pick-up current, minimum phase fault current, minimum ground pick-up current, and minimum ground fault current. This is a study mode-only display.

This display can be accessed from the Protection Validation display.

For the base PRV application, two sets of data are shown on the tabular displays and output pop-ups: maximum fault current data and minimum fault current data.

- **Maximum Fault Current Results**

  The maximum fault current results are based on simulating faults placed on the load side of the breakers and reclosers. The PRV calculated fault current is compared to the maximum interruption current provided in the model; if the fault current is larger than the maximum interruption current, a violation has occurred. The maximum capacity ratio (calculated fault current divided by the maximum interruption current) is also calculated ("good" values are less than 100%). The calculated fault current, maximum capacity ratio, and violation results are shown in the tabular display and in the results displays for breakers and reclosers.

- **Minimum Fault Current Results**

  The minimum fault current results are based on simulating phase-to-phase-to-ground and phase-to-ground faults at the highest impedance distance from the head of the protection zone. The PRV calculated fault current is compared to the minimum pick-up current provided in the data model (multiplied by a phase or ground trip safety factor); if the fault current is less than the minimum pick-up current, a violation has occurred. The pick-up ratio (pick-up current divided by the calculated fault current) is also calculated ("good" values are less than 100%). The calculated fault current, pick-up ratio, and violation results are shown in the tabular display and in the results displays for breakers and reclosers.

Additional output data is available for study mode-only protection validation presence and protection coordination options. If fuses are included for analysis in these options, more PRV zones are created in the analysis.

- **Protection Presence Results**

  The protection presence results include the protection validation status (protected or not protected), the phase and ground fault protective device, the model-specified minimum pick-up current, and the calculated fault current. These results are shown in a tabular display and in the device pop-ups for breakers and reclosers.

- **Protection Coordination Results**

  The protection coordination results indicate if a coordination error has occurred (coordinated or not coordinated, based on device minimum trip point or rating and the largest fuse that coordinates values), and also the upstream device that can provide the proper protection. These results are shown in a tabular display and in the device pop-ups for breakers, reclosers, and fuses (fuses are shown only in study mode and if the Check Fuses option is enabled).

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

Individual device results displays are available for fuses, breakers, and reclosers via output displays.

A PRV Results output file is generated whenever PRV is triggered. This file provides a solution summary, application configurations, and protection validation results.

RAVEN - Protection Validation Results (Execution Time: 6/...

Energizing Station	Energizing Feeder	Fault Impedance Evaluation	Fault Impedance-Bolted Fault (R+jX ohm)	Fault Impedance-User Configured	Device Name	Device Type	Locate	Device Status	Operation Mode
RAVEN	--	Bolted Fault...	0.00 + j0.00	--	8810	Breaker		Closed	--
RAVEN	--	Bolted Fault...	0.00 + j0.00	--	8811	Breaker		Closed	--
RAVEN	F8861	Bolted Fault...	0.00 + j0.00	--	8861F	Breaker		Closed	--
RAVEN	F8862	Bolted Fault...	0.00 + j0.00	--	8862F	Breaker		Closed	--
RAVEN	F8863	Bolted Fault...	0.00 + j0.00	--	8863F	Breaker		Closed	--
RAVEN	F8864	Bolted Fault...	0.00 + j0.00	--	8864F	Breaker		Closed	--
RAVEN	F8865	Bolted Fault...	0.00 + j0.00	--	8865F	Breaker		Closed	--

Figure 31. Station PRV Results Display

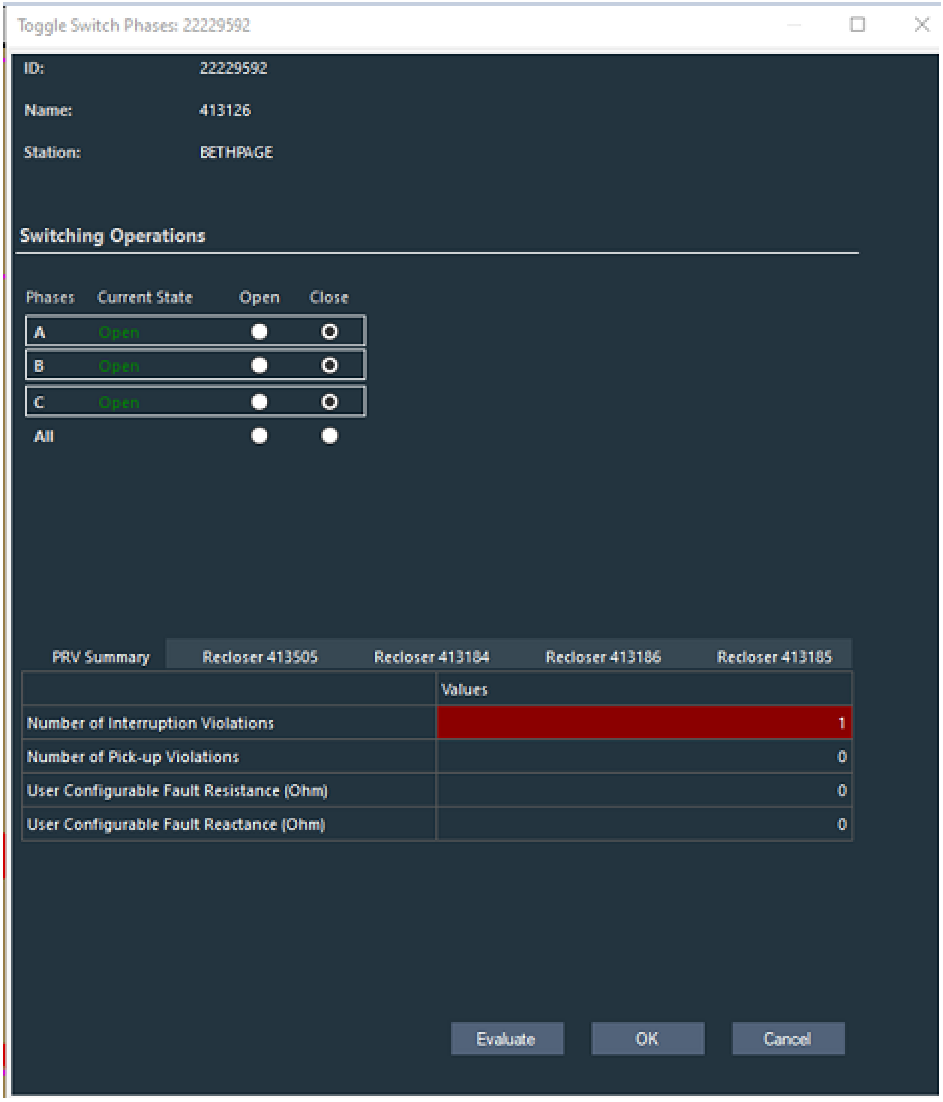
4.4. PRV Check Before Operate

The Check Before Operate (CBO) functionality issues notifications related to potential protection settings violations when the operator attempts to issue a switching action.

If the Enable Protection Validation option is enabled in the Viewer Configuration Editor, then the ADMS client sends a request to the ADMS server to solve the PRV application for the network configuration resulting from the proposed switching action. The ADMS server solves PRV and notifies the operator if a relay setting violation could occur because of the switching action. The CBO solution also reports the following details:

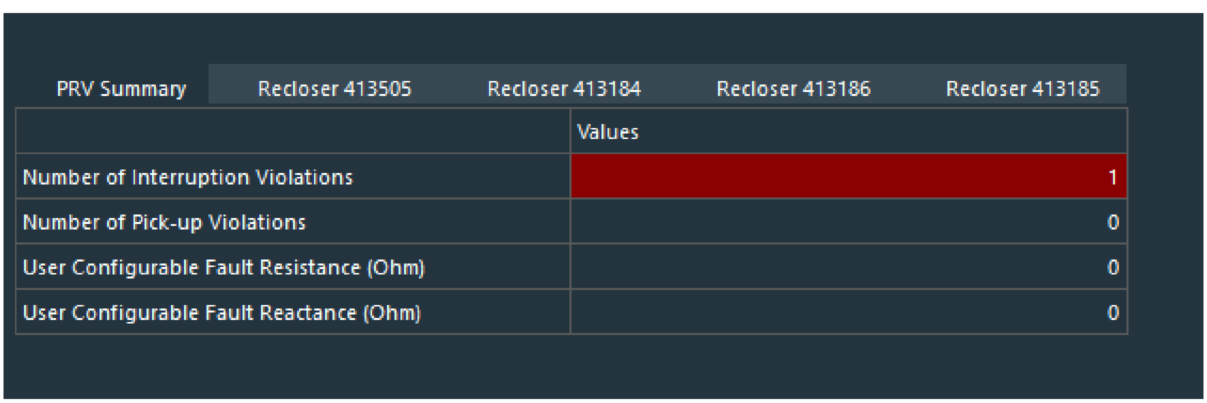
- All protective devices impacted due to the switching action.
- For each impacted protective device, whether the following violations were detected:
  - Maximum interruption violation and associated fault current.
  - Minimum phase pick-up violation and associated fault current.
  - Minimum ground pick-up violation and associated fault current.

The PRV analysis of the post-operation conditions is supported for switching operations, tap changing transformer tap position changes, and capacitor status changes. To run this check, click Evaluate on the device state change dialog.



The PRV CBO Results are organized into multiple tabs displayed under Notifications:

- **PRV Summary:** The number of interruption violations, number of pick-up violations, user-configurable fault resistance, and user-configurable fault reactance are displayed. Violations are displayed as red in color.



- **Device tab:** A separate tab is displayed for each of the protective devices affected because of the switching action. The values include maximum fault currents and capacity ratios, minimum phase fault currents along with their associated ratios and fault sensitivities, and minimum ground fault currents along with their associated ratios and fault sensitivities for each protective device.

Breaker 4137F | Recloser 413505 | Recloser 413184 | Recloser 53222 | Recloser 5172

	Values
Maximum Interruption Current (Amps)	6000.00
Maximum Fault Current with Bolted Fault (Amps)	2222.30
Maximum Fault Current with Configured Fault Impedance (Amps)	0.00
Maximum Capacity Ratio with Bolted Fault (%)	37.04
Maximum Capacity Ratio with Configured Fault Impedance (%)	0.00
Minimum Phase Pick-up Current with Safety Factor (Amps)	1725.73

Breaker 4137F | Recloser 413505 | Recloser 413184 | Recloser 53222 | Recloser 5172

Minimum Phase Pick-up Current with Bolted Fault (Amps)	1725.73
Minimum Phase Fault Current with Configured Fault Impedance (Amps)	0.00
Minimum Phase Pick-up Ratio with Bolted Fault (%)	46.94
Minimum Phase Pick-up Ratio with Configured Fault Impedance (%)	0.00
Phase Fault Sensitivity with Bolted Fault (%)	213.05
Phase Fault Sensitivity with Configured Fault Impedance (%)	0.00
Minimum Ground Pick-up Current with Safety Factor (Amps)	450.00

Breaker 4137F | Recloser 413505 | Recloser 413184 | Recloser 53222 | Recloser 5172

Minimum Ground Pick-up Current with Safety Factor (Amps)	450.00
Minimum Ground Fault Current with Bolted Fault (Amps)	1427.62
Minimum Ground Fault Current with Configured Fault Impedance (Amps)	0.00
Minimum Ground Pick-up Ratio with Bolted Fault (%)	0.00
Minimum Ground Pick-up Ratio with Configured Fault Impedance (%)	31.52
Ground Fault Sensitivity with Bolted Fault (%)	317.25
Ground Fault Sensitivity with Configured Fault Impedance (%)	0.00

4.5. Protection Validation Manual Inputs

Manual input parameters for PRV can be entered as described in Application Input Displays.

4.6. Protection Validation Configuration

The PRV configuration attributes are described in the "Protection Validation" chapter in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

5. Short Circuit Calculation

The Short Circuit Calculation (SCC) application simulates short-circuit conditions for a selected distribution location and calculates the resulting fault currents. A distribution location is a primary network switching device (upstream of distribution transformers), transformer (power, regulator, step, zig-zag, and auto), or line.

The SCC application uses the same analysis engine as the DPF, but it includes faults in the model to simulate the placement of faults in the network.

5.1. Running Short Circuit Calculation

The FL application can be executed in real-time mode or study mode. The SCC solution can only be triggered manually, as described in Executing DNAF Applications Manually.

5.2. Short Circuit Calculation Results

The Short Circuit Calculation results can be viewed on the SCC Summary tabular display, which can be accessed from the Station Output Summary display and from the DNAF Solution display.

For the SCC application, fault currents are shown on the SCC Summary tabular display and output displays, including station name, device ID, node ID, fault type, and fault currents (Phase A, Phase B, and Phase C) based on the simulated fault types.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

An SCC Results output file is generated whenever SCC is triggered. This file provides a solution summary, application configurations, and short circuit calculation results.

RAVEN - Short Circuit Calculation (Execution Time: 9/4/2019 4:09:18 AM Solution Time: 9/4/2019

Station	Device ID	Node ID	Type of Fault	Fault Impedance	Phase A Fault Current (Amps)	Phase B Fault Current (Amps)	Phase C Fault Current (Amps)	Phase A Post Fault Voltage	Phase B Post Fault Voltage	Phase C Post Fault Voltage
RAVEN	33480...	33480...	AG	Bolted Fault	3655.7	0	0	0.00 (120.00V)	139.48 (120.00V)	148.69 (120.00...
RAVEN	33480...	33480...	BG	Bolted Fault	0	3734.7	0	147.38 (120.0...	0.00 (120.00V)	139.01 (120.00...
RAVEN	33480...	33480...	CG	Bolted Fault	0	0	3768.2	138.48 (120.0...	148.95 (120.00V)	0.00 (120.00V)
RAVEN	33480...	33480...	ABG	Bolted Fault	4981.9	4733	0	0.00 (120.00V)	0.00 (120.00V)	149.99 (120.00...
RAVEN	33480...	33480...	AB	Bolted Fault	4645.2	4645.2	0	59.28 (120.00V)	59.28 (120.00V)	121.65 (120.00...
RAVEN	33480...	33480...	BCG	Bolted Fault	0	5064.7	4777.3	149.43 (120.0...	0.00 (120.00V)	0.00 (120.00V)
RAVEN	33480...	33480...	BC	Bolted Fault	0	4691	4691	120.12 (120.0...	60.82 (120.00V)	60.82 (120.00V)
RAVEN	33480...	33480...	ACG	Bolted Fault	4684.9	0	5103.3	0.00 (120.00V)	150.20 (120.00V)	0.00 (120.00V)
RAVEN	33480...	33480...	AC	Bolted Fault	4718	0	4718	61.14 (120.00V)	120.57 (120.00V)	61.14 (120.00V)
RAVEN	33480...	33480...	ABC	Bolted Fault	5374.2	5420	5434.1	1.58 (120.00V)	1.59 (120.00V)	1.58 (120.00V)
RAVEN	33480...	33480...	ABCG	Bolted Fault	5354.7	5409.5	5463.9	0.00 (120.00V)	0.00 (120.00V)	0.00 (120.00V)

Figure 32. Station SCC Results Display

5.3. Short Circuit Calculation Manual Inputs

Manual input parameters for SCC can be entered as described in Application Input Displays.

5.4. Short Circuit Calculation Configuration

The SCC configuration attributes are described in the "Short Circuit Calculation" chapter in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

6. Fault Location

The Fault Location (FL) application uses fault Amp measurements and faulted phase status measurements on a device to determine potential fault locations along distribution feeders.

The FL application generates a list of potential fault locations, which are summarized in a tabular display. Graphically, the FL results are shown by halos that are drawn over the line sections. These halos are called "fault location halos". There are three levels of fault location halos, where Level 1 are the most likely locations and Level 3 are the least likely locations.

Also, fault detector halos are drawn over line sections that are downstream from set fault detectors (fault detectors that have sensed fault current). The client Network Topology Processor (NTP) function provides this tracing capability. These halos are called "fault indicator halos". After the fault is cleared or isolated and NTP detects that downstream lines are re-energized, the fault indicator halos are reset. The reset time is defined by the NTP_FAULTIND_RESET_TIME parameter in the simulator server configuration file.

Fault indicator halos may be preserved after the fault indicator status has been reset if the HoldFaultIndicationCapable field is set to True for the upstream breaker object in the model. Fault indication halos are maintained on the user interface for a configured time period after the fault indication status is reset or until the server is restarted. The time periods for holding fault indicator halos are configured by the FAULT_INDICATION_HELD_NORMAL_DURATION and FAULT_INDICATION_HELD_MAXIMUM_DURATION parameters in the `NetServerMainConfig.txt` configuration file.

The portions of the distribution feeder where fault location halos and fault indicator halos overlap are the most likely fault locations.

The existing fault location results (data on the Fault Location tabular display and halos on the geographic display) are cleared every half hour. You can specify the minimum halo lifetime in the DNAF Configuration Editor to prevent clearing the halos too soon (the default time period is five minutes). New triggered execution of a fault analysis for the same feeder, for line segments downstream of the previously faulted segments, also clears previous fault location results. Manual execution of the FL application for a station also resets the results from the previous FL executions for station feeders.

6.1. Running Fault Location in Real-Time Mode

The FL application is an event-triggered application. By default, the application solves after a valid analog fault current measurement is received for the triggering device. In addition, FL may require Set fault indicator measurements for the same device to trigger (this option is defined by the DNAF_FL_REQUIRE_FIND_TRIGGER parameter in the DNAF server configuration file).

The FL application can be executed manually after setting the fault current analog values via the FL manual entry display.

6.2. Running Fault Location in Study Mode

In study mode, the user can execute the FL application manually after setting the fault current analog and fault indicator status values through the FL manual entry display. As long as the FL application is executed manually, FL results and halos from previous FL executions are deleted and reset.

6.3. Fault Location Results

The FL application results can be shown on tabular displays and the geographic user interface. The following FL tabular displays are available:

- **Fault Location Display:** This display shows the triggered devices that have a current FL solution. This display also indicates the number of potential fault locations and provides navigation to the Potential Fault Locations display.
- **Potential Fault Locations Display:** This display shows calculated fault currents (per phase) and the difference (in percent) between the calculated fault currents and the measured (or manually entered) fault currents at potential fault locations. You can also navigate to a potential fault location from this display.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

ROYAL TROON - Fault Locations (Fault Current in Amps: Maximum: 2072.8 A: 207 2054.

Exact Location	Band	Station Name	Branch Device Id	Phase A Calculated Fault Current (A)	Difference A (%)	Phase A Calculated Fault Voltage (pu)	Phase B Calculated Fault Current (A)	Difference B (%)	Phase B Calculated Fault Voltage (pu)
Yes	1	ROYAL TRO...	343480632	2076.80	-0.19	0.68	2068.70	-0.05	0.69
Yes	1	ROYAL TRO...	342885759	2072.10	0.04	0.69	2083.80	-0.77	0.69
Yes	1	ROYAL TRO...	343445113	2081.00	-0.40	0.68	2063.50	0.20	0.69
Yes	1	ROYAL TRO...	344101820	2066.80	0.29	0.69	2087.20	-0.93	0.69
Yes	1	ROYAL TRO...	343043919	2049.90	1.12	0.69	2107.40	-1.88	0.69
No	1	ROYAL TRO...	343467764	2074.40	-0.08	0.68	2065.10	0.13	0.69
No	1	ROYAL TRO...	342372141	2073.50	-0.03	0.68	2063.90	0.18	0.69
No	1	ROYAL TRO...	22224120	2073.50	-0.03	0.68	2063.90	0.18	0.69
No	1	ROYAL TRO...	189544241	2080.10	-0.35	0.68	2069.60	-0.09	0.69
No	1	ROYAL TRO...	22262420	2077.10	-0.21	0.68	2051.80	0.77	0.69
No	1	ROYAL TRO...	343478854	2087.50	-0.70	0.68	2075.10	-0.36	0.69

Figure 33. Station Potential Fault Locations Display

The FL results can be viewed graphically by enabling the fault location and fault indicator halos. In the above image, the fault location halo toolbar button and the halos are displayed. In this image, there are two levels of fault indicator halos (a maximum of three levels is available).

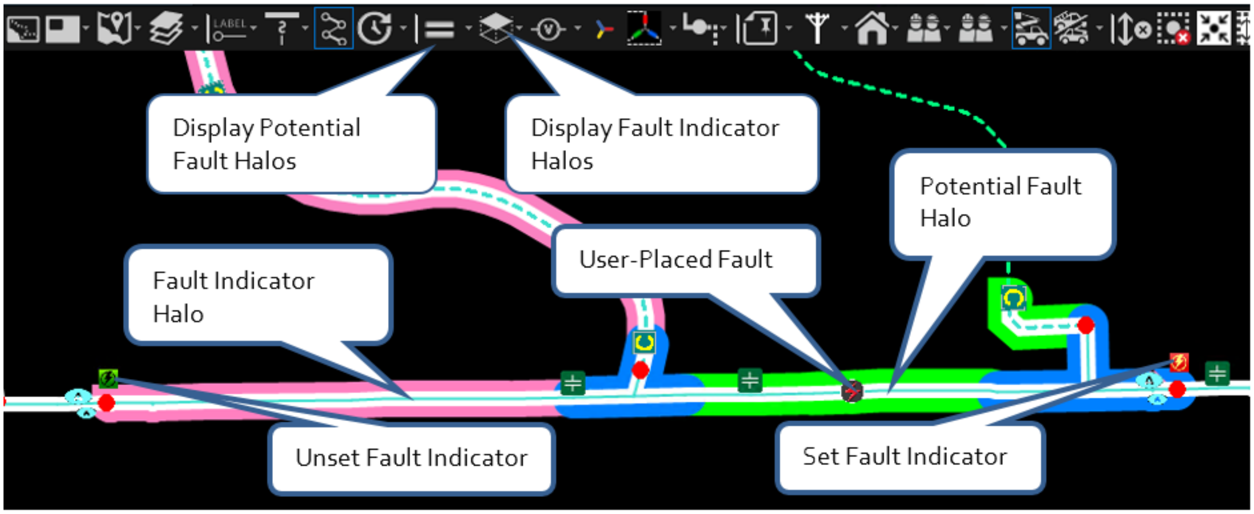


Figure 34. Display of Fault Location Halos and the Likely Position of the Fault

The fault indicator halo toolbar button and halos are also shown in the preceding image. This halo extends downstream from set fault indicators, and stops at unset fault indicators or network end points.

An FL Results output file is generated whenever FL is triggered. This file provides a solution summary, application configurations, equivalent system impedance values, and fault information.

6.4. FL Manual Inputs

Fault location data can be set on tabular displays and device data-entry displays as described in Application Input Displays.

In addition to changing power flow-related manual entries, the fault location data needs to be entered any time FL is manually executed, either in real-time mode or in study mode. These input values are entered through the Fault Location Input display:

- **Fault Location Input:** This display allows fault location fault current Amp values and faulted phases to be entered for each potential trigger device in the selected station.

6.5. FL Configuration

The FL configuration attributes are described in the "Fault Location" chapter in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

7. Load and Volt/VAR Management

Load and Volt/VAR Management (LVM) application generates a number of control actions to improve system operations. For example, the plans can be aimed to keep the nodal voltages inside operational limits, to change the operation point to satisfy power factor requirements at specific locations in the network, or to reduce total demand or losses in the network.

The LVM control actions include tap position adjustment of power transformers and regulators, on/off switching of shunt capacitors, and VAR output adjustment of available generators and DERs.

The following approaches to LVM are supported:

- **Optimization LVM:** This approach uses DPF results to determine the optimal series of control actions to meet a specified objective. An accurate network model and high-quality DPF results are required for accurate optimization LVM solutions.
- **Heuristic LVM:** This approach uses real-time measurements, the dynamic network connectivity, and the impedance model instead of DPF results to determine violations and control sensitivities, such as voltage and power mismatches. It generates one control action at a time (a capacitor switching move, a tap position adjustment, or a voltage setpoint adjustment) to achieve a specified objective.
- **Fixed Regulation LVM:** This approach is based on a specified voltage reduction percentage for all regulating transformers (LTCs and regulators) in the network selection. It can be used to obtain the desired voltage reduction in an emergency voltage reduction situation or in cases where optimization LVM would not give good results due to a low-quality power flow solution. The voltage reduction LVM solution is essentially topology ordering. Plan Manager issues controls to regulating transformers in a topology-based order.

For more details about LVM solution approaches, refer to the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

7.1. Running Load and Volt/VAR Management in Real-Time Mode

In real-time mode, the LVM application can be configured to solve for event triggers, such as a switch status change or plan implementation failure due to a device control failure. If event triggers are enabled, the LVM application solves the impacted station and the connected surrounding portions of the network.

In addition to event triggers, a periodic trigger can also be configured. If a solution has not been performed within the maximum periodic time interval, a forced solution is executed for the station.

Also, a deadband for periodic triggers is implemented to avoid excessive triggering. If LVM was last solved for a station within a configurable time frame, the LVM periodic trigger for that station is ignored. For more information, refer to the `NetServerDnafConfig.txt` section in the *ADMS Software Installation and Maintenance Guide*.

An LVM solution can be manually requested at any time, as described in Executing DNAF Applications Manually.

7.2. Running Load and Volt/VAR Management in Study Mode

After study mode has been initialized and the desired changes have been made, the LVM solution can only be triggered manually, as described in Executing DNAF Applications Manually.

7.3. Problem Formulations

For the advanced applications, a problem formulation (also called an "objective function") describes a pre-defined optimization goal. A problem formulation includes the objective or goal that is to be achieved, available controls that are allowed, the constraints including the device limits that are to be considered, and parameters required by the analysis engine.

By changing the options and scaling factors of the problem formulation through the DNAF Configuration Editor, different problem formulations can be defined. A problem formulation is defined by an analyst or engineer.

For LVM, the following problem formulations are currently defined. Additional problem formulations can be created using the DNAF Configuration Editor.

- **Minimize Demand**

  The Minimize Demand problem formulation minimizes the total demand (loads plus loss) in the distribution system. LVM attempts to reduce the load voltages, lowering the overall demand (the load-to-voltage dependency data has a great impact on the results). Based on the available controls and the existing voltage profile, it may not always be possible to reduce the total demand.

  This problem formulation applies only to the optimization LVM approach. This problem formulation is similar to the "Voltage Reduction (Heuristic)" objective using heuristic LVM.

- **Loss Minimization**

  The Loss Minimization problem formulation is used to reduce the total losses in the distribution system. These losses include line losses, reactor losses, and transformer losses including station transformers, auto-transformers, step-transformers, regulators, and distribution transformers. Based on the available controls and the existing voltage profile, it might not be possible to minimize losses.

  This problem formulation applies only to the optimization LVM approach.

- **Reactive Area Support**

  The Reactive Area Support problem formulation is used to maintain a target power factor on power transformers within a user-configured range. A power factor target range is always evaluated at the source or the load side of the station power transformers.

This problem formulation applies only to the optimization LVM approach. This problem formulation is similar to the "Power Factor Control (Heuristic)" objective using heuristic LVM.

- **Reduce Violations**

  The Reduce Violations problem formulation is the simplest LVM problem formulation. It checks for voltage violations in the base case. The goal of this problem formulation is to remove or at least reduce any existing flow or voltage violations. It may not be possible to eliminate all violations, depending on the current system state and the available control devices.

  Most other LVM problem formulations generally incorporate the goals of this problem formulation because it is a basis on which other problem formulations are built.

  This problem formulation applies only to the optimization LVM approach.

- **Voltage Reduction**

  This problem formulation can be used when the heuristic approach is selected for LVM. This problem formulation is similar to the "Minimize Demand" objective using optimization LVM.

- **Power Factor Control**

  This problem formulation can be used when the heuristic approach is selected for LVM. This problem formulation is similar to the "Reactive Area Support" objective using optimization LVM.

- **Voltage Reduction 1**

  The purpose of this problem formulation is to perform the level 1 emergency voltage reduction mode to lower the effective voltage band-center by 2.9%.

  When LVM solves in this mode, all eligible LTCs/regulators with a control measurement mapped for the associated voltage reduction mode are included in an LVM plan, where voltage reduction is set to a common value of 2.9%.

  This problem formulation applies only to the fixed regulation LVM approach.

- **Voltage Reduction 2**

  The purpose of this problem formulation is to perform the level 2 emergency voltage reduction mode to lower the effective voltage band-center by 5%.

  When LVM solves in this mode, all eligible LTCs/regulators with a control measurement mapped for the associated voltage reduction mode are included in an LVM plan, where voltage reduction is set to a common value of 5%. In addition, enabling this mode also sends a control to all capacitor banks to turn off their high/low voltage regulation (disable capacitor voltage rails) so that low voltages are permitted.

  This problem formulation applies only to the fixed regulation LVM approach.

- **No Optimization**

The purpose of this problem formulation is to disable voltage reduction, effectively disabling LVM for all stations, power transformers, or feeders using this problem formulation. The voltage reduction value is also reset to the value given in the Default Voltage Reduction (%) field, but voltage reduction is disabled. This problem formulation may be utilized when switching is being performed in the field.

When LVM solves in this mode, all eligible LTCs/regulators with a control measurement mapped for the associated voltage reduction mode are included in an LVM plan, where voltage reduction is set to a common default value of 2.9%.

This problem formulation applies only to the fixed regulation LVM approach.

7.4. LVM System Monitor

> **Note**
>
> For using the LVM System Monitor, the LVM System Monitor plug-in must be enabled and the Forecast Server (LVM Analysis Server) must be running.

The LVM System Monitor Overview display shows the current LVM summary and the evaluated conditions for a specific profile (a set of problem formulations) for the entire system, a station optimization group (area), or a station optimization division.

From this display, you can generate LVM solutions for profiles, implement solutions, and implement transitions between profiles.

> **Note**
>
> There is also a read-only version of the LVM System Monitor Overview display, where you can independently generate and study different LVM scenarios, without the ability to implement them.

Figure 35. LVM System Monitor - Overview Display

The LVM System Monitor display includes the following fields and controls:

- **Overview:**
  - **System Element:** Choose a system element from the menu (entire system, a station optimization group, or a station optimization division) for the LVM summary and evaluation.
  - **Implement:** Click this button to show the New Implementation display where you can start implementation for an evaluated solution.
  - **Forecast Profile:** Choose a problem formulation profile from the drop-down menu for the evaluation.

    > **Note**
    >
    > When another LVM System Monitor instance generates an evaluation for a system element, the evaluated profile is synchronized with the evaluated profile at all instances that have the same system element (or a sub-element) selected.

  - **Forecast Status:** Shows the solution evaluation status for the selected system element and the selected profile. Note that this area of the display shows a progress bar when evaluation is in progress.
  - **Forecast Age:** Shows the average and oldest evaluation ages of successfully evaluated plans. Evaluated solutions that do not have any plan steps do not impact this display.
- **Current Conditions:** Current system conditions.
  - **Power:** Shows the current total real power (MW), reactive power (MVAR), and power factor values as calculated by the DPF. Station power values are calculated at the bulk supply

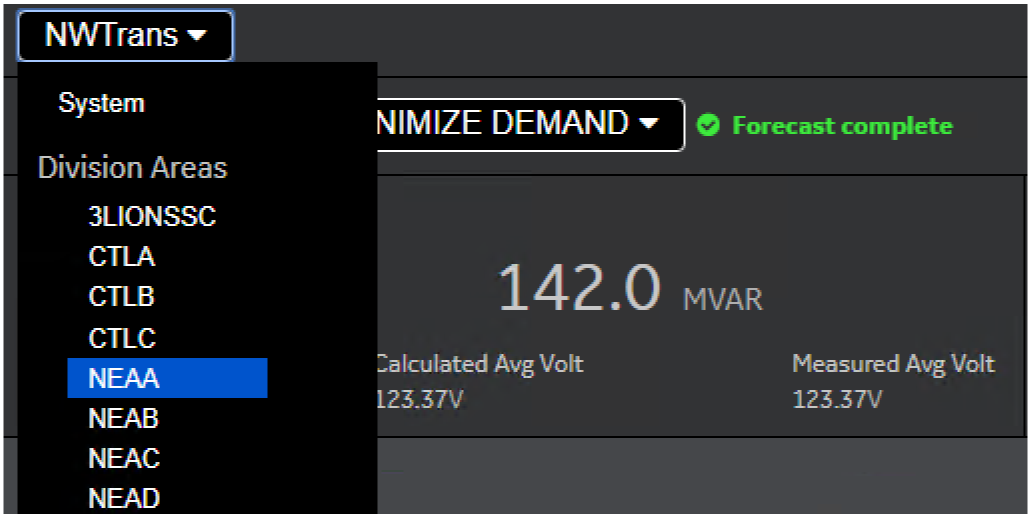
points (high side / primary voltage).

  - **Voltage:** Shows the current calculated and measured voltages (in Volts).
    - Average voltage is the average voltage at each power transformer and regulator weighted by kW power.
    - Measured average voltage is calculated at power transformers and regulators with valid power measurements.
- **Forecast Conditions:** Evaluated post-implementation system conditions.
  - **Power:** Shows the evaluated total real power (MW), reactive power (MVAR), and power factor values as calculated by the DPF using post-LVM system conditions. Station power values are calculated at the bulk supply points (high side / primary voltage).
  - **Voltage:** Shows the calculated average voltage, which is the average voltage at each power transformer and regulator weighted by kW power.
  - **Forecast Period:** Shows the evaluation period, in minutes. Station solution evaluations are updated under the latest system conditions on this schedule.
- **Demand Trend:** Shows the chart for the real and reactive power demand, real and reactive power rate of change, or average voltage for the specified time period (by default, 60 minutes). Note that limits for the rate of change chart are only shown if they are active for the currently selected system area and evaluated profile.
- **Station Problem Formulations:** Shows evaluated station problem formulations for the selected system element.
- **Problem Formulation Locks:** Shows problem formulations locks for stations, power transformers, and feeders that are associated with the selected system element.
- **LVM Devices:** Shows the LVM device eligibility summary for the selected system element. Excluded devices are those that appear on the exclusion list and are also ineligible. The ineligible device count includes those that are ineligible but not on the exclusion list. Eligible devices are those that can potentially be used for LVM plans.
- **Station Plan Summary:** Shows the current state of each station evaluation for the selected system element.

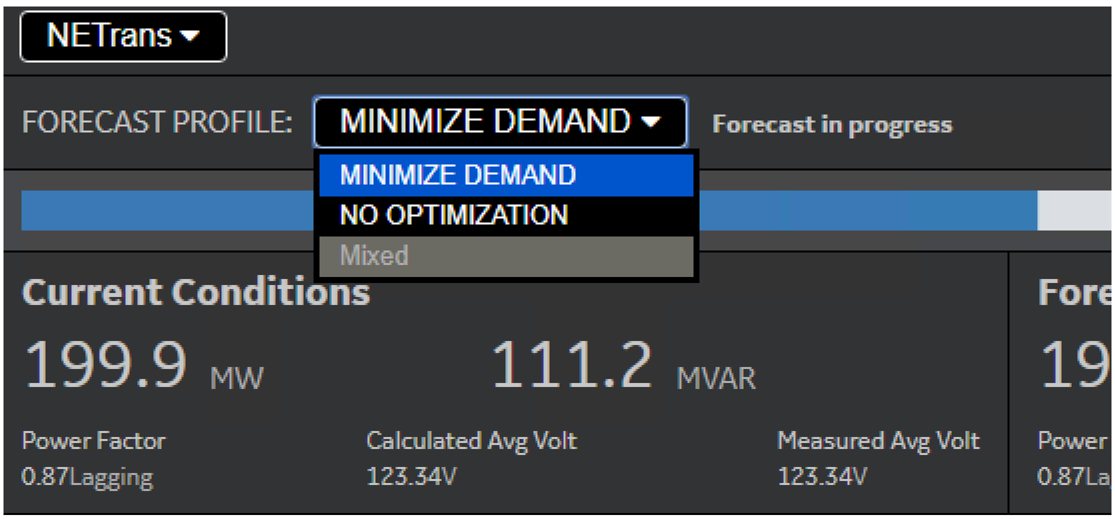
7.4.1. Generating an Evaluation

To generate an evaluation for a problem formulation profile:

1. Select the required system element from the menu at the upper-left corner of the display.



2. Select the required problem formulation profile from the Forecast Profile drop-down list.



The status field shows "Forecast in progress" and a progress bar is shown in that section of the display.

3. Wait until the Forecast Server (LVM Analysis Server) completes the evaluation for all stations.

   The status field shows "Forecast complete".

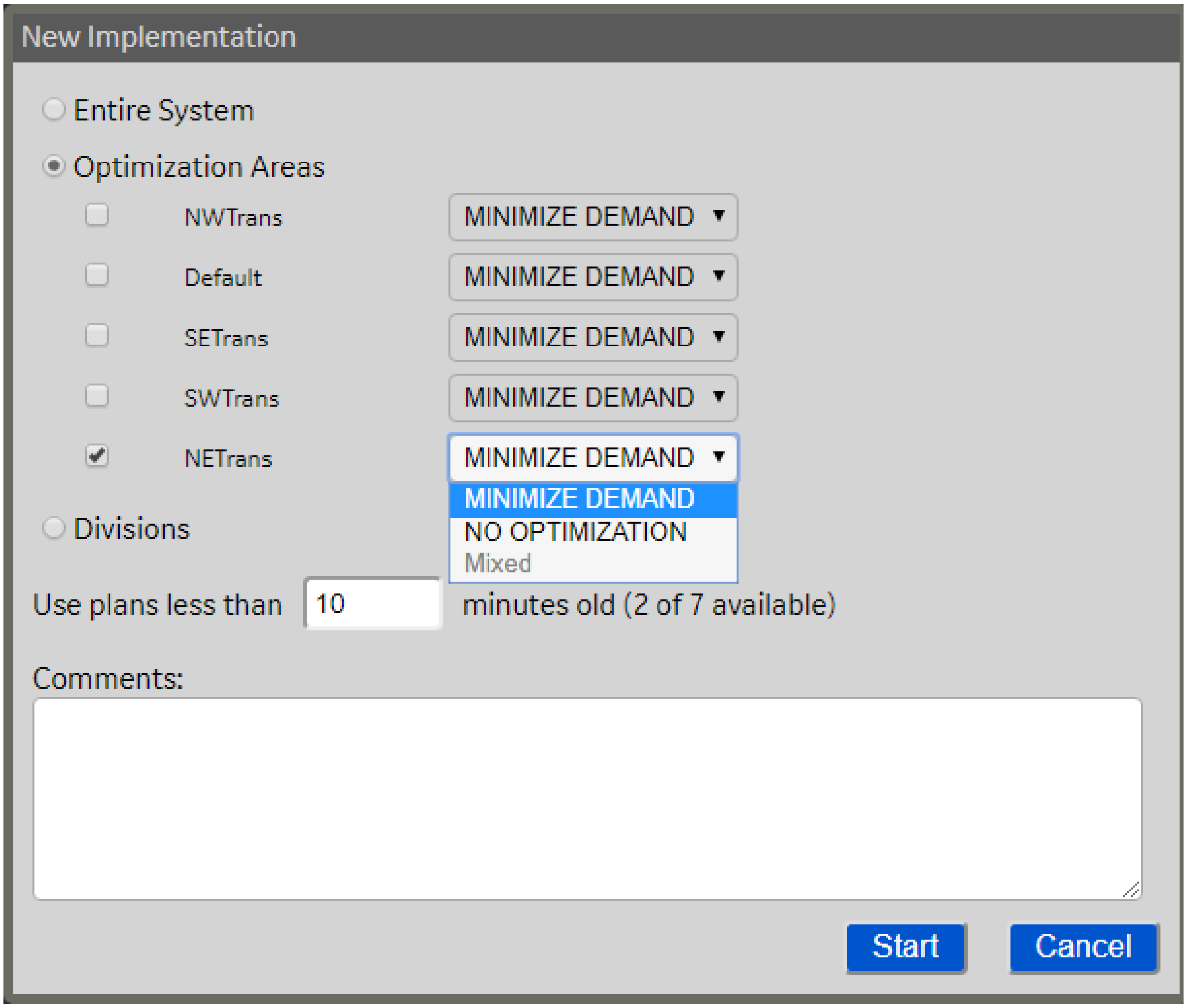
At any time, you can recalculate the evaluation for a specific system area using a different problem formulation profile.

7.4.2. Implementing an Evaluated Solution and Profile Transition

To start implementing an evaluated solution:

1. Click the Implement button at the upper-right corner of the display.

   The New Implementation pop-up display appears.



2. Select the required system elements and their target profiles.
3. Optionally, set the plan age. Plans that are older than the specified time (in minutes) are not used. Instead, they are reevaluated under the latest system conditions before being sent to the plan manager for implementation.
4. Click Start.

   The solution implementation and profile transition start. The station problem formulations (as seen in the DNAF System Configuration display) change to the problem formulations associated with the profiles set in the New Implementation dialog box. To see the implementation status, navigate to the Implementation Details display.

> **Note**
>
> The problem formulation changes will not be implemented for system elements with a locked problem formulation and their descendants (stations, power transformers, and feeders).

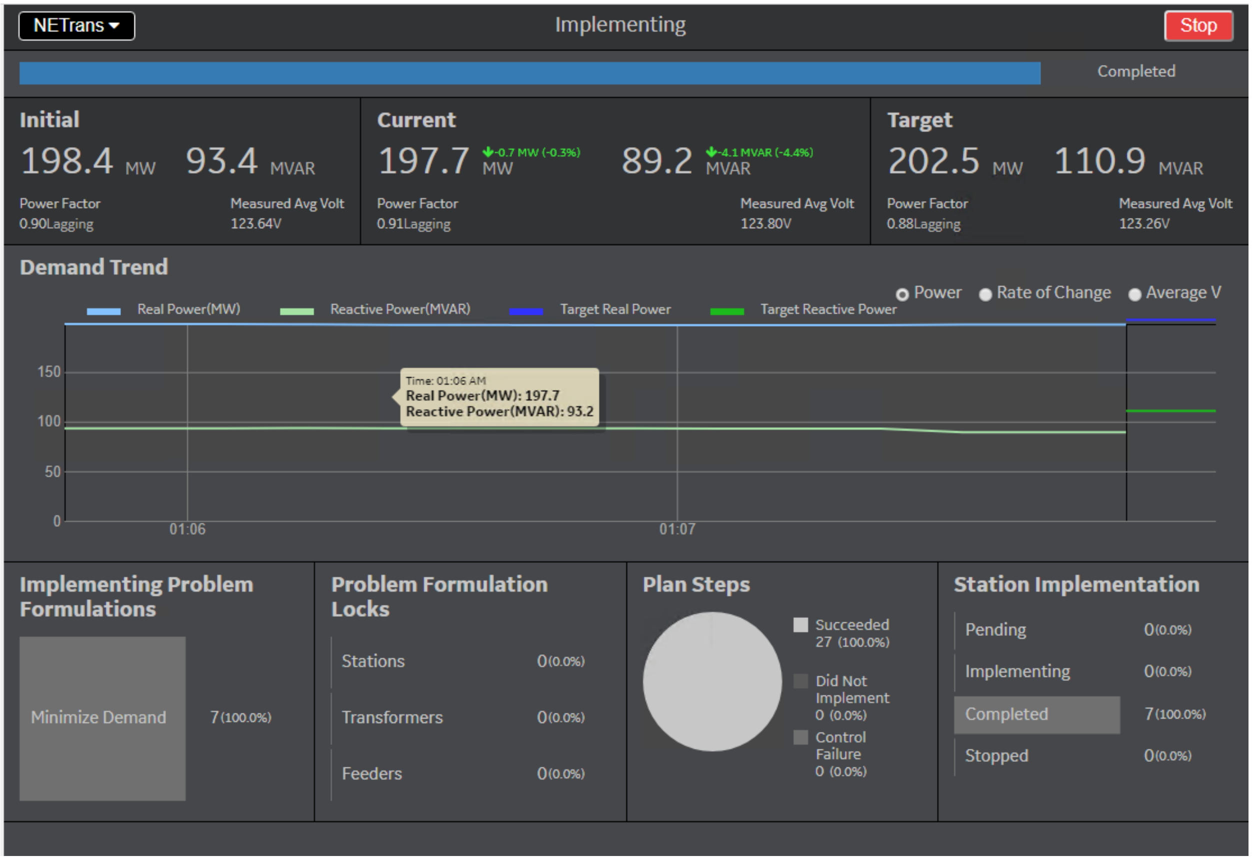


Figure 36. LVM System Monitor - Implementation Details Display

The plan implementation status and details are shown on the Implementation Details display.

You can click Stop to stop the current plan implementation, as appropriate.

Note

The Stop action operation only stops (prevents) implementation of plans where implementation has not yet started. If a plan is sent for implementation and at least one step has started, the plan will be completed to prevent partial implementation.

Plan implementation may be slower when:

- Throttling limits cause the system to delay plan implementation. In this case, the Stop action applies to stations where plans are queued, but not yet implemented.
- The implementation starts before all plans are computed. The Stop action prevents implementing plans that are not yet generated.

The Implementation Details display includes the following fields and controls:

- **Overview:**
  - **System Element:** Choose a system element from the menu (entire system, a station optimization group, or a station optimization division) for the LVM implementation.

- Implementation Status: Shows the solution implementation status for the selected system element. Note that a progress bar appears when implementation is in progress.
- **Stop:** Stops the current plan implementation for the selected system element. The station problem formulations are reset to the ones used before the implementation.

- **Initial Conditions**: System conditions at the beginning of the implementation.
  - **Power:** Total real power (MW), reactive power (MVAR), and power factor values are composed of measured values if available. Calculated values are applied if measurements are currently unavailable or invalid.
  - **Voltage:** Measured average voltage is calculated at power transformers and regulators with valid power measurements and is weighed by kW power.
- **Current Conditions:** System conditions during the implementation.
  - **Power:** Total real power (MW), reactive power (MVAR), and power factor values are composed of measured values if available. Calculated values are applied if measurements are currently unavailable or invalid.
  - **Voltage:** Measured average voltage is calculated at power transformers and regulators with valid power measurements and is weighed by kW power.
- **Target Conditions:** Evaluated post-implementation system conditions.
  - **Power:** Shows the evaluated total real power (MW), reactive power (MVAR), and power factor values as calculated by the DPF using post-LVM system conditions.
  - **Voltage:** Shows the evaluated calculated average voltage, which is the average voltage at each power transformer and regulator weighted by kW power.
- **Demand Trend:** Shows the chart for the real and reactive power demand, real and reactive power rate of change, or average voltage for the specified time period (by default, 60 minutes). Note that limits for the rate of change chart are only shown if they are active for the currently selected system area and evaluation profile.
- **Implementing Problem Formulations:** Shows the station problem formulations that are being implemented.
- **Problem Formulation Locks:** Shows problem formulations locks for stations, power transformers, and feeders that are associated with the selected system element.
- **Plan Steps:** Shows the plan step implementation summary.
- **Station Implementation:** Shows the station LVM solution implementation summary.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

7.4.3. Viewing Evaluation and Implementation Results

To see the LVM plan evaluation and implementation summary for stations, navigate to the LVM System Monitor Stations display.

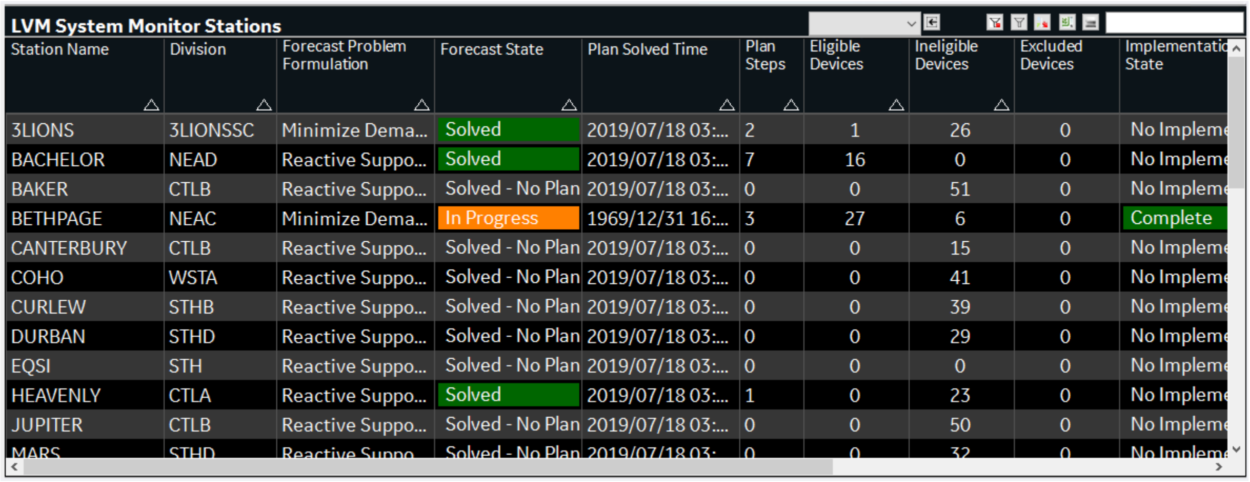


Figure 37. LVM System Monitor - LVM System Monitor Stations Display

7.4.4. Managing Profiles

Use the LVM System Monitor Profiles display to define profile-based problem formulations for system elements (stations, transformers, and feeders). These definitions can be used for evaluating and implementing LVM solutions and transitioning between problem formulations using the LVM System Monitor. For more information about LVM System Monitor profiles, refer to the *DNAF Advanced Applications Analyst Guide.*

On this display, you can also change active problem formulations for system elements based on the defined profiles.

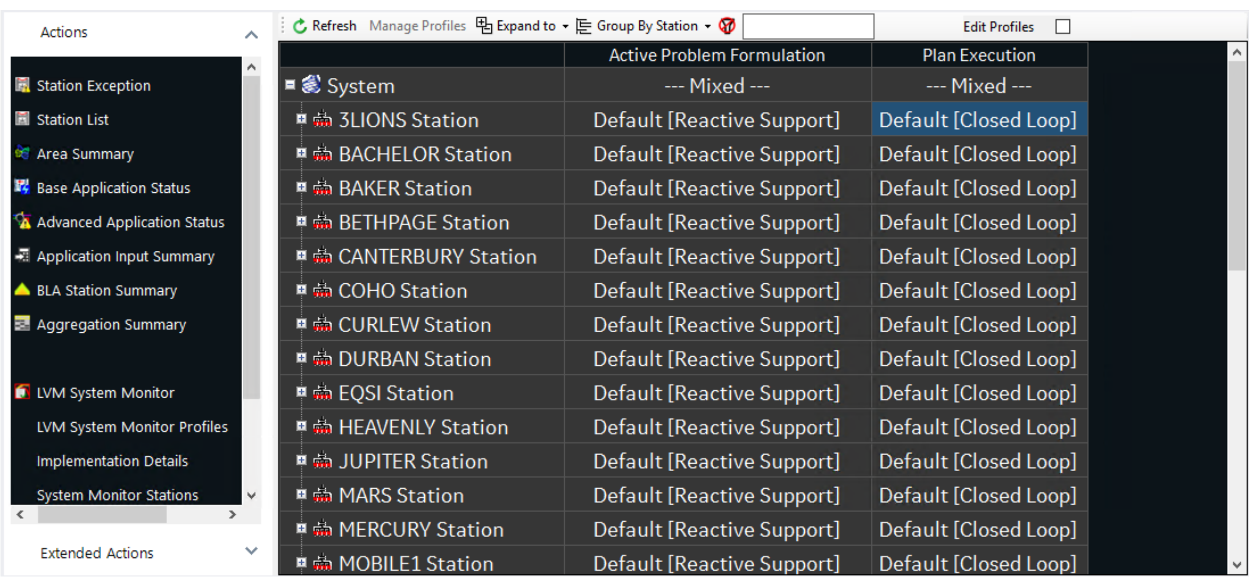
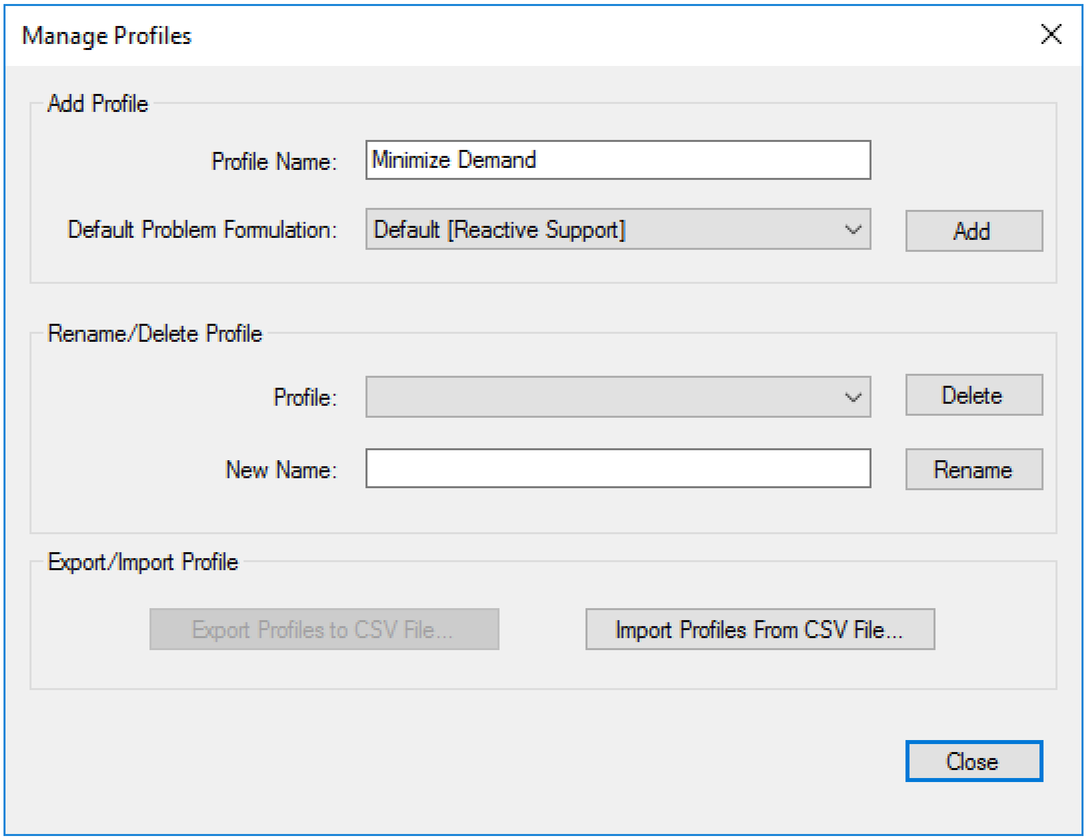


Figure 38. LVM System Monitor Profiles

7.4.4.1. Defining a Profile

1. Select the Edit Profiles option.
2. Click Manage Profiles.
3. Enter a profile name.
4. Select the default problem formulation.



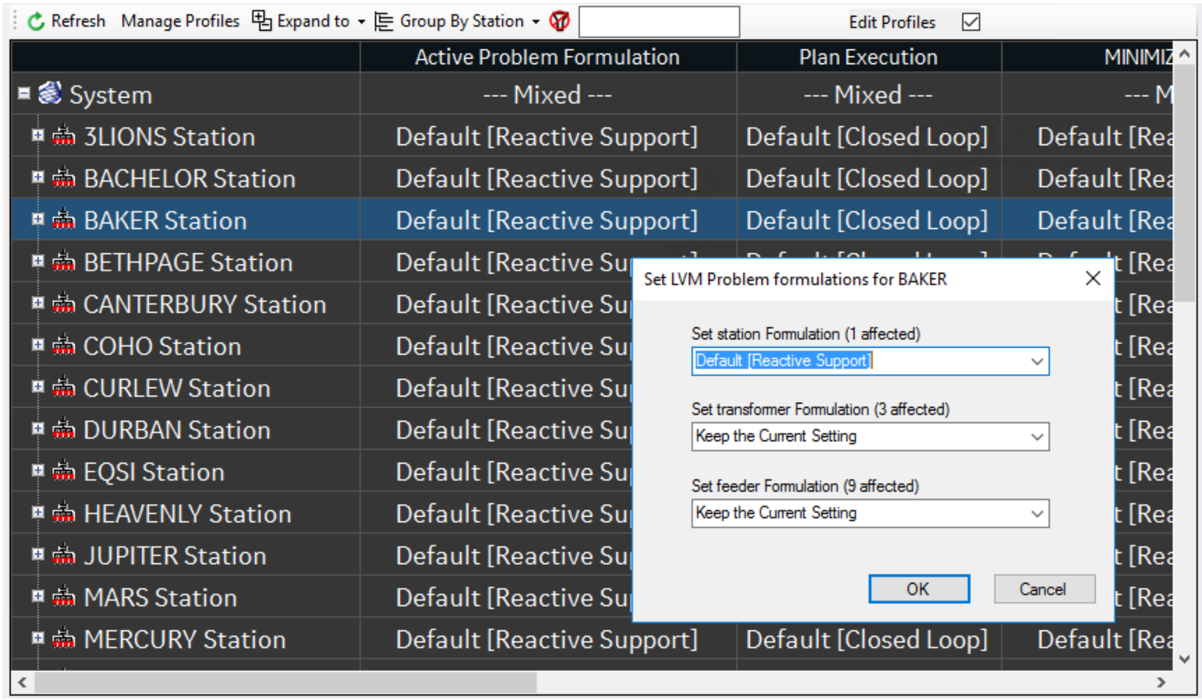
5. Click Add.
6. Add more profiles, as appropriate.
7. Click Close.

7.4.4.2. Modifying Problem Formulations for a Profile

1. Select the Edit Profiles option.
2. Click the profile's problem formulation field for a system element.

   The Set LVM Problem Formulations dialog box appears.

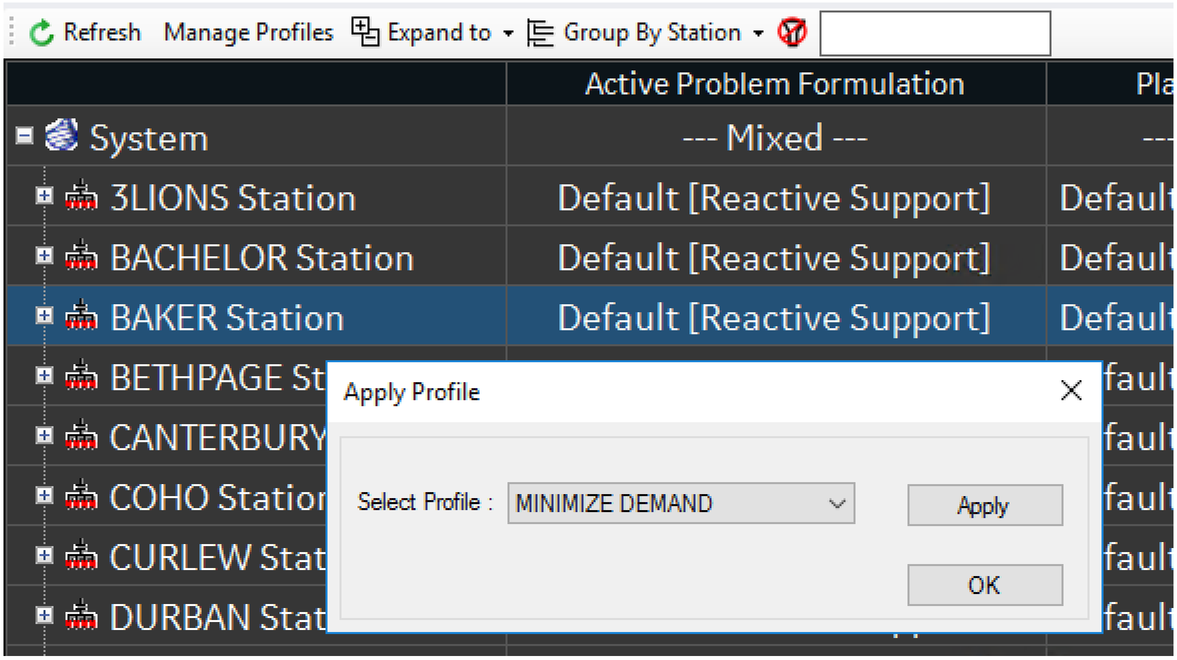
3. Set problem formulations for the system element and its sub-elements.



4. Click OK.
5. Modify problem formulations for other profiles, as appropriate.

7.4.4.3. Applying Profile Problem Formulations to System Elements

1. Select the Edit Profiles option.
2. Click the Active Problem Formulation field for a system element.
3. Select the required profile, and then click Apply.

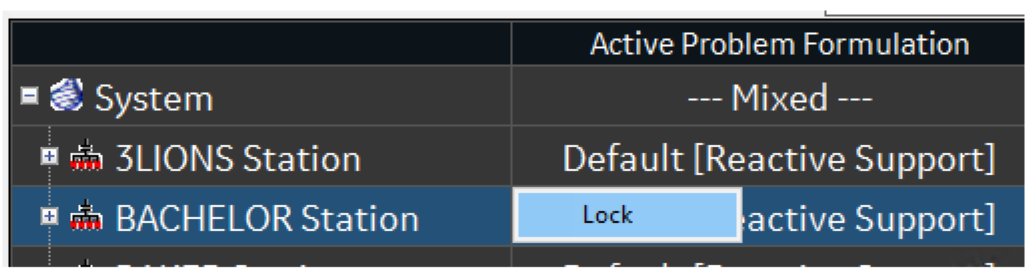


The selected profile's problem formulations are applied to the specified system element and its sub-elements accordingly. The problem formulations on the DNAF System Configuration display are updated accordingly.

7.4.4.4. Locking Profile Problem Formulations

To lock a profile's problem formulations for system elements:

1. Right-click the Active Problem Formulation field for a system element.



2. Click Lock.

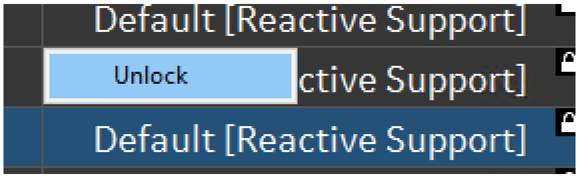
   The Lock Information dialog box appears.

3. Enter a lock comment.
4. Click OK.

   The problem formulation settings for the system element and its sub-elements are locked. Locked settings cannot be modified until unlocked. Locked settings are shown with a black lock icon .

To unlock a profile's problem formulations for system elements:

1. Right-click the locked problem formulation field.



2. Click Unlock.

7.4.4.5. Exporting and Importing Profile Data

To export profile data to a .CSV file:

1. Select the Edit Profiles option.
2. Click Manage Profiles.

   The Manage Profiles display appears.

3. Click Export.
4. Specify the file path and name, and then click Save.

   The profile data is saved to the specified file.

To import profile data from a .CSV file:

1. Click Manage Profiles.

   The Manage Profiles display appears.

2. Click Import.
3. Navigate to the file, and then click Open.

   The profile data is imported and loaded.

> **Warning**
>
> When you import profile data, the imported data overwrites your current profile data.

7.5. Load and Volt/VAR Management Results

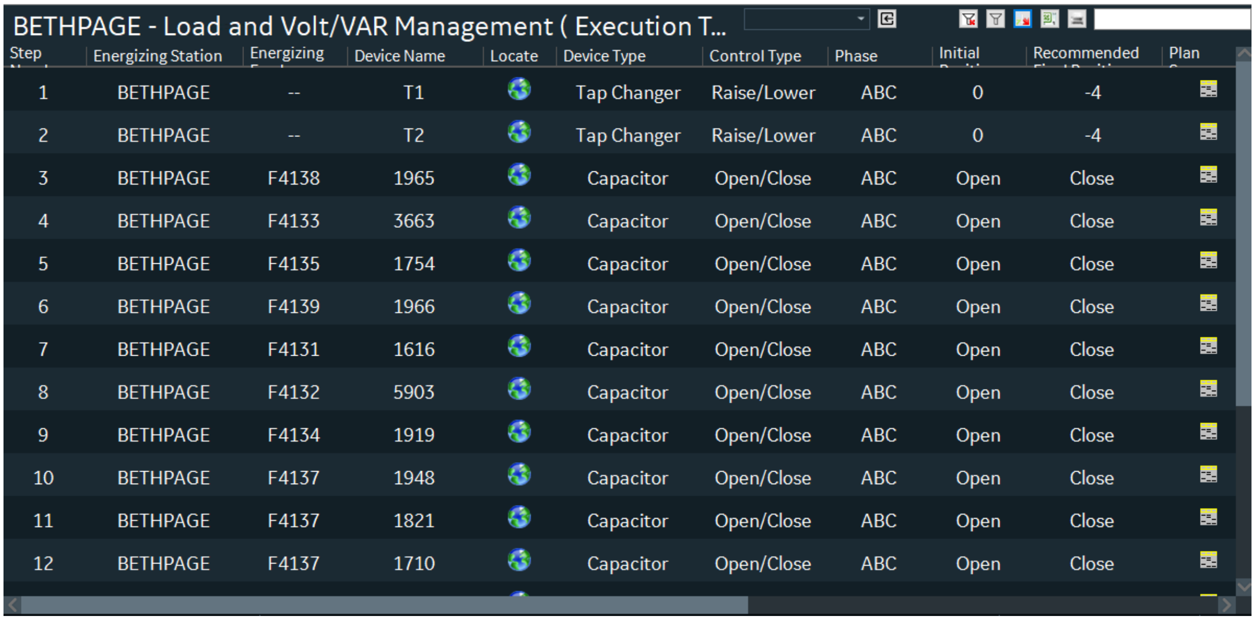
The LVM results can be viewed from the following tabular displays:

- **LVM Station Summary:** This display provides an overview of all the stations with respect to their LVM status. This display shows key parameters such as whether a station is LVM enabled, the active problem formulation, the problem formulation's type, and the time LVM was last solved with a valid plan. In addition, watchdog counters indicate the number of capacitor operations and the number of tap changer operations in the monitoring interval. Also, the number of consecutive infeasible, daily infeasible, daily discarded, and consecutively discarded LVM solutions are displayed.

This display can be accessed from the RT menu. In provides navigation to the LVM Plan Control Moves, the Capacitor Control Display, and the LTC/Regulator Control Display is also provided via the LVM Solution Details, Capacitor Summary, and LTC and Regulator Summary columns, respectively.

- **LVM Plan Control Moves:** This display lists the recommended control actions, including capacitor switching, regulator/LTC tap positions, and power factor and VAR output recommendations for generators and DERs. This display shows the initial and recommended steps and includes a Locate button. Navigation to other LVM displays is provided. In addition, the ability to implement and restore a plan is available in study mode.
- **LVM Plans Summary:** This display shows the currently selected problem formulation and the LVM station group. It also provides navigation to the LVM Plan Statistics and LVM Plan Weighted Performance Indexes displays.
- **LVM Plan Statistics:** This display shows key plan statistics in two horizontal panes, including the number of control moves, real and reactive power demand and demand reduction, minimum and maximum power factor, maximum segment loading, and low and high voltage values. It also provides navigation to the LVM Bus Voltages display and to the low and high voltage devices.
- **LVM Plan Weighted Performance Indexes:** This display shows LVM-related problem formulation performance indices in two horizontal panes.
- **LVM Bus Voltages:** This display provides a summary of all telemetered bus voltages in the station (measured and calculated values).
- **LVM Violations:** This display shows a summary of pre-plan and post-plan flow violations for the selected LVM plan.
- **LVM Control Failure Devices:** This display provides a summary of devices that failed to respond to LVM controls. It also provides the ability to remove the "LVM Control Failure" flag from the devices.
- **LVM System Monitor Stations:** This display shows the LVM evaluation and implementation summary for stations in the system. This display can be accessed from the DNAF Display plug-in.

An LVM Results output file is generated whenever LVM is triggered. This file contains information pertinent to the LVM run, including a solution summary, configuration parameters, measurement values, eligible devices, and plan details.


BETHPAGE - Load and Volt/VAR Management (Execution T...

Step	Energizing Station	Energizing	Device Name	Locate	Device Type	Control Type	Phase	Initial	Recommended	Plan
1	BETHPAGE	--	T1		Tap Changer	Raise/Lower	ABC	0	-4	
2	BETHPAGE	--	T2		Tap Changer	Raise/Lower	ABC	0	-4	
3	BETHPAGE	F4138	1965		Capacitor	Open/Close	ABC	Open	Close	
4	BETHPAGE	F4133	3663		Capacitor	Open/Close	ABC	Open	Close	
5	BETHPAGE	F4135	1754		Capacitor	Open/Close	ABC	Open	Close	
6	BETHPAGE	F4139	1966		Capacitor	Open/Close	ABC	Open	Close	
7	BETHPAGE	F4131	1616		Capacitor	Open/Close	ABC	Open	Close	
8	BETHPAGE	F4132	5903		Capacitor	Open/Close	ABC	Open	Close	
9	BETHPAGE	F4134	1919		Capacitor	Open/Close	ABC	Open	Close	
10	BETHPAGE	F4137	1948		Capacitor	Open/Close	ABC	Open	Close	
11	BETHPAGE	F4137	1821		Capacitor	Open/Close	ABC	Open	Close	
12	BETHPAGE	F4137	1710		Capacitor	Open/Close	ABC	Open	Close	

Figure 39. Station LVM Results Display

7.6. Load and Volt/VAR Management Manual Inputs

Device parameters and eligibility for reconfiguration applications can be set on tabular displays and device data-entry displays, as described in Application Input Displays.

In real time, the operator can override the key LVM parameters, such as LVM eligibility at various hierarchical levels (station or station power transformer or feeder), and power factor targets. This can be done from the Manual Input Parameter Entry display and the DNAF Real-Time System Configuration display from the RT DMS drop-down menu on the geographical display.

The following parameters for system/station/power transformer/feeder can be modified from the DNAF Real-Time System Configuration display:

- LVM enable/disable status
- LVM problem formulation
- LVM plan execution mode
- LVM alarms enable/disable status

The following parameters relevant to LVM can also be overridden from the Manual Input Parameter Entry display:

- Feeder minimum/maximum power factor target
- Bus voltage limits

7.7. Load and Volt/VAR Management Configuration

The LVM configuration parameters are described in the "Load and Volt/VAR Management Configuration" section of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

8. Fault Isolation and Service Restoration

The Fault Isolation and Service Restoration (FISR) application generates switching plans to isolate faulted sections and to restore service to non-faulted sections.

Similar to the other advanced applications, FISR uses a problem formulation that defines one or more operational objectives, such as minimizing unserved kW, minimizing the number of switching operations, or minimizing voltage drop along the reconfigured feeders. The problem formulation also specifies the network and operational constraints, including branch flow limits, load, and bus voltage limits, and relay settings.

When triggered, FISR places a fault based on the fault location results or downstream of the most downstream switching device with the Set fault indicator status.

The FISR application starts by determining which side of the fault to isolate and restore first. Priority is given to the side with an automatic recloser (if the Use Automatic Switching Devices option is enabled in the problem formulation in the DNAF Configuration Editor) and secondary consideration is given to the number of customers de-energized on each side of the fault. Once the priority is determined, the application creates a plan to isolate each side of the fault separately.

The isolation switching is done to avoid closing in on the fault. To restore loads upstream of the fault, FISR always closes the upstream protective devices that tripped after the isolation switching is done. More advanced logic is used for generating the restoration plans for loads that are downstream of the fault. Details about this logic can be found in chapter 4 Switching Reconfiguration Applications in the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

8.1. Running Fault Isolation and Service Restoration in Real-Time Mode

In real-time mode, FISR can be configured to solve for event triggers:

- Fault indicator status is updated from "Not Set" to "Set".
- Fault indicator status is updated from "Not Set" to "Set" and the Fault Location application is calculated. This option is configured by the FISR FL Event Trigger option on the FISR page in the DNAF System Configuration display.
- Loss of voltage event occurs (voltage drops to nearly zero volts).

These triggers can be enabled and disabled in the DNAF server configuration file. For more information, refer to the `NetServerDnafConfig.txt` section in the *ADMS Software Installation and Maintenance Guide*.

FISR can also be run manually after a fault is placed (either automatically or manually by the operator).

8.2. Running Fault Isolation and Service Restoration in Study Mode

To execute the FISR application in study mode, a fault object must first be inserted on a de-energized line, transformer, or bus on the geographic display. After the fault has been placed, FISR can be executed manually, as described in Executing DNAF Applications Manually. If no faults are placed, the application cannot be solved.

The "Only Solve for the Triggering Fault" option in the DNAF Configuration Editor specifies whether FISR only generates a plan that restores load for the triggering faults.

- If this option is selected, FISR solves only the triggering fault (which is inserted in the selected location), but other previously inserted faults in the network selection are respected.
- If this option is not selected, FISR solves the triggering fault and all other faults in the network selection.

8.3. Problem Formulations

For the advanced applications, a problem formulation (also called an "objective function") describes a pre-defined optimization goal. A problem formulation includes the objective or goal that is to be achieved, available controls that are allowed, the constraints including device limits that are to be considered, and parameters required by the analysis engine.

By changing the options, performance index weights, and other parameters of the problem formulation through the DNAF Configuration Editor, different problem formulations can be defined. A problem formulation is generally defined by an analyst or engineer.

For FISR, the following problem formulations are currently defined. Additional problem formulations can be created using the DNAF Configuration Editor.

- **Minimum Unserved kW (Auto + Manual)**

  The Minimum Unserved kW (Auto + Manual) problem formulation finds restoration plans that restore as much load (kW) as possible while respecting the user-specified limit sets. Both automated and manual switching devices can be used in the restoration plans.

- **Minimum Unserved kW (Auto Only)**

  The Minimum Unserved kW (Auto Only) problem formulation finds restoration plans that restore as much load (kW) as possible while respecting the user-specified limit sets. Only automated switching devices can be used in the restoration plans.

- **Minimum Customer Minutes Interrupted**

  The Minimum Customer Minutes Interrupted (CMI) problem formulation finds restoration plans that minimize the total number of customer minutes interrupted. The calculated CMI value is based on manual and automatic switch operation times, which can be configured in the DNAF

Configuration Editor.

- **Minimum Loss of Customer Value of Uninterrupted Service**

  The Minimum Loss of Customer Value of Uninterrupted Service problem formulation finds restoration plans that minimize the loss of customer value of uninterrupted service. The customer priorities are assigned based on the costs of uninterrupted service that are given for each load category.

- **Minimum Number of Switching Operations**

  The Minimum Number of Switching Operations problem formulation finds restoration plans that minimize the total number of switching operations.

- **Minimum Voltage Drop Along the Reconfigured Feeders**

  The Minimum Voltage Drop along the Reconfigured Feeders problem formulation finds restoration plans that minimize the voltage drop between the nodes with the highest voltage and the lowest voltage along the feeder.

8.4. Fault Isolation and Service Restoration Results

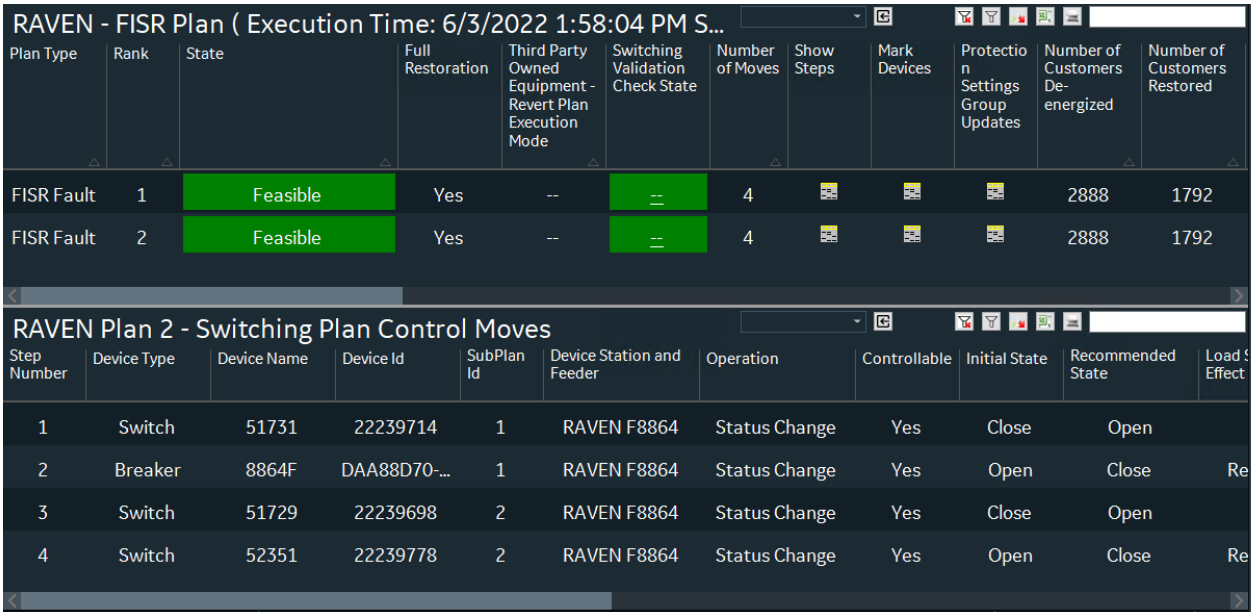
The FISR results are shown on tabular displays. These tabular results include an ordered list of recommended switching device actions.

The following tabular displays are available for the FISR application:

- **FISR Results Summary:** This display shows the number of plans and evaluation and execution times. It also provides navigation to other FISR-related displays that show the plans, statistics, and performance indices. For FISR triggered by permanent fault, the triggering device information (station name, device name, device id, device type and tripping time) is also displayed (first tripped device only).
- **FISR Plan:** This display shows the plan details, including the number of customers and distribution transformers not restored, maximum segment loading, problem formulation and low and high voltage locations. It also provides navigation to the FISR Protection Validation (PRV) results displays.
- **FISR Plan Details:** This display shows the individual plan details, including the performance indices.
- **FISR Switching Plan Control Moves:** This display shows the ordered list of recommended switching actions. From this display, each switching device can be marked and located on the user interface. Where modeled for the recloser, the Settings Group Template is displayed as a link to access a display named Settings Group Template that lists the Settings Groups in the template.
- **FISR Plan Weighted Performance Indexes:** This display provides plan performance characteristics based on performance indexes for number of moves, dropped customer count, implementation cost, voltage and flow violations, losses, etc.

- **FISR Plan Statistics:** This display shows key plan statistics in two horizontal panes, including the number of control moves, the number of customers not restored, maximum segment loading, and low and high voltage values. It also provides navigation to the protection validation results and the low and high voltage devices.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.



RAVEN - FISR Plan (Execution Time: 6/3/2022 1:58:04 PM S...

Plan Type	Rank	State	Full Restoration	Third Party Owned Equipment - Revert Plan Execution Mode	Switching Validation Check State	Number of Moves	Show Steps	Mark Devices	Protection Settings Group Updates	Number of Customers De-energized	Number of Customers Restored
FISR Fault	1	Feasible	Yes	--	--	4				2888	1792
FISR Fault	2	Feasible	Yes	--	--	4				2888	1792

RAVEN Plan 2 - Switching Plan Control Moves

Step Number	Device Type	Device Name	Device Id	SubPlan Id	Device Station and Feeder	Operation	Controllable	Initial State	Recommended State	Load Effect
1	Switch	51731	22239714	1	RAVEN F8864	Status Change	Yes	Close	Open	
2	Breaker	8864F	DAA88D70-...	1	RAVEN F8864	Status Change	Yes	Open	Close	Re
3	Switch	51729	22239698	2	RAVEN F8864	Status Change	Yes	Close	Open	
4	Switch	52351	22239778	2	RAVEN F8864	Status Change	Yes	Open	Close	Re

Figure 40. FISR Plan Display

Protection settings are validated and changed as required to suit new configuration. The active protection settings group is selected by finding the settings group with the highest phase trip current that is below the minimum phase fault current on the protected zone (the settings group's phase trip current is multiplied by the appropriate safety factor), and that is greater than the estimated load current (also multiplied by the appropriate safety factor).

FISR plans are optionally validated by the Protection Validation application (the protection validation is executed if the protection validation performance index weight in the used problem formulation is larger than 0). To show the PRV results for a switching plan, click the Protection Validation Results button on the FISR Plan display. For the detailed description of PRV displays, refer to Protection Validation Results.

A FISR Results output file is generated whenever FISR is triggered. This file contains information pertinent to the FISR run including input data, configuration parameters, measurement values, switch eligibility, and control moves.

In addition, a number of alarms associated with the FISR application signal various events associated with FISR runs. These alarms can be enabled and disabled via the DNAF System Configuration display. For more details about the alarms, refer to the "FISR Alarms" section in the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide.*

8.5. Integration with Switching Orders

After a FISR plan has been generated, the plan can be imported into the Switching Orders application by clicking the Switch Order or Daily Log button next to any of the switching steps listed on the FISR Switching Plan Control Moves display. After the Switch Order or Daily Log button is clicked, the Switch Order application form appears. From this display, the switch order or daily log can be modified, validated, and executed. For more information about switch orders' functionality, consult the *Switching Operations User Guide*.

8.6. FISR Manual Inputs

Device parameters and eligibility for reconfiguration applications can be set on tabular displays and device data-entry displays, as described in Application Input Displays.

8.7. FISR Configuration

The FISR configuration attributes are described in the "Switching Reconfiguration Applications" and "Fault Isolation and Service Restoration" chapters of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

9. Automatic Feeder Reconfiguration

The Automatic Feeder Reconfiguration (AFR) application provides switching plans to optimally unload overloaded segments or return all switches within a substation to their normal state, dependent on the problem formulation. The application takes into account the peak loading condition within a user-configured time window.

Similar to the other advanced applications, AFR utilizes a problem formulation that defines one or more operational objectives, including load shedding, return to normal, and reducing overloads and voltage violations.

9.1. Running Automatic Feeder Reconfiguration in Real-Time Mode

The AFR application can be configured to trigger automatically by event. In this case, after DPF is run for a station, the results are analyzed for any overloads on feeder main lines or power transformers. If any are found, AFR is executed to eliminate these violations.

The AFR application can be run manually in real-time mode, as described in section Executing DNAF Applications Manually. In some cases, no plans may be generated by the AFR (for example, if overloads are to be reduced but there are no violations).

9.2. Running Automatic Feeder Reconfiguration in Study Mode

The AFR application can be executed manually in study mode, as described in Executing DNAF Applications Manually.

9.3. Problem Formulations

For the optimization applications, a problem formulation (also called an "objective function") describes a pre-defined optimization goal. A problem formulation includes the objective or goal that is to be achieved, available controls that are allowed, the constraints including the device limits that are to be considered, and parameters required by the analysis engine.

By changing the options and performance index weights of the problem formulation through the DNAF Configuration Editor, different problem formulations can be defined. A problem formulation is generally defined by an analyst or engineer.

For AFR, the following problem formulations are currently defined. Additional problem formulations can be easily created using the DNAF Configuration Editor.

-

Reduce Overloads and Violations

The Reduce Overloads and Violations problem formulation is the nominal problem formulation that is used to relieve violations, taking into account the peak load within the user-configured time window of interest.

- **Load Shedding**

  The Load Shedding problem formulation performs the action of shedding load to reduce overloads in transformers, lines, and breakers within the time window of interest.

  Only non-critical loads are considered for shedding, which is determined by whether the $/kW value of the load (defined in the load model) is greater than the lowest non-zero $/kW value in the network selection.

- **Return to Normal**

  The Return to Normal problem formulation finds a set of switching plans that return all switching devices in their normal state for a small contiguous group of substations in the distribution network. This set of switching plans minimizes the number of customer interruptions while also considering voltage and flow constraints.

9.4. Automatic Feeder Reconfiguration Results

The AFR results are shown on tabular displays. These tabular results include an ordered list of recommended switching device actions.

The following tabular displays are available for the AFR application:

- **AFR Results Summary:** This display shows the number of plans and evaluation and execution times, and it provides navigation to other AFR-related displays that show the plans, statistics, and performance indices.
- **AFR Plan:** This display shows the plan details, including the number of de-energized customers post-plan, maximum segment loading, problem formulation and low and high voltage locations. It also provides navigation to protection validation results displays.
- **AFR Plan Details:** This display shows the individual plan details, including the performance indices.
- **AFR Switching Plan Control Moves:** This display shows the ordered list of recommended switching actions. From this display, each switching device can be marked and located on the user interface.
- **AFR Plan Weighted Performance Indexes:** This display provides plan performance characteristics based on performance indexes for number of moves, dropped customer count, implementation cost, voltage and flow violations, losses, etc.
- **AFR Plan Statistics:** This display shows key plan statistics in two horizontal panes, including the number of control moves, the number of customers not restored, maximum segment loading, and low and high voltage values. It also provides navigation to the protection validation results

and the low and high voltage devices.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

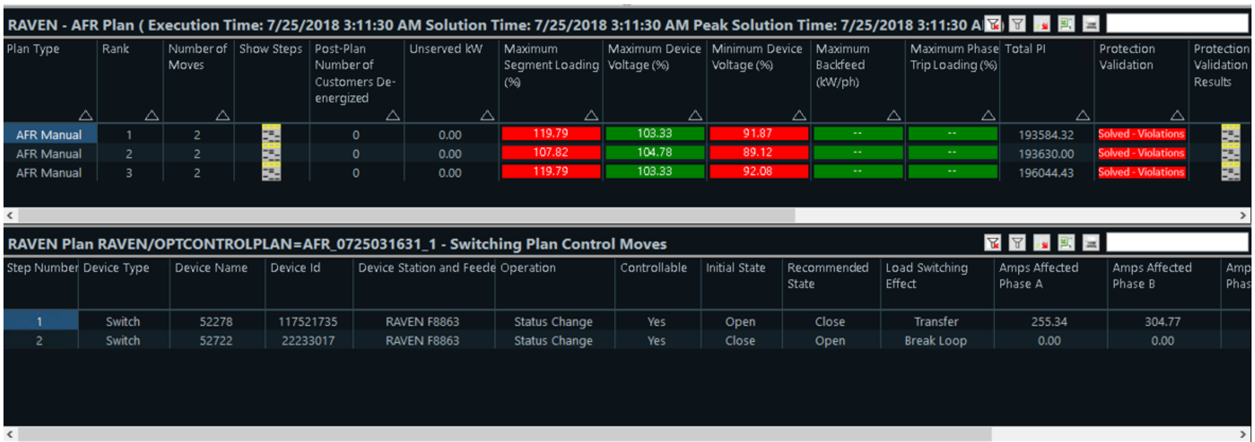


Figure 41. AFR Plan Display

The reconfiguration in AFR may require adjusting of protection settings. The active protection settings group is selected by finding the settings group with the highest phase trip current that is below the minimum phase fault current on the protected zone (the settings group's phase trip current is multiplied by the appropriate safety factor), and that is greater than the estimated load current (also multiplied by the appropriate safety factor).

AFR plans are validated by the Protection Validation (PRV) application (PRV is executed if the protection validation performance index weight in the used problem formulation is larger than 0). To show the PRV results for a switching plan, click the Protection Validation Results button on the AFR Plan display. For the detailed description of PRV displays, refer to Protection Validation Results.

An AFR Results output file is generated whenever AFR is triggered. This file contains information pertinent to the AFR run including input data, configuration parameters, measurement values, switch eligibility, and control moves.

In addition, a number of alarms associated with the AFR application signal various events associated with an AFR run. These alarms can be enabled and disabled via the DNAF System Configuration display. For more details about the alarms, refer to the "AFR Alarms" section in the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide.*

9.5. Integration with Switching Orders

After an AFR plan has been generated, the plan can be imported into the Switching Orders application by clicking the Switch Order or Daily Log button next to any of the switching steps listed on the AFR Switching Plan Control Moves display. After the Switch Order or Daily Log button is clicked, the Switch Order application form appears. From this display, the switch order or daily log can be modified, validated, and executed. For more information about switch orders' functionality, consult the *Switching Operations User Guide*.

9.6. AFR Manual Inputs

Device parameters and eligibility for reconfiguration applications can be set on tabular displays and device data-entry displays, as described in Application Input Displays.

9.7. AFR Configuration

The AFR configuration attributes are described in the "Switching Reconfiguration Applications" and "Automatic Feeder Reconfiguration" chapters of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

10. Planned Outage Study

The Planned Outage Study (PLOS) application generates plans to isolate a component, minimizing the number of interrupted customers. PLOS is a study-mode-only application.

The PLOS application uses methods similar to the FISR application for generating the switching plans. PLOS also utilizes the same problem formulations as FISR (for those problem formulations, see Fault Isolation and Service Restoration). The only major difference is that PLOS first transfers the affected load and then isolates the component to be taken out of service to prevent the unnecessary dropping of load.

10.1. Running Planned Outage Study in Study Mode

To execute the Planned Outage Study application in study mode, first select a device that is to be taken out of service. This device can be a line section, a station transformer, or any other topology device. PLOS generally creates a temporary loop, and then breaks the loop by isolating the smallest switchable section. In some cases, such as offloading a heavily loaded line or a power transformer, additional switching operations may be required to reconfigure the system so that violations can be avoided.

10.2. Planned Outage Study Results

The PLOS results are shown on tabular displays. These tabular results include an ordered list of recommended switching device actions.

The following tabular displays are available for the PLOS application:

- **Planned Outage Service:** This display contains summaries of PLOS plans. It shows the number of plans and evaluation and execution times, and it provides navigation to other PLOS-related displays that show the plans, statistics, and performance indices.
- **PLOS Plan:** This display shows the plan details, including some important plan statistics including the number of customers not restored, maximum segment loading, and low and high voltage values. It also provides navigation to protection validation results and plan details displays.
- **PLOS Plan Detail Display:** This display shows the individual plan details, including the performance indices.
- **PLOS Switching Plan Control Moves:** This display shows the ordered list of recommended switching actions. From this display, each switching device can be marked and located on the user interface.
- **PLOS Plan Weighted Performance Indexes:** This display provides plan performance characteristics based on performance indexes for number of moves, dropped customer count, implementation cost, voltage and flow violations, losses, etc.
-

PLOS Plan Statistics: This display shows key plan statistics in two horizontal panes, including the number of control moves, the number of customers not restored, maximum segment loading, and low and high voltage values. It also provides navigation to the protection validation results and the low and high voltage devices.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

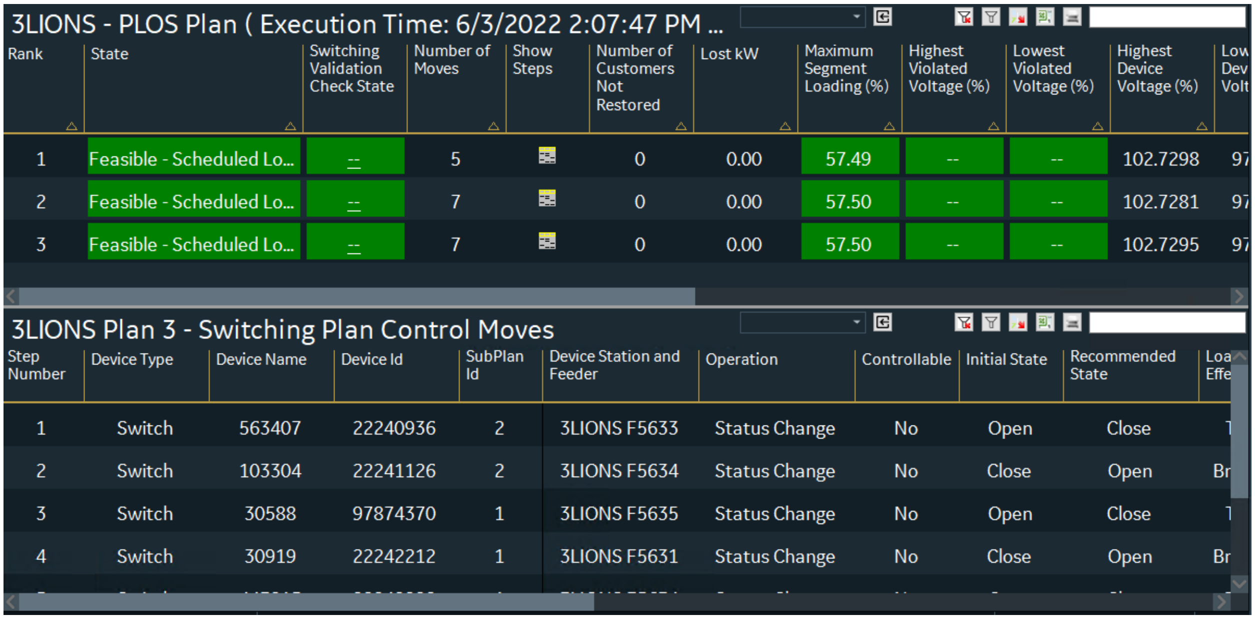


Figure 42. PLOS Plan Display

Protection settings are validated for the network configuration in PLOS. The active protection settings group is selected by finding the settings group with the highest phase trip current that is below the minimum phase fault current on the protected zone (the settings group's phase trip current is multiplied by the appropriate safety factor), and that is greater than the estimated load current (also multiplied by the appropriate safety factor).

PLOS plans can also be validated by the Protection Validation (PRV) application. PRV is executed if the protection validation performance index weight in the used problem formulation is larger than 0. To show the PRV results for a switching plan, click the Protection Validation Results button on the PLOS Plans display. For the detailed description of PRV displays, refer to Protection Validation Results.

A PLOS Results output file is generated whenever PLOS is triggered. This file contains information pertinent to the PLOS run, including input data, configuration parameters, measurement values, switch eligibility, and control moves.

10.3. Integration with Switching Orders

After a PLOS plan has been generated, the plan can be imported into the Switching Orders application by clicking the Switch Order or Daily Log button next to any of the switching steps listed on the PLOS Switching Plan Control Moves display. After the Switch Order or Daily Log button is clicked, the Switch Order application form appears. From this display, the switch order or daily log can be modified, validated, and executed. For more information about switch orders' functionality, consult the *Switching Operations User Guide*.

10.4. PLOS Manual Inputs

Device parameters and eligibility for reconfiguration applications can be set on tabular displays and device data-entry displays, as described in Application Input Displays.

10.5. PLOS Configuration

The PLOS configuration attributes are described in the "Switching Reconfiguration Applications" and "Planned Outage Study" chapters of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

11. Fault Isolation and Service Restoration Contingency Analysis

The Fault Isolation and Service Restoration Contingency Analysis (FISRCA) application executes a FISR analysis for every switchable section in a substation. A "switchable section" is the portion of the network that is bounded by feeder main switching devices (most commonly disconnect switches, reclosers, and breakers). FISRCA is a study-mode-only application.

FISRCA packages up a group of FISR executions, sequentially placing a fault on each switchable section, and then solving FISR to determine if a valid restoration plan can be found that restores power to all customers connected to the non-faulted portion of the network.

FISRCA uses the same problem formulations and methodology as the FISR application, as described in Fault Isolation and Service Restoration. Depending on the size and number of switchable sections in the solution network, the total computational time may be high.

11.1. Running FISRCA in Study Mode

To run FISRCA in study mode, select a component and bring up the DNAF Solution display. After you select the FISRCA application and the desired problem formulation, the FISRCA application internally runs a FISR analysis on every switchable section for the selected station.

11.2. FISRCA Results

The FISRCA results are shown on tabular displays. These tabular results include the top FISR switching plan for each switchable section.

The following tabular displays are available for the FISRCA application:

- **FISRCA Solution Overview:** This display shows plan execution and plan solution time. It also shows the total number of customers that cannot be restored for all evaluated switchable sections. This display provides navigation to the FISRCA Results Summary display.
- **FISRCA Results Summary:** This display shows the summary of every switchable section, including boundary switches, the availability of a good FISR plan, and the number of customers not restored. This display also provides navigation to FISRCA Plan, FISRCA Plan Statistics, and FISRCA Weighted Performance Indexes displays for any switchable section.
- **FISRCA Plan:** This display shows summary information for each plan, including the number of control moves, number of customers not restored, unserved kW and maximum/minimum load voltages. It also provides navigation to FISRCA Switching Plan Control Moves, FISRCA Plan Details, and Protection Validation Results displays. This display is available for every plan for every switchable section.
- **FISRCA Plan Details:** This display shows the individual plan details, including the performance indices. This display is available for every plan for every switchable section.

- **FISRCA Switching Plan Control Moves:** This display shows the ordered list of recommended switching actions. From this display, each switching device can be marked and located on the user interface. This display is available for every plan for every switchable section.
- **FISRCA Weighted Performance Indexes:** This display provides plan performance characteristics based on performance indexes for number of moves, dropped customer count, implementation cost, voltage and flow violations, losses, etc.
- **FISRCA Plan Statistics:** This display shows key plan statistics in two horizontal panes, including the number of control moves, the number of customers not restored, maximum segment loading, and low and high voltage values. It also provides navigation to the protection validation results and the low and high voltage devices.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

Protection settings are validated for the network configuration in FISRCA. The active protection settings group is selected by finding the settings group with the highest phase trip current that is below the minimum phase fault current on the protected zone (the settings group's phase trip current is multiplied by the appropriate safety factor), and that is greater than the estimated load current (also multiplied by the appropriate safety factor).

FISRCA plans are validated by the Protection Validation (PRV) application (PRV is executed if the protection validation performance index weight in the used problem formulation is larger than 0). To show the PRV results for a switching plan, click the Protection Validation Results button on the FISRCA Plans display. For the detailed description of PRV displays, refer to Protection Validation Results. These displays are available for every plan for every switchable section.

A FISRCA Results output file is generated whenever FISRCA is triggered. This file contains information pertinent to the FISRCA run, including input data, configuration parameters, measurement values, switch eligibility, and control moves.

11.3. FISRCA Manual Inputs

Device parameters and eligibility for reconfiguration applications can be set on tabular displays and device data-entry displays as described in Application Input Displays.

11.4. FISRCA Configuration

The FISRCA configuration attributes are described in the "Switching Reconfiguration Applications" and "Fault Isolation and Service Restoration Contingency Analysis" chapters of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

12. Faulted Section Determination

The Faulted Section Determination (FSD) function provides switching plans to identify a fault and isolate faulted sections. The FSD function applies a switching-based approach with step-by-step opening and closing switching cycles. At stage 1, a faulted feeder downstream of a faulted power transformer is identified and isolated. At stage 2, the faulted section is identified and isolated, and the fault is placed in the faulted section.

12.1. Running FSD

Note

For using the FSD application, the FSD plug-in must be enabled.

FSD can be run only in the real-time mode.

You can run FSD manually for the downstream sections of the set fault indicators. FSD can also solve for the following event triggers:

- Power transformer arc suppression coil (ASC or Peterson coil) fault indicator status is updated from "Not Set" to "Set".
- Feeder fault indicator status is updated from "Not Set" to "Set".

You can configure FSD event triggers on the FSD page in the DNAF Configuration Editor.

Use the Faulted Section Determination window to run FSD and review the FSD summary. This window can be opened from the breaker or power transformer right-click menu (for power transformers with ASC).

Running FSD Stage 1

You can manage the FSD Stage 1 execution using the FSD Stage 1 tab on the Faulted Section Determination window.

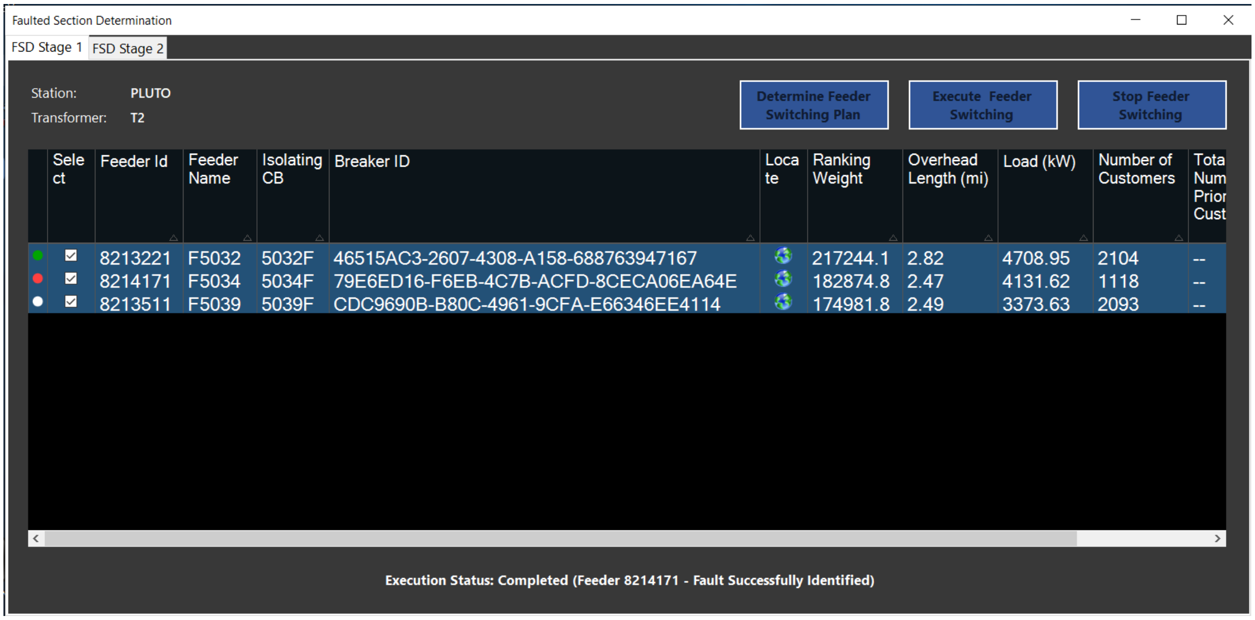


Figure 43. FSD Stage 1 Display

The following fields and controls appear on the FSD Stage 1 display:

- **Station**: Station name.
- **Transformer**: Power transformer name.
- **Determine Feeder Switching Plan:** Click to rank feeders and create an FSD Stage 1 switching plan.
- **Execute Feeder Switching:** Click to execute the feeder switching plan. Feeders will be opened and closed one-by-one according to their ranking to determine the faulted feeder.
- **Stop Feeder Switching:** Click to stop the feeder switching.

The following information is shown on the FSD Stage 1 grid tabular display:

- **Status**: FSD Stage 1 processing status for the feeder breaker (None, Ineligible, Pending, Executing, Not Faulted, or Faulted).
- **Select:** Click to select a feeder.
- **Feeder ID:** Displays the feeder ID.
- **Feeder Name:** Displays the feeder name.
- **Isolating CB:** Displays the feeder head circuit breaker name.
- **Breaker ID:** Displays the feeder head circuit breaker ID.
- **Locate:** Locates the feeder head circuit breaker on the network view.
- **Ranking Weight:** Displays the feeder weight ranking.
- **Overhead Length:** Displays the total feeder overhead conductor length (in miles).
- **Load:** Displays the total load (in kW) connected to the feeder.
-

Number of Customers: Displays the total number of customers on the feeder.

- **Total Number of Priority Customers:** Displays the total number of priority customers on the feeder.
- **Generation:** Displays the total generation (in kW) connected to the feeder.

Running FSD Stage 2

You can manage the FSD Stage 2 execution using the FSD Stage 2 tab on the Faulted Section Determination window.

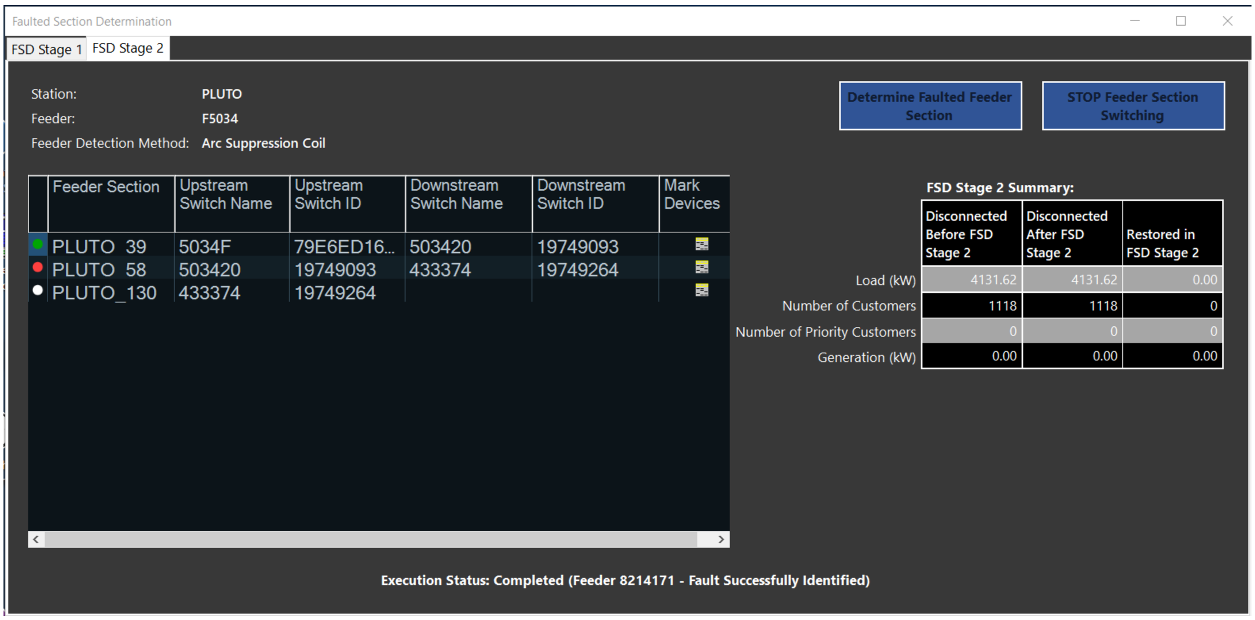


Figure 44. FSD Stage 2 Display

The following controls appear on the FSD Stage 2 display:

- **Determine Faulted Feeder Section:** Click to determine the faulted feeder section by step-by-step opening and closing switching of the feeder breaker and SCADA-controllable field switching devices on the faulted feeder.
- **STOP Feeder Section Switching**: Click to stop feeder section switching.

The following information and controls are shown on the FSD Stage 2 Summary grid tabular display:

- **Status**: FSD Stage 2 processing status for the feeder automated switch (None, Ineligible, Pending, Executing, Not Faulted, or Faulted).
- **Feeder Section:** Displays the feeder section ID.
- **Upstream Switch Name:** Displays the upstream automated switching device name for the feeder section.
- **Upstream Switch ID:** Displays the upstream automated switching device ID.
- **Downstream Switch Name:** Displays the downstream automated switching device name for

the feeder section.

- **Downstream Switch ID:** Displays the downstream automated switching device ID.
- **Mark Devices:** Marks feeder section's upstream and downstream automated switching devices on the network view.

The FSD Stage 2 Summary grid shows the FSD disconnection and restoration summary for load, customers, and generation.

12.2. Problem Formulations

For the advanced applications, a problem formulation (also called an "objective function") describes a pre-defined optimization goal. A problem formulation includes the objective that is to be achieved, available controls that are allowed, the constraints including device limits that are to be considered, and parameters required by the analysis engine.

By changing the options and weighting factors of the problem formulation through the DNAF Configuration Editor, different problem formulations can be defined. A problem formulation is defined by an analyst or engineer.

For FSD, in the currently defined problem formulation, feeders with the set fault indicators have higher ranking so that they are processed first during the FSD switching cycles. Additional problem formulations can be easily created using the DNAF Configuration Editor.

12.3. FSD Configuration

The FSD configuration attributes are described in the "Switching Reconfiguration Applications" and "Faulted Section Determination" chapters of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.

13. Surgical Load Shed

The Surgical Load Shed functionality is part of a combined ADMS and Scada application that prioritizes and sheds circuits to obtain a desired target reduction in system load.

- The DMS Surgical Load Shed (SLS) functionality controls configuration to enable or disable circuits for load shed. It defines the order in which circuits should be shed.
- The Rotating Load Shed (RLS) Scada application controls the actual load shedding operations, such as opening and closing breakers and reclosers and the shedding timing. In addition, it automatically places tags on breakers and reclosers that have been tripped as part of a load shed event.

The RLS sheds and restores load automatically from a group of loads on a rotating basis at an operator-specified cycle time. For more information, refer to the *Rotating Load Shed Operator's Guide*, which is part of the Scada product.

Unlike the other DNAF applications, the SLS application is not an analysis that is run to obtain a set of results. Instead, it continually updates Scada with a set of enabling and ordering data to be used by the RLS application.

13.1. Circuit Classification

The circuits in the system are classified by the importance and availability for load shed by certain conditions, such as the number of critical customers, amount of DER generation, and load category. The classification value is assigned to a breaker or recloser that feeds the circuit to indicate the device priority for load shed.

Classification values are defined in the DNAF Configuration Editor. The default classification values are as follows:

- **Do Not Shed (1):** Never include these circuits in load shed.

  Circuits that supply downtown underground feeders.

- **Critical (2)**: Require operator command to include these circuits.

  Circuits that supply customers vital to society - major airports, major water treatment facilities, major hospitals, major emergency services, and communications facilities.

- **Medium (3)**: Require operator command to include these circuits.

  Circuits that supply customers vital to society - regional airports, large water treatment facilities, and regional hospitals.

- **Low (4)**: First circuits to be shed.

  Circuits with mostly residential load and with no critical customers or downtown underground systems supplied.

A device is not considered for load shed if it or a downstream device has a Do Not Shed tag applied.

When calculating the circuit classification for a device, the most restrictive value of the following is applied:

- Circuit classification value of the applied rule set
- Circuit classification value of the associated condition (for example, the number of critical customers)
- Nominal circuit classification value defined in the static model file for the device
- Nominal circuit classification value defined in the static model file for the device feeder

In addition, all downstream devices (for example, field reclosers and devices that belong to sub-fed substations) are checked, even if such devices are below another load shed device. In such cases, the restrictive load shed conditions are propagated up to the upstream devices.

To define the load shed order for the circuits of the same classification, the topology inclusion rules and load drop ordering settings are applied.

The circuit classification is automatically recalculated for the following events:

- Client restart.
- Backbone or lateral topology changes.
- New station model loading.
- DPF executed for a station. For sub-fed stations, the power flow must be run on both the upstream and downstream stations.
- SLS parameter changes in the DNAF Configuration Editor.
- SLS parameter changes on the Real-Time DNAF System Configuration display.

13.2. Rule Sets

Rule sets combine sets of circuit classification and topology inclusion rules that are applicable to certain network conditions. A rule set has its own circuit classification value. In addition, it includes rules to specify the circuit classifications for special conditions that may occur on the circuit and specify how a device is selected for load shed if two or more devices are in series.

Several rule sets may be defined for the system - for example, one for normal circumstances and the others for emergency situations to consider more devices for shedding.

Different rule sets can be assigned to different stations. A rule set associated with a station can be defined at run time using the Real-Time DNAF System Configuration display.

13.3. Surgical Load Shed Results

The Surgical Load Shed summary is provided on tabular displays:

- **Load Shed Overview:** This display shows the load shed status for all stations in the system. The data includes the number of applicable, disabled, and currently shed devices. From this display, the user can navigate to the Load Shed Station Details display for a particular station.
- **Load Shed Station Details:** This display shows the details for the load shed devices associated with the selected station. The data includes the device type, load shed status, circuit classification, and Cold Load Pickup (CLPU) grade.
- **Load Shed Sort Values:** This display shows the load shed sorting values for all devices that are eligible for load shed. The data includes the device type, load shed status, load shed class, and sort value.

For more information about tabular displays, refer to the *ADMS Display Reference Manual*.

13.4. SLS Configuration

The SLS configuration attributes are described in the "Surgical Load Shed" chapter of the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*.